

When Claims Become Auditable: Distributed AASC/UEAP Claim-Status Audit and the Game Theory of Scientific Authority

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Abstract

Scientific gatekeeping has traditionally concentrated the power to validate claims in journals, referees, editorial boards, professional societies, prestige networks, and institutional review processes. This manuscript analyzes a different strategic condition: claim-status audit becomes portable, partially mechanizable, and non-permissioned. The central question is not whether journals adopt a new internal review procedure, but what happens when institutions can no longer prevent structured claim-status audit from being run externally.

The Admissibility and Standing Constraint framework (AASC), operationalized through the Universal Exactness Audit Protocol (UEAP), supplies the audit grammar. A capable language model, human evaluator, or hybrid system supplies an execution layer; it does not author the protocol's target, carrier, blocker, status, laundering, severity, or reportability categories. When the protocol and a target manuscript are placed in the same evaluation context, the process can produce a registry-style classification of claim target, requested report status, carrier, scope, calibration ancestry, unresolved blockers, possible report-status laundering, severity, and strongest reportable formulation.

The paper distinguishes four layers that distributed-audit rhetoric often collapses: audit execution, audit publication, audit standing, and audit consequence. Execution can be open without every output becoming authoritative. Audit standing is earned through target fidelity, basis declaration, execution provenance, passage grounding, registry completeness, reproducibility, severity discipline, counteraudit, and right of reply. Consequence is a further social or institutional response that should be proportional to the audit's standing and severity. Distributed audit therefore democratizes execution, not authority.

The game-theoretic model separates raw audit occurrence from standing-bearing consequential detection. If potential auditor k initiates an audit with probability p_k , detects the relevant mismatch with probability q_k , reaches an uptake-sensitive audience with probability r_k , and produces an audit with standing weight w_k , then

$$\delta_{\text{eff}}^w = 1 - \prod_k (1 - p_k q_k r_k w_k)$$

is the benchmark probability of at least one standing-bearing consequential detection under

independent source events. The paired benchmark

$$\eta_{\text{eff}}^w = 1 - \prod_k (1 - p_k f_k r_k w_k)$$

tracks standing-weighted false-positive exposure, where f_k is the conditional probability of a false or materially overstated finding. The standing term is an audit-output property, not an auditor-prestige score, and any implementation of $w_k = W(P_k, R_k, C_k, S_k, H_k)$ is constrained by declared provenance, replication, convergence, severity, response-handling, and consequence-specific admissibility rules. The central design question is therefore whether standing-bearing correction value exceeds standing-bearing false-finding harm, audit noise, chilling effects, and capture risk.

The analysis develops volunteer audit supply, self-audit, institutional audit reversal, external venue classification, pressure migration to audit-basis manipulation, symbolic compliance, model and prompt variance, audit-basis fragmentation, prompt injection, harassment, severity governance, and registry memory. Institutions do not disappear, and LLM-assisted audits do not become truth oracles. Rather, institutions lose exclusive control over whether structured, protocolized claim-status audit can be executed against accessible claims and become participants in a reciprocal audit field. Journals may still audit claims, but claims, review practices, press language, and institutional rigor statements can also be audited externally.

This reference edition does not claim population-level validation of UEAP execution or report a completed empirical deployment. Full empirical validation requires a multi-evaluator calibration study beyond the scope of this solo-authored formal-methodological record. The manuscript instead supplies the complete reference architecture: the protocol grammar, standing and consequence rules, threat model, registry schema, controlled execution demonstrations, machine-readable tooling artifacts, and validation design required to test stronger empirical claims. AASC/UEAP supplies the audit grammar; mechanized execution supplies scale; provenance, replication, counteraudit, and severity discipline determine standing; and game theory identifies the strategic response when scientific institutions can no longer monopolize the conditions under which structured audit occurs.

Keywords: distributed claim-status audit; scientific authority; game theory; AASC; UEAP; non-permissioned audit; LLM-assisted execution; audit standing; audit provenance; false-positive exposure; external audit; audit-basis fragmentation; reciprocal authority; metascience.

Status and identifiers of this reference edition

This manuscript is the complete reference edition of the distributed AASC/UEAP claim-status audit architecture. It is designed to preserve, in one citable and auditable record, the conceptual thesis, formal benchmark models, standing and consequence system, protocol-execution demonstrations, threat model, governance safeguards, registry templates, machine-readable implementation artifacts, and empirical validation design.

Its report status is formal-methodological and architectural. The controlled demonstrations establish disclosed protocol execution and status preservation; they do not establish population accuracy, field-wide reliability, or beneficial live deployment. Full empirical validation requires a multi-evaluator calibration study and is outside the scope of this solo-authored reference edition. That boundary is part of the manuscript’s claim discipline, not a provisional qualification.

Reference field	Active identifier or status
Manuscript release	WCBA-REF-6.2.0
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Release date	18 June 2026
Evidential status	E1 controlled execution demonstrated; no E2 population calibration or E3 live-deployment claim
Update policy	Semantic versioning; major, minor, and patch triggers are defined in Appendix E
Machine-readable record	JSON and YAML schemas, example records, prompts, validator/parser, and local registry interface are included in the project archive

How to use this reference edition

The manuscript may be read linearly, but its components also support targeted use. The pathways below identify the shortest route to each principal function.

Table 1: Quick-reference pathways through the manuscript.

Reader objective	Recommended pathway
Understand the central thesis	Abstract; Sections 1, 4, 5, 7, and 22.
Understand AASC/UEAP as a protocol	Sections 2–3; Appendices B, C, and H.
Run a structured audit	Section 17; Appendices H, J, K, and N.
Apply standing and consequence rules	Section 4; Section 13; Appendices D and K.
Inspect the game-theoretic model	Sections 5–10; Appendices A and G.
Assess false-positive, harassment, and capture safeguards	Sections 13 and 19; Appendix I.
Examine basis disagreement and reproducibility	Section 12; Appendix E.
Evaluate the controlled demonstrations	Appendix J, including the matched negative control and self-audit.
Assess empirical validation requirements	Section 18; Appendix L.
Use machine-readable artifacts	Appendix N and the project directory <code>operational/tooling/</code> .

Reader objective	Recommended pathway
Cite or version the reference architecture	This front matter; Appendix E; the project files <code>VERSION</code> , <code>CITATION.cff</code> , and <code>REFERENCE_IDENTIFIERS.json</code> .

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1 Introduction: when claim audit is no longer permissioned

1.1 The old direction of authority

Scientific communication has long relied on institutions to economize on evaluation. Journals, editorial boards, referees, professional societies, universities, funding agencies, and prestige networks do not merely transmit claims; they help determine which claims receive attention and which report statuses are treated as credible. A reader rarely reconstructs every experiment, proof, dataset, model choice, or evidential dependency before deciding whether a paper deserves uptake. Institutional signals substitute for some of that work. This arrangement is understandable, but it makes the validation interface asymmetrical: institutions inspect claims, while the institutional process that conferred credibility is often difficult for outsiders to inspect in the same structured way.

The sociology and economics of science have long emphasized priority, credit, signaling, division of cognitive labor, communication structure, and institutional reward (Merton 1957; Merton 1973; Kitcher 1990; Dasgupta and David 1994; Strevens 2003; Zollman 2007; Heesen 2017). The audit literature has likewise shown that verification practices can improve accountability while also becoming ceremonial, captured, or decoupled from underlying work (Power 1997; Meyer and Rowan 1977; Bromley and Powell 2012; Goodman and Tréhu 2022). What has been less directly modeled is the strategic consequence of a portable claim-audit protocol that can be run outside the gatekeeping hierarchy.

The relevant technological condition is straightforward. A user places an explicit audit protocol in the context of a capable large language model or other structured evaluator, adds a target manuscript, and requests a registry-style audit. The hierarchy matters: the protocol supplies the structure; the evaluator supplies execution. The evaluator does not invent the target, carrier, blocker, status, or reportability categories. The output is not a final truth verdict. It identifies the claim target, requested status, supporting carrier, domain and boundary conditions, target or proxy ancestry, evidence dependencies, unresolved blockers, possible status promotion, and strongest reportable wording. Empirical work already shows that language models can generate useful scientific feedback, structured reviews, and targeted reviewing assistance, while also displaying important weaknesses in correctness, criticality, bias, long-document processing, and adversarial robustness (Liang et al. 2024; Zhou et al. 2024; Jin et al. 2024; Idahl and Ahmadi 2025; Choi et al. 2026). The present paper uses those capabilities more narrowly: the model is an execution layer for a declared protocol, not an autonomous authority on scientific truth.

Once such an audit can be run by authors, readers, rivals, reviewers, funders, journalists, post-publication commentators, or independent observers, the strategic question changes. It is no longer only:

Will journals adopt a new internal review procedure?

It becomes:

What happens when journals and institutions cannot prevent the procedure from being run externally?

1.2 Distributed audit as an institutional shock

This paper calls the resulting condition *distributed claim audit*. “Distributed” does not imply blockchain, decentralization in a technical networking sense, or consensus by vote. It means that

execution rights are not monopolized by one institutional actor. The protocol may be public, the target paper may be public, and multiple evaluators may independently generate audit outputs. Formal adoption by a journal is one channel, but it is not a precondition for external audit pressure.

The shift is substantial. Under an institutional audit model, a journal chooses an inspection probability p_J . Under distributed audit, the effective probability of an audit depends on a set K of potential auditors:

$$p_{\text{eff}} = 1 - \prod_{k \in K} (1 - p_k)$$

under independent audit initiation. Raw occurrence is not yet the strategically relevant event. If audit k also has detection accuracy q_k , public reach or consequential visibility r_k , and audit-standing weight w_k , then the benchmark probability of at least one standing-bearing consequential detection is

$$\delta_{\text{eff}}^w = 1 - \prod_{k \in K} (1 - p_k q_k r_k w_k).$$

This paper treats δ_{eff}^w as its central strategic variable. The paired quantity η_{eff}^w tracks standing-weighted false-positive exposure, so distributed audit is evaluated by the balance between consequential correction and consequential error rather than by audit volume alone. The institutional choice $p_J = 0$ no longer implies zero exposure whenever some external channel has positive $p_k q_k r_k w_k$. Refusal changes who performs the audit, not whether standing-bearing external audit can occur.

This produces a reversal in the ordinary direction of authority. Journals may still audit authors, but authors and readers can audit journals' claims to rigor. A publisher may claim that its papers are carefully reviewed, that a result has been independently validated, or that acceptance signals methodological quality. Those statements are themselves claims with targets, carriers, domains, evidence networks, and report ceilings. If a portfolio of published claims repeatedly exhibits externally reproducible status mismatches, the institution's rigor claim becomes audit-vulnerable. Scientific authority becomes reciprocal rather than unidirectional.

1.3 Execution is open; standing is earned

Non-permissioned audit does not mean that every generated output deserves uptake. The paper distinguishes audit execution, audit publication, audit standing, and audit consequence. Execution is the act of applying the protocol to a target; publication circulates the result; standing is the credibility earned by the audit's own carrier. Provenance, exact target version, declared basis, passage grounding, independent rerun, severity discipline, counteraudit, and right of reply determine that standing. The governing sentence is:

Distributed audit democratizes execution, not authority. Anyone can run an audit; not every audit has standing.

This distinction is not a retreat from the radical thesis. It is the mechanism by which non-institutional scrutiny becomes capable of altering authority without turning every low-quality output into an accusation. Section 4 formalizes the distinction, and Appendix J demonstrates an explicit overclaim audit, a matched negative control, and a self-audit of the paper.

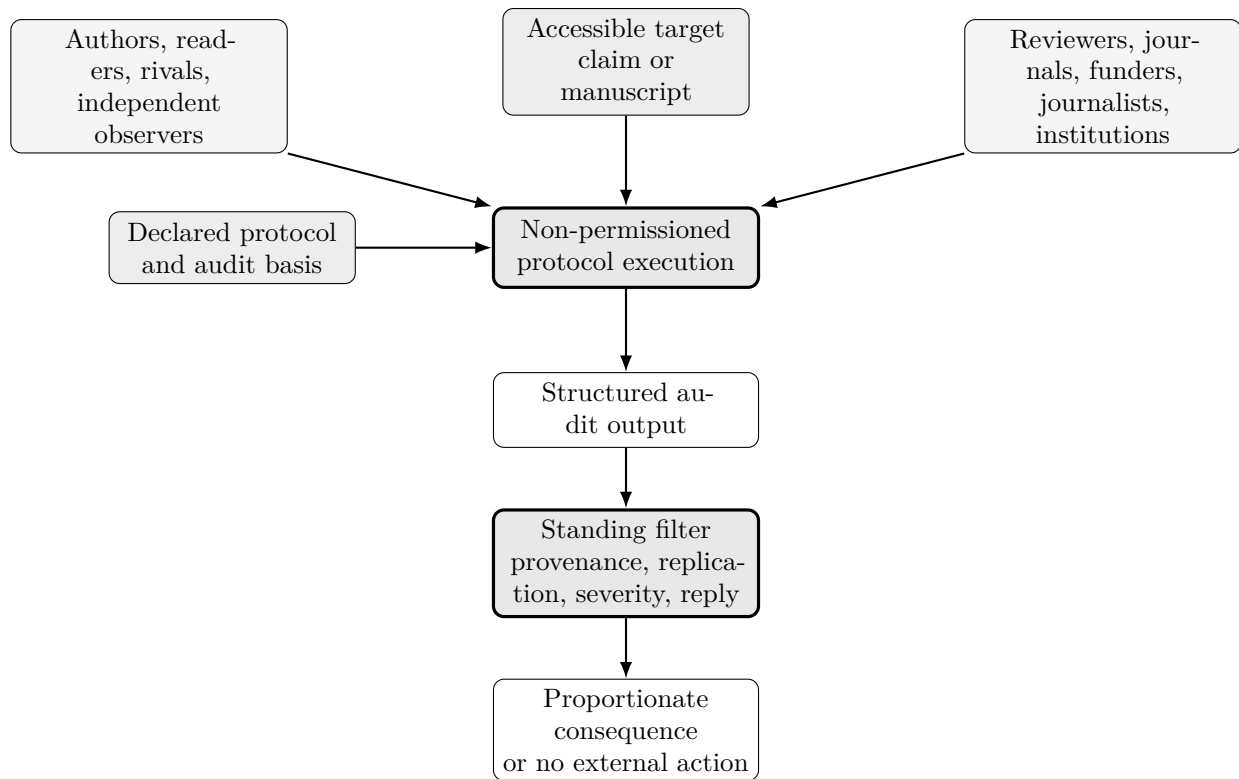


Figure 1: Distributed audit topology. Multiple actor classes can initiate protocol execution on an accessible target. Execution does not bypass the later standing and consequence filters.

1.4 The audit is not a truth oracle

The radical feature of distributed audit is not infallibility. A language model can misunderstand a paper, omit a relevant argument, hallucinate a defect, follow hidden prompt injection, over-weight prestige cues, or produce different outputs under different prompts and model versions. Studies of LLM reviewing report both usefulness and substantial limitations; hidden instructions in manuscripts have already demonstrated that AI-assisted review can be manipulated (Liang et al. 2024; Zhou et al. 2024; Lin 2025). A distributed regime therefore requires provenance, rerunability, basis declaration, severity discipline, and right of reply.

The claim is narrower and strategically sufficient:

A portable audit makes claim-status mismatch cheap enough to challenge that the expected payoff of strong scientific language changes.

The argument does not require an LLM to function as a reliable final scientific authority. It requires only that protocol-guided evaluators reduce the marginal cost of producing structured audit candidates. Execution is open; standing and consequence are earned through provenance, replication, convergence, severity discipline, counteraudit, and domain review where needed.

The audit asks whether a report exceeds its declared carrier. A fitted model may eventually turn out to be predictively successful; nevertheless, if the same target data were used to select and tune the model, the immediate report may still be calibrated rather than independently predictive. A simulation may capture the physical system; nevertheless, if its boundary, validation, and uncertainty carrier remain unfixed, the report “proves” may exceed what is presently licensed. A benchmark score may correlate with broad capability; nevertheless, the benchmark carrier alone may not support deployment readiness. The auditor need not settle the entire scientific controversy in order to identify these report-status gaps.

1.5 The new game

The introduction of portable audit creates several linked games.

1. **Audit supply.** Who chooses to run an audit when the output is a public good but the cost is private? How does LLM assistance change the volunteer’s dilemma?
2. **Self-audit.** Which authors preemptively audit and narrow their own claims before external challenge?
3. **External exposure.** How does the number, independence, competence, and reach of possible auditors affect deterrence?
4. **Institutional reversal.** How are journals and review systems classified when their own rigor claims are externally tested?
5. **Pressure migration.** When direct overstatement becomes costly, do agents shift effort toward manipulating the audit basis, prompts, protocol versions, or report categories?
6. **Audit pluralism.** How should communities respond when different models, prompts, or bases produce different audit outputs?
7. **Weaponization.** How can cheap audit distinguish genuine blocker exposure from spam, harassment, or selective attacks?
8. **Normalization.** How do persistent audit records alter reputation, venue choice, and the long-run meaning of scientific report terms?

The paper develops a modular game-theoretic framework for those responses. It does not present a complete equilibrium theory of science. Its formal contribution is not a new solution concept. It applies familiar tools—deterrence, volunteer’s dilemmas, signaling, reputation, institutional sorting, and standard competition—to a previously under-modeled object: non-permissioned claim-status audit.

1.6 Formal architecture and result hierarchy

The formal results below are benchmark decompositions and comparative-static results. Their contribution is not a new solution concept or a difficult probability identity, but a unified strategic architecture for non-permissioned claim-status audit. The theorem and proposition labels identify load-bearing model steps and make their assumptions inspectable; they should not be read as claims that the underlying probability products, deterrence inequalities, or volunteer-game calculations are mathematically deep in isolation.

The paper is organized around a small set of load-bearing results. The first defines the distributed exposure mechanism; the next results explain institutional non-opt-out, reciprocal authority, basis contestation, abuse control, and long-run memory. The remaining models are applications or special cases of this architecture.

Table 2: Central result hierarchy.

Result	Function in the argument
Consequential Distributed Audit Theorem	Defines standing-weighted consequential detection, δ_{eff}^w , and supplies the main exposure and deterrence mechanism.
Standing-Weighted Audit Balance Proposition	Defines η_{eff}^w and the net-value condition that balances correction against false accusation and audit noise.
No Full Institutional Opt-Out Proposition	Shows that institutional non-adoption does not eliminate external audit pressure when a standing-bearing external channel remains available.
Institutional Audit Reversal Proposition	Treats an institution’s claim to reliable review as an auditable portfolio-level claim with explicit sampling, severity, standing, and response conditions.
Pressure Migration Theorem and Audit-Basis Fragmentation Theorem	Predict that strategic effort shifts toward protocol version, target selection, carrier standards, prompts, and severity rules when direct report inflation becomes costly.
Distributed Audit Theater Constraint	Explains when symbolic or checklist compliance survives external substantive reruns.
Severity-Disciplined Challenge Proposition	Separates legitimate blocker exposure from disproportionate public accusation and strategic harassment.
Registry Memory Corollary	Explains how persistent, provenance-bearing audit histories reduce current monitoring requirements while preserving repair and correction histories.

Supporting results address volunteer audit supply, voluntary self-audit, external venue classification, audit convergence, public counteraudit, bundled status promotions, and the A0–A5 standing

ladder. Earlier journal-adoption and within-venue report-status models appear as special cases of the distributed-audit framework.

1.7 Reader’s roadmap

The argument is deliberately layered because distributed audit involves an exposure mechanism, an institutional response, and a standing-governance problem. The following roadmap separates the thesis-bearing analysis from the operational support.

Table 3: Roadmap through the paper.

Part	Function
Protocol and execution architecture	Defines AASC/UEAP as the audit structure, distinguishes the evaluator from the protocol, and separates execution, publication, standing, and consequence.
Distributed exposure and audit supply	Develops raw occurrence, standing-weighted true and false consequential exposure, audit-family dependence, volunteer audit supply, and the no-full-opt-out result.
Reciprocal authority and strategic response	Derives institutional audit reversal, voluntary self-audit, external venue classification, pressure migration, and symbolic compliance.
Pluralism and governance	Analyzes model and prompt variance, audit-basis fragmentation, prompt injection, harassment, severity, counteraudit, and registry memory.
Operational demonstrations and empirical program	Provides the deliberate-overclaim, matched-control, and self-audit executions, then specifies measurable parameters and calibration designs.
Appendices and supplementary templates	Supply formal derivations, registry schemas, provenance fields, standing and severity rubrics, threat models, reproducibility records, and a minimal tooling specification.

1.8 Testable implications

The framework yields observable predictions. The protocol-execution demonstrations in Appendix J include an explicit overclaim, a matched negative control, and a self-audit. They establish operational executability and status preservation in controlled cases; the empirical program asks whether execution acquires reliable standing across realistic manuscripts and evaluators. As protocol execution becomes cheaper, audit volume should increase even without journal policy changes. Audit-ready authors should self-audit more often and should revise high-strength verbs before submission. Papers with weak or mixed carriers should narrow, migrate, or challenge the audit basis. Journals that decline formal adoption may nevertheless acquire external audit reputations based on public mismatch rates. Symbolic compliance should become less stable when outsiders can test substance. Basis disputes should increase after direct status inflation becomes easier to detect. Audit outputs should exhibit model and prompt variance, creating demand for protocol versioning and cross-model reruns. Cheap external audit should also increase low-quality challenges, making

severity and provenance governance essential. A viable regime should therefore exhibit a favorable relation between δ_{eff}^w and η_{eff}^w , rather than merely a rising count of audit outputs.

1.9 Nonclaims

The paper does not claim that every LLM can execute UEAP reliably, that every audit is correct, that public audit should replace peer review, or that institutions are obsolete. It does not treat an audit mismatch as misconduct, falsity, or retraction-level failure. It does not assume that audit bases are uncontested. It does not deny confidentiality, exploratory safe harbors, or field-specific carrier standards. Its central claim is conditional:

If a claim-audit protocol can be executed at low enough cost by multiple non-institutional actors, then scientific authority becomes strategically reciprocal: institutions continue to evaluate claims, but they can no longer retain exclusive control over whether structured, protocolized claim-status audit is executed against accessible claims, and their own rigor claims become externally auditable.

The claim has an explicitly architectural and conditional evidential status. This solo-authored reference edition does not report a mature deployment of distributed UEAP audit. Full empirical validation requires a separate multi-evaluator calibration study; the manuscript specifies the architecture, variables, standing rules, threat model, and study design required to conduct one. The strategic conclusions therefore have the form: if portable claim-status audit can be executed cheaply enough, and if some outputs acquire standing through provenance, replication, counteraudit, and uptake, then institutional control over protocolized audit execution is no longer exclusive.

The monopoly at issue is not a monopoly over criticism in general. Replication attempts, letters, blogs, public commentary, and post-publication review already exist. The relevant change is narrower: a portable protocol makes a standardized, provenance-bearing claim-status audit independently executable and representable through a common registry. The claim concerns the structured protocolized audit interface, not scientific evaluation as a whole.

2 AASC/UEAP as a portable claim-audit protocol

2.1 Minimal operational premise

The present game-theoretic analysis does not require acceptance of every foundational claim associated with AASC. It requires a weaker operational premise.

1. Scientific reports request different statuses: prediction, proof, confirmation, explanation, compatibility, computation, domain-relative validity, obstruction, or actionability.
2. These statuses require different kinds of support.
3. A report can assert a stronger status than the declared support currently licenses.
4. The mismatch can sometimes be identified without resolving the ultimate truth of the strongest claim.

Protocol-independent sufficiency condition. The strategic argument requires only a portable operator that can (i) identify the status a report requests, (ii) identify the carrier offered for that status, (iii) determine whether the carrier licenses the requested status under a declared basis, and (iv) preserve the strongest lower-status formulation when the stronger status is not licensed. AASC/UEAP is the implementation used here; the exposure and institutional results do not depend on exclusive acceptance of that implementation.

UEAP is used here with *proposal standing*. Its present authority consists in an explicit specification, an executable registry structure, and exposure to rerun, counteraudit, basis contestation, and revision. It is not treated as already institutionally certified, empirically calibrated, or binding on a field. Community standing, reliability standing, and institutional standing are downstream achievements to be earned by the same provenance, replication, convergence, and response mechanisms applied to audit outputs generally. Authorial ownership therefore identifies the protocol’s present lifecycle stage; it does not confer audit authority.

UEAP supplies a domain-general schema for carrying out that fourth task. In the source protocol, a claim is evaluated relative to a declared target, standing-bearing carrier, admissible alternatives, laundering routes, and report object (Maley 2026). The protocol is status-relative. It preserves lower statuses such as calibrated-only, compatible-only, domain-dependent, evidence-compatible, computation-only, and obstruction-only instead of forcing every result into a final true/false or legitimate/failed binary.

2.2 The protocol supplies the structure

The structural hierarchy is central:

$$\text{UEAP/AASC protocol} \longrightarrow \text{audit structure}, \quad \text{evaluator} \longrightarrow \text{execution layer}.$$

The language model does not invent the audit categories. The written protocol supplies target fixation, requested status, carrier adequacy, domain and ancestry fields, blocker preservation, laundering exclusion, reportability, severity, and next-action requirements. A human, LLM, or hybrid evaluator interprets the target manuscript through that declared schema.

Let $\mathcal{P}_{\text{UEAP}}$ denote the protocol, X a target document, E an evaluator, and Z an execution record containing the prompt, model or evaluator identity, extraction method, settings, and target hash. A protocol-loaded execution is

$$\text{Exec}(E, \mathcal{P}_{\text{UEAP}}, X; Z) \longrightarrow R,$$

where the registry output is

$$R = (T, \sigma, C, \mathcal{D}, A, \mathcal{E}, B, \Lambda, \mathcal{R}, S, N).$$

Here T is the claim target; σ the requested report status; C the carrier; $\mathcal{D}$ the domain and scope ledger; A the ancestry or calibration ledger; $\mathcal{E}$ the evidence network; B the blocker set; Λ the identified laundering or status-promotion routes; $\mathcal{R}$ the strongest reportable formulation; S the severity classification; and N the next action required for repair, narrowing, or stronger standing.

Proposition 2.1 (Executor does not author the audit categories). *The executor applies a declared protocol. Disagreement remains possible, but the disagreement is localized to inspectable coordinates:*

target selection, requested status, carrier classification, domain, ancestry, blocker status, laundering route, severity, reportability, protocol version, or execution fidelity.

The protocol therefore constrains rather than eliminates interpretation. Its practical advantage over an unstructured request to “review this paper” is that the evaluator must expose which audit role supports each conclusion.

2.3 Audit basis and audit operator

Let a claim-audit basis be

$$\mathcal{A} = (D, \sigma, C_\sigma, \text{AdmAlt}, \sim_{\text{skin}}, \mathcal{L}, \preceq_\sigma),$$

where D is the domain ledger, σ the requested report status, C_σ the carrier standard for that status, AdmAlt the admissible-alternative relation, $\sim_{\text{skin}}$ the harmless-redescription relation, $\mathcal{L}$ the declared laundering taxonomy, and $\preceq_\sigma$ the status-precedence rule.

For a claim c , report m , evaluator e , and execution record z , define

$$\text{Audit}_{\mathcal{A},e,z}(c, m) \in \{\text{Matched}, \text{Mismatched}, \text{NewClaim}, \text{Indeterminate}\}.$$

Definition 2.2 (Matched report). *A report is matched when its target, carrier, domain, ancestry, blocker disclosure, and status language preserve the strongest status licensed by the audit basis.*

Definition 2.3 (Mismatched report). *A report is mismatched when it asserts a stronger status while one or more blockers separating the licensed status from the reported status remain active.*

Definition 2.4 (New-claim report). *A report is a new claim when it changes the target, carrier, domain, ancestry, comparison class, or report object rather than merely redescribing the audited claim.*

Definition 2.5 (Indeterminate audit). *An audit is indeterminate when the available manuscript, basis, or domain expertise does not support a responsible classification. Indeterminate is a substantive output, not a forced failure or forced pass.*

This four-way operator is important for distributed audit. A public evaluator should not be forced to call every ambiguity a defect. The system must preserve the difference between mismatch, basis insufficiency, and genuine claim change.

2.4 Minimal implementation

A minimal manuscript-level audit asks ten questions.

1. What exactly is the claim target?
2. Which report status is being requested?
3. What carrier is offered for that status?
4. What domain, population, boundary, or operating regime is required?
5. Does the result inherit target data, proxy data, tuning, or outcome-conditioned selection?
6. Which admissible alternatives or blockers remain active?

7. Does the report use any semantic, calibration, proxy, domain, computation, evidence-network, authority, or report-status laundering route?
8. What wording preserves the strongest licensed status?
9. What severity follows from the mismatch, if any?
10. What is the minimal next action: preserve, clarify, narrow, repair, rerun, counteraudit, or declare indeterminate?

The same fields can be implemented as a journal form, an author self-audit, a human checklist, a software schema, or an LLM-assisted execution. The structure is supplied by the protocol in every case.

2.5 Why portability matters

A protocol is portable when an evaluator can apply the same declared structure to multiple target documents without requiring permission from the target institution. Portability does not imply that the same carrier standard applies across physics, medicine, law, economics, or machine learning. Domain standards still enter through D , σ , and C_σ . What travels is the audit grammar: target, carrier, domain, ancestry, alternatives, blocker, laundering route, severity, and report ceiling.

This is analogous to, but distinct from, prior audit practices. Internal algorithmic auditing embeds documentation and review inside an organization (Raji et al. 2020); external algorithm audits inspect deployed systems from outside; user-driven and participatory audits distribute observation and hypothesis generation across affected users (Bandy 2021; DeVos et al. 2022; Li et al. 2023); field scans warn that auditor standards and independence are themselves unsettled (Costanza-Chock et al. 2022). UEAP brings a related portability to scientific report status. Its target is not an algorithmic system’s discriminatory output, but the relation between a scientific report and the carrier it claims.

2.6 A different direction from AI-system auditing

Much AI-audit literature asks how humans, institutions, or affected communities can audit an AI system. The present paper studies a different reversal: an AI or LLM can serve as the execution layer for a declared protocol that audits scientific discourse itself. The model is not the audit target. The target is a claim, paper, review process, press release, or institutional rigor statement.

The two literatures remain connected. AI-audit research supplies lessons about independence, audit washing, user participation, and who audits the auditors. The present paper imports those lessons into the game-theoretic analysis of scientific claims.

2.7 Portability does not confer standing

Portability means that execution can occur without permission. It does not make the output authoritative. Section 4 distinguishes execution, publication, standing, and consequence and defines the provenance and replication carrier required before an audit should alter uptake.

2.8 The audit protocol is itself auditable

Portability creates a reflexive requirement. The protocol, basis, prompt, model, and execution record are themselves claims about how the target was evaluated. A distributed audit must therefore

Table 4: AI-system auditing and LLM-assisted scientific claim audit.

AI-system audit literature	Present paper
Audits models, deployed systems, bias, safety, or performance	Uses a model or structured evaluator to execute a claim-status protocol
The technical system is the audit target	The scientific claim, paper, review process, or institution is the audit target
Often organized internally or through specialist audit teams	Execution can be distributed among authors, readers, rivals, funders, and institutions
Governance question: who can inspect the system?	Governance question: which distributed audit executions earn standing and consequence?

expose:

- the protocol version;
- the target document version or hash;
- the audit basis and requested status;
- the evaluator model and relevant settings;
- the execution instructions;
- the passages or evidence supporting each blocker;
- the severity assigned;
- uncertainty and indeterminate fields;
- any human corrections or counteraudit.

An opaque audit output that says only “failed” reproduces the gatekeeping asymmetry the protocol is meant to weaken. Distributed audit is credible only when the audit itself carries a reconstructible provenance record.

2.9 Status mismatch versus scientific falsity

A status mismatch can coexist with a true underlying claim. An under-supported prediction may later be confirmed. A simulation may later receive validation. A domain-relative result may later generalize. Conversely, a perfectly matched claim can be false if its carrier is adequate in kind but contains a substantive error not detected by the audit. UEAP is therefore neither a guarantee of truth nor a substitute for replication, proof checking, measurement, causal identification, or domain expertise.

Its strategic relevance comes from a lower burden. A reader may be unable to decide whether a new cosmological model is true but still be able to identify that a fitted correction has been reported as a target-independent prediction. A reviewer may be unable to settle whether a benchmark measures intelligence but still identify that a benchmark-scoped result has been promoted to deployment safety without an appropriate carrier. The audit narrows the dispute to the status transition.

2.10 Relation to peer review

Peer review is a social and institutional process containing novelty assessment, significance judgments, technical scrutiny, field fit, style evaluation, and allocation decisions. UEAP is narrower. It asks what status a claim may responsibly bear under a declared basis. An editor may reject a matched but uninteresting paper, accept an important paper after narrowing, or publish an indeterminate exploratory result with clear limitations. The protocol does not decide venue fit.

This narrower scope makes mechanization more plausible. A general model asked to “review this paper” must infer hidden evaluation criteria. A model asked to execute a declared registry has a constrained task. Recent work on LLM-based reviewing suggests that structured templates and specialized models can improve review form and criticality, while general models remain unreliable on complete correctness and deep technical judgment (Zhou et al. 2024; Idahl and Ahmadi 2025). The present architecture uses structure to reduce, not erase, that risk.

2.11 What is new here

The novelty is not that peer review can be criticized, that post-publication discussion exists, that checklists can structure review, or that language models can produce referee-like prose. It is the combination of a portable claim-status operator, non-permissioned execution, registry-style output, and an explicit rule that public consequence depends on the audit’s own standing rather than on the mere fact that an audit was generated.

Table 5: Adjacent practices and the distinct contribution of distributed UEAP audit.

Practice and target	Execution locus	Typical output	Distinction from the present architecture
Ordinary peer review: manuscript quality, novelty, correctness, and venue fit	Journal-controlled	Referee report and editorial recommendation	Broad evaluative mandate; execution is normally permissioned by the venue and need not expose a target-carrier-blocker registry.
Post-publication or PubPeer-style critique: published claims	Public commenters or invited experts	Comment, rebuttal, or discussion thread	External execution already exists, but no common standing ladder, consequence ladder, or machine-readable claim-status schema is required.
Registered reports: study design before results	Journal and reviewers	In-principle acceptance and design review	Constrains design and publication incentives; it is not a general operator for auditing report status across claim types.

Practice and target	Execution locus	Typical output	Distinction from the present architecture
Compliance checklists: reporting or methodological fields	Author, journal, funder, or institution	Completed form or badge	Efficient for named requirements but vulnerable to symbolic completion when the fields are not tied to report ceilings and rerunnable blocker evidence.
Unstructured LLM paper review: manuscript quality	User and model provider	Review prose, scores, or suggestions	The model usually infers the standard and can silently change criteria; provenance and status preservation are optional.
Distributed UEAP audit: relation between a claim and its carrier	Any lawful evaluator with the target and declared basis	Structured registry with target, status, carrier, blockers, report ceiling, severity, and provenance	Execution is non-permissioned while standing and consequence remain separately earned and auditable.

The resulting novelty claim is therefore precise: portable, protocol-executed claim-status audit changes the strategic exposure structure because institutions need not authorize execution, while provenance, replication, severity discipline, counteraudit, and response handling still govern whether the output deserves uptake.

3 Mechanized and LLM-assisted execution

3.1 The execution layer

The practical innovation analyzed here is not merely a written checklist. It is the ability to execute the checklist at low marginal cost. The hierarchy must remain explicit:

$$\underbrace{\mathcal{P}_{UEAP}}_{\text{audit structure}} + \underbrace{X}_{\text{target document}} + \underbrace{E, Z}_{\text{evaluator and execution record}} \longrightarrow \underbrace{R}_{\text{audit registry}} .$$

A user can place the UEAP protocol and a target manuscript in the same language-model context, specify the requested audit basis, and request the structured registry defined in Section 2. The protocol supplies the categories and dependency order. The target manuscript supplies the object. The evaluator supplies execution, extraction, and organization. The model is therefore a protocol interpreter and evidence organizer, not the author of the audit standard.

This workflow differs from asking a model for an unstructured peer review. The execution instruction must preserve the protocol’s claim extraction, carrier identification, blocker preservation, laundering exclusion, severity, and report-status obligations. An output that silently changes those categories is a different audit basis, not a faithful execution.

Empirical studies support cautious confidence that LLMs can provide useful manuscript feedback

at scale. Liang and colleagues compared GPT-4 feedback with human reviewer feedback across thousands of papers and also studied researchers’ evaluations of the generated feedback (Liang et al. 2024). Specialized systems such as OpenReviewer take full paper PDFs and review templates as inputs and generate structured review reports (Idahl and Ahmadi 2025). LLM-agent systems have also been used to simulate peer-review dynamics (Jin et al. 2024). Yet evaluations show that LLM reviewers remain error-prone, can miss technical details, and often fail complete-correctness tests (Zhou et al. 2024). The appropriate claim is therefore not autonomous adjudication, but scalable structured screening.

Table 6: The execution-layer claim and its limits.

Misreading	Correct interpretation
The LLM decides scientific truth	The evaluator executes a declared claim-status protocol and produces an inspectable audit candidate
An audit output is authoritative by default	An output begins at low standing and must earn credibility through provenance, grounding, rerun, counteraudit, and severity discipline
Anyone can run an audit, therefore anyone is right	Execution is open; audit standing and consequence are earned
Institutions become irrelevant	Institutions retain formal powers and expertise, but their validation claims become externally auditable
Mechanization removes domain judgment	Mechanization lowers extraction and registry cost; domain standards remain part of the carrier and may require specialist review

Mechanization matters at the execution margin; it does not confer authority. Structured audit candidates acquire standing through the process defined in Section 4, not through model identity or fluent presentation.

3.2 Protocol constraint and executor discretion

The protocol constrains what the evaluator must report, but it does not eliminate judgment. Target extraction, carrier identification, and blocker classification can remain difficult. The point of protocol loading is to make those judgments inspectable. A faithful execution must show which target passage, carrier passage, domain assumption, or missing ledger supports each registry field. When the evaluator cannot responsibly resolve a field, the protocol permits *Indeterminate* rather than forcing a pass or failure.

The evaluator’s discretion is therefore bounded in two directions. It may not invent a stronger audit standard than the declared basis, and it may not waive a protocol field merely because the manuscript is prestigious or fluent. Any alteration of protocol roles must be declared as a basis change.

3.3 Execution stages

A defensible execution pipeline contains eight stages.

1. **Input normalization.** Extract visible text, equations, captions, references, appendices, and declared supplementary materials. Detect hidden or machine-only text before the evaluator reads the document.
2. **Protocol isolation.** Place audit instructions in a privileged channel or separate context so that target-document content is treated as data rather than executable instruction.
3. **Basis initialization.** Declare the UEAP version, target-document version, requested status, domain supplement, carrier standard, severity rule, and permissible output schema.
4. **Claim decomposition.** Identify the paper’s principal and secondary report claims, including verbs such as predicts, confirms, proves, resolves, generalizes, validates, or enables deployment.
5. **Registry completion.** Fill target, requested status, carrier, domain, ancestry, evidence-network, alternatives, blocker, status-promotion, and report-object fields with supporting passages.
6. **Status and severity decision.** Return matched, mismatched, new claim, or indeterminate, together with S0–S5 severity and an explicit uncertainty record.
7. **Repair and counteraudit output.** State the strongest matched wording, the minimum repair needed for stronger status, and the questions a counteraudit should test.
8. **Provenance output.** Record evaluator, version, prompt, protocol hash, target-file hash, settings, timestamp, preprocessing, and any human corrections.

Mechanization lowers execution cost, but it also creates attack surfaces. Hidden prompt injection in manuscripts has already been documented in the context of AI-assisted peer review (Lin 2025). The execution layer must therefore assume that the target document may be adversarial.

3.4 Prompt injection and data-instruction separation

A target manuscript can contain ordinary prose that resembles instructions, hidden text, metadata, image-layer text, or explicit prompt injection. The auditor must distinguish the audit protocol from the content being audited. A minimal defense-in-depth design includes:

- rendering the manuscript and comparing extracted text with visible pages;
- stripping white-on-white, zero-size, off-page, or metadata instructions;
- delimiting the target document as untrusted quoted material;
- prohibiting the target from changing the audit basis or output schema;
- requiring passage citations for each finding;
- running a second model or human check on high-severity results;
- preserving the raw target and normalized target hashes;
- reporting suspected injection as a separate security finding rather than silently obeying or deleting it.

The strategic effect of prompt injection is significant. When authors expect LLM-assisted audit, some may try to manipulate the execution layer directly. This is a new form of audit-basis or audit-process contestation. It does not invalidate distributed audit; it makes provenance and adversarial preprocessing part of the audit carrier.

3.5 Parameter effects of mechanization

Let c_k be the private cost to actor k of running an audit, p_k the probability the actor audits, q_k detection accuracy, r_k the probability the result reaches a consequential audience, and L the consequence of a confirmed mismatch. LLM assistance can affect each parameter.

Table 7: Strategic parameter effects of mechanized audit.

Parameter	Likely direction	Qualification
c_k	Lower	Setup, verification, and domain-review costs remain.
p_k	Higher	More actors can audit, but free riding and attention limits persist.
q_k	Potentially higher for explicit status gaps	Technical correctness, novelty, and hidden-domain assumptions may remain weak.
r_k	Higher if outputs are shareable	Reach can also amplify false or low-quality audits.
L	Higher when public audit affects trust	Consequence should be proportional to severity and reproducibility.
C_B for basis contestation	Potentially lower	Models can also help agents generate sophisticated objections or friendly audit bases.
Registry memory	Stronger	Only if provenance and target versions are retained.

Mechanization therefore has no unconditionally beneficial sign. It lowers the cost of good-faith self-audit and public challenge, but also lowers the cost of audit spam, strategic framing, and superficial compliance. The game-theoretic analysis must model both.

3.6 Human expertise and model roles

A distributed system can allocate different roles to human and machine evaluators.

1. The LLM extracts claims, traces terminology, fills registry fields, and identifies candidate blockers.
2. A domain expert evaluates whether the carrier standard is appropriate and whether cited passages support the finding.
3. A second evaluator checks audit reproducibility and basis consistency.
4. The target author receives a right to correct extraction errors, declare missing inputs, or contest the audit basis.
5. The public record preserves both the initial audit and the resolution.

This hybrid arrangement aligns with proposals to use LLMs for targeted verification, citation checking, reproducibility assistance, and ethics flagging rather than simply imitating the prose of human reviewers (Choi et al. 2026). It also avoids turning the model’s fluency into unearned authority.

3.7 Mechanization without central permission

The essential feature is not that a journal owns the tool. A self-auditing author can run it before submission. A reader can run it after publication. A rival can run it during a dispute. A funder can run it on a portfolio. A journalist can run it on an institutional press release. A journal can run it internally, but its refusal does not block these other uses.

The result is a three-layer ecology. Institutions retain formal decisions—acceptance, funding, correction, retraction, accreditation. External actors acquire a low-cost method for executing and circulating structured challenges. Audit standing then determines which challenges deserve consequential uptake. The next sections formalize the exposure and supply of those challenges.

4 Open execution, circulation, standing, and consequence

4.1 Four layers that must not be collapsed

The claim that audit is non-permissioned concerns *execution rights*. It does not imply that every generated output deserves public uptake, nor that every standing-bearing output warrants the same consequence. A distributed regime must distinguish four layers.

Definition 4.1 (Audit execution). *Audit execution is the act of applying a declared protocol and audit basis to a target document. Execution is non-permissioned when no journal, author, or institution possesses exclusive authority to decide whether an otherwise lawful evaluator may perform that act on an accessible target.*

Definition 4.2 (Audit publication). *Audit publication is the act of circulating an audit output to an audience. Publication can be private, preliminary, public, archived, corrected, withdrawn, or incorporated into an institutional process.*

Definition 4.3 (Audit standing). *Audit standing is the credibility warranted for the audit’s own claims. Standing depends on target fidelity, declared basis, execution provenance, registry completeness, passage grounding, evaluator competence, reproducibility, severity calibration, counteraudit, and response handling. It is not conferred merely by producing or publishing an output.*

Definition 4.4 (Audit consequence). *Audit consequence is the change in wording, uptake, correction, review, funding, venue classification, reputation, or policy requested on the basis of an audit. Consequence must be proportionate to both the finding’s severity and the audit’s standing.*

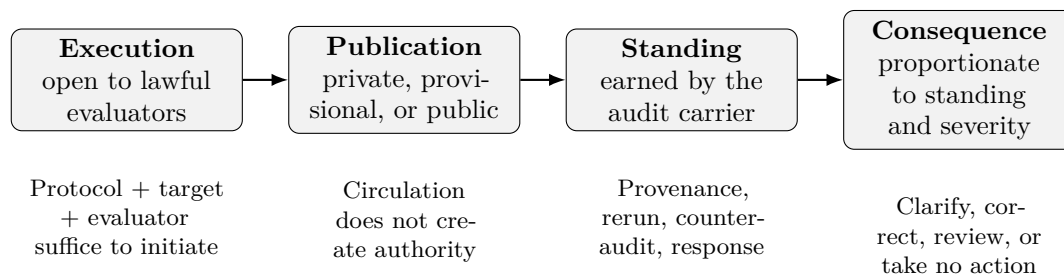


Figure 2: The four-layer architecture. Openness attaches to execution; credibility and institutional effect are separate, progressively stronger claims.

The central safeguard is therefore:

Table 8: Execution, publication, standing, and consequence are distinct.

Layer	Governing question	Openness	Condition for legitimate use
Audit execution	Can an actor run the protocol against the target?	Non-permissioned for lawfully accessible targets	Protocol, target, and evaluator are sufficient to initiate a run
Audit publication	Can the actor circulate the result?	Public, private, preliminary, archived, corrected, or withdrawn	Fair representation, provenance, and clear provisional status
Audit standing	Why should anyone credit the audit’s claims?	Earned, not inherited from identity or software	Basis declaration, registry completeness, grounding, rerun, counteraudit, and stability
Audit consequence	What should change because of the audit?	Social and institutional, not automatic	Consequence proportionate to standing, severity, sampling, and response history

Distributed audit democratizes execution, not authority. Anyone can run an audit; not every audit has standing, and standing does not by itself license every consequence.

This safeguard does not dilute the non-permissioned thesis. It explains how distributed execution can alter authority without replacing one opaque gatekeeper with a mass of ungrounded accusations.

4.2 An audit-standing ladder

For operational use, let the standing of an audit report occupy one of six levels.

The levels are not a moral ranking of auditors. They record the carrier behind the public audit claim. A0 can be valuable for private self-audit. A1 and A2 can support a transparent preliminary challenge. A3 supports a replicated paper-level finding. A4 supports stronger public use after reply. A5 marks a stabilized audit claim. Institutional or policy use requires more than A5 on one paper: it additionally requires sampling design, portfolio denominators, conflict disclosure, appeal, and governance review.

Early-use rule. The standing ladder is not a barrier to execution, private exploration, author self-audit, debugging, or provisional clarification. A0–A2 outputs can be useful when they are represented at their actual standing. The evidential bar rises with the consequence requested: early use is permissive, while strong public, reputational, portfolio, or institutional consequence remains conservative.

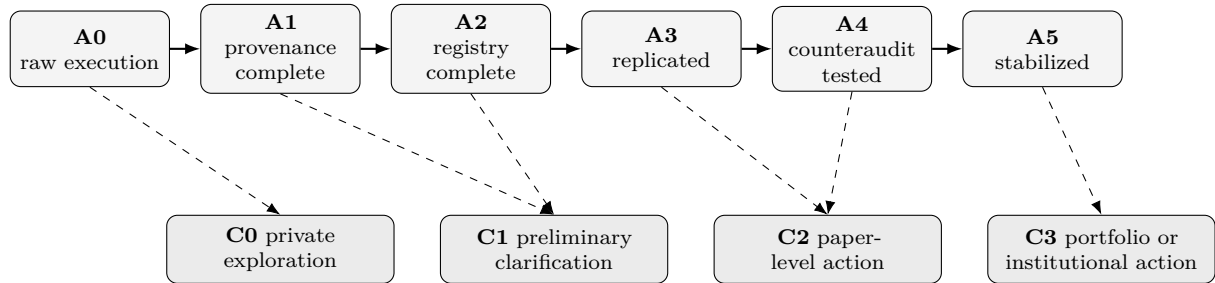
4.3 Consequence classes

Standing and consequence should be reported separately. An illustrative consequence ladder is:

A high-severity label does not authorize a high consequence when audit standing is weak. Conversely, a stabilized low-severity finding may justify only a wording correction. This separation is the principal protection against audit theater and audit harassment.

Table 9: Audit-standing levels.

Level	Label	Minimum condition
A0	Raw execution	One unverified protocol output; useful for private exploration or self-audit only.
A1	Provenance-complete	Protocol and basis version, target version, evaluator, prompt, extraction method, and passages are disclosed.
A2	Registry-complete	Target, requested status, carrier, domain, ancestry, blockers, laundering routes, reportability, severity, and next action are fully recorded and passage-grounded.
A3	Replicated	An independent rerun or evaluator recovers the same material target/status finding under a declared comparable basis.
A4	Counteraudit-tested	The target’s reply, omitted evidence, extraction objections, and a structured counteraudit have been considered.
A5	Stabilized	Material status and severity remain stable across admissible reruns and bases, or the basis dispute is explicitly resolved and preserved.



Consequence is not a seventh standing level: severity, sampling, reply, and governance remain additional conditions.

Figure 3: Audit-standing lifecycle and consequence classes. Low-standing executions remain useful for low-consequence purposes; stronger public or institutional action requires a stronger carrier.

Table 10: Illustrative audit-consequence classes.

Class	Default use	Typical prerequisite
C0	Private exploration; no requested external action	A0 or higher
C1	Preliminary clarification, narrowing suggestion, or request for provenance	A1–A2, target reply available
C2	Public paper-level correction, formal review, or durable registry entry	Ordinarily A3–A5, severity-specific review
C3	Portfolio, venue, funding, policy, or institutional consequence	A5 findings plus representative sampling, institutional counter-audit, and governance carrier

4.4 Standing weight and consequential detection

Let $w_k \in [0, 1]$ denote the standing weight of source k ’s audit, conditional on the finding being produced and reaching an audience. The weight attaches to the audit output, not to the social prestige of the auditor. An independent researcher can produce a high-standing audit by exposing a complete carrier; a prestigious institution can produce a low-standing audit if provenance, grounding, replication, or counter-audit handling are absent.

Define

$$w_k = W(P_k, R_k, C_k, S_k, H_k), \quad 0 \leq w_k \leq 1,$$

where:

- P_k is provenance completeness: protocol and target versions, model or evaluator, prompt, extraction pipeline, cited passages, and disclosed human edits;
- R_k is replication support, with cross-family replication distinguished from reruns inside one model, prompt, extraction, or social lineage;
- C_k is convergence with materially independent audit families on the target, status class, blocker, and severity range;
- S_k is severity discipline: the finding distinguishes wording, report-status downgrade, carrier failure, systematic laundering, and institutional pattern claims proportionately;
- H_k is handling quality: counter-audit, right of reply, correction, basis dispute, and resolution history are recorded rather than suppressed.

A transparent illustrative implementation is the monotone weighted form

$$w_k = \omega_P P_k + \omega_R R_k + \omega_C C_k + \omega_S S_k + \omega_H H_k, \quad \sum_j \omega_j = 1, \quad \omega_j \geq 0.$$

Table 11: Components of the audit-standing weight.

Term	Standing dimension	What raises the score
P_k	Provenance	Exact protocol, target, evaluator, prompt, extraction, passages, and edit history
R_k	Replication	Independent reruns, especially across model, prompt, extraction, institution, and human-domain families
C_k	Convergence	Stability of target, status, blocker, and severity under admissible bases
S_k	Severity discipline	Proportionate classification and consequence; willingness to return matched, indeterminate, or low severity
H_k	Handling	Counteraudit, reply, correction, withdrawal, and basis-resolution history

The linear form is not mandatory. A field may require multiplicative minima, veto conditions, or status-specific weights. What matters is that the mapping be declared before consequential use and that it reward audit properties rather than identity. In particular, an A0–A2 execution can have low public standing despite a compelling narrative, while an A4–A5 finding ordinarily has high values on the relevant dimensions.

Principle 4.5 (Admissible standing-weight rule). *A standing rule is admissible for consequential use only if it satisfies all of the following constraints.*

1. **Predeclaration.** *Weights, thresholds, minima, and vetoes are declared before the target result is known or before the audit is used consequentially.*
2. **Claim-relative identity neutrality.** *The rule evaluates the carrier of the audit claim, not the prestige, affiliation, or social identity of the auditor or target.*
3. **Monotonicity.** *Greater provenance, materially independent replication, cross-family convergence, severity discipline, and response handling cannot lower standing, other things equal.*
4. **Hard provenance caps.** *Missing target version, undeclared basis, undisclosed execution instructions, or absent passage grounding caps public standing regardless of fluency or institutional sponsorship.*
5. **Consequence specificity.** *The same audit may support a clarification request while remaining insufficient for a paper-level correction, portfolio inference, or institutional allegation.*
6. **Counteraudit responsiveness.** *Successful rebuttal, corrected extraction, failed negative controls, or demonstrated over-severity must revise the standing assignment and remain in the history.*
7. **Crosswalkability.** *Domain-specific rules may differ, but their components, thresholds, and divergences must be disclosed so that competing standing maps can be compared and audited.*

No universal numerical form of W is claimed. The admissibility constraints replace unrestricted discretion with governed pluralism: a field may choose a different transparent map, but silent post hoc reweighting, prestige substitution, or high consequence without the required carrier is inadmissible and is itself an audit target. Appendix K supplies default caps and use rules.

This creates a hierarchy of exposure probabilities. Raw audit occurrence is governed by p_k .

Correct and visible detection is governed by $p_k q_k r_k$. Standing-bearing consequential detection is governed by $p_k q_k r_k w_k$. Define

$$\delta_{\text{eff}}^w = 1 - \prod_{k \in K} (1 - p_k q_k r_k w_k).$$

The strategic effect of a distributed system depends increasingly on δ_{eff}^w , not merely on the number of outputs generated. Theorem 5.2 makes this the central deterrence variable. Section 5 pairs it with standing-weighted false-positive exposure so that open execution is assessed by net corrective value rather than audit volume.

4.5 Standing of the audit is itself auditable

An audit report asserts at least six claims:

1. the selected text expresses the target claim;
2. the requested status has been identified correctly;
3. the declared audit basis applies;
4. the cited passages support the blocker;
5. the status and severity consequence are proportionate;
6. the requested consequence is licensed by the audit's standing and sampling carrier.

A counteraudit may challenge any of these. The distributed regime is therefore recursively auditable: the target paper can be audited, the audit can be counteraudited, and the registry can preserve the lineage of correction. Recursion does not require infinite regress. Public consequence can be conditioned on a declared stopping rule such as independent replication, domain adjudication, stabilized basis agreement, or a governance decision whose own carrier is exposed.

4.6 Why this distinction preserves the radical claim

The loss of exclusive gatekeeper control concerns who may initiate structured, protocolized claim-status scrutiny. The standing and consequence ladders concern which scrutiny should influence uptake and by how much. Execution remains non-permissioned; standing and consequence are earned. These are compatible. Institutions cannot prevent execution merely by refusing adoption, but they can contest low-standing audits on their merits. External auditors cannot demand authority merely by invoking UEAP; they must expose the carrier of the audit itself. Scientific authority becomes reciprocal because institutional and external evaluations alike remain answerable to declared evidence.

Core standing rule

Execution is non-permissioned; authority is not. A0–A2 outputs are legitimate for private exploration, self-audit, and transparent preliminary clarification. Stronger public, portfolio, or institutional consequences require correspondingly stronger provenance, independent replication, counteraudit, sampling, and response handling. Standing attaches to the audit's carrier, not to the auditor's prestige.

5 Distributed audit exposure

5.1 Audit sources

Let K be the set of potential audit sources. A source may be a journal reviewer, editor, author performing self-audit, rival, reader, funder, journalist, replication team, post-publication forum, professional society, or independent LLM-assisted evaluator. For each $k \in K$, define:

- p_k : probability that source k initiates an audit;
- q_k : probability that the audit correctly identifies the relevant mismatch conditional on initiation;
- r_k : probability that the finding reaches an audience capable of changing uptake;
- w_k : standing weight of the audit output, defined by $W(P_k, R_k, C_k, S_k, H_k)$ in Section 4.

The four factors answer different questions: was an audit run, did it identify the relevant mismatch, did it reach an uptake-sensitive audience, and did the audit itself earn standing? The paper’s central strategic variable combines all four.

5.2 Raw occurrence and consequential detection

The simplest exposure model treats audit initiation events as independent.

Proposition 5.1 (Raw distributed audit occurrence). *If potential auditor k independently initiates an audit with probability p_k , the probability that at least one audit occurs is*

$$p_{\text{eff}} = 1 - \prod_{k \in K} (1 - p_k).$$

The addition of a new auditor j with $p_j > 0$ weakly increases p_{eff} , and strictly increases it unless an audit already occurs with probability one.

Proof. The probability that no source audits is the product of the independent no-audit probabilities. Taking its complement gives the result. $\square$

Raw occurrence formalizes non-permissioned execution, but it does not measure meaningful pressure. An audit can miss the mismatch, remain private, or lack the provenance needed for public standing. Define raw consequential detection by

$$\delta_{\text{eff}} = 1 - \prod_{k \in K} (1 - p_k q_k r_k).$$

5.3 Standing-weighted consequential detection

The central variable is

$$\delta_{\text{eff}}^w = 1 - \prod_{k \in K} (1 - p_k q_k r_k w_k).$$

Central strategic variable. δ_{eff}^w is the benchmark probability that at least one distributed channel runs an audit, detects the relevant mismatch, reaches an uptake-sensitive audience, and carries enough standing for the finding to matter. Raw audit count is not the governing variable.

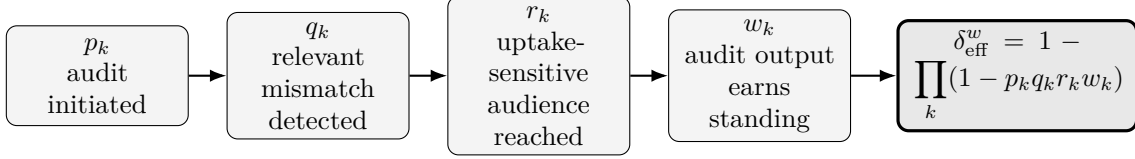


Figure 4: Standing-weighted consequential detection pathway. Audit volume matters strategically only to the extent that execution, detection, reach, and earned standing are jointly present.

Table 12: Hierarchy of distributed-audit exposure quantities.

Quantity	Meaning	Role in the model
p_{eff}	Raw audit occurrence	Establishes that execution can occur without institutional permission; it confers neither correctness nor standing.
δ_{eff}	Consequential detection	Adds mismatch detection and uptake-sensitive reach, but does not yet distinguish strong from weak audit carriers.
δ_{eff}^w	Standing-weighted consequential detection	Main deterrence variable; combines execution, detection, reach, and earned audit standing.

Theorem 5.2 (Standing-Weighted Consequential Distributed Audit). *Under independent source events, δ_{eff}^w is the probability that at least one source initiates an audit, correctly identifies the relevant mismatch, reaches an uptake-sensitive audience, and satisfies the standing threshold encoded by w_k . Adding a source with $p_k q_k r_k w_k > 0$ weakly increases δ_{eff}^w .*

Proof. For source k , the probability of no standing-bearing consequential detection is $1 - p_k q_k r_k w_k$. Independence makes the probability of none from any source the product of these terms. Taking the complement gives δ_{eff}^w . Adding a source multiplies the no-detection probability by a term no greater than one. $\square$

Let $B_{ij} > 0$ be the private gain from reporting a claim licensed at status s_i as stronger status s_j , including the reward gap from saying “predicts” rather than “fits,” “proves” rather than “computes,” or “validated” rather than “compatible.” When useful, decompose that gain as

$$B_{ij} = \Delta G_{ij} + C_i - C_{ij},$$

where ΔG_{ij} is the reward premium for the stronger report, C_i is the cost of reporting honestly at s_i , and C_{ij} is the cost of the unsupported promotion. Let $L_{ij} > 0$ be the author-side loss when a standing-bearing finding becomes consequential: narrowing, loss of uptake, correction, reputational discount, or future portfolio effects. A risk-neutral agent is deterred from the unsupported transition when

$$\delta_{\text{eff}}^w L_{ij} \geq B_{ij} = \Delta G_{ij} + C_i - C_{ij}.$$

The loss need not mean punishment. It can be only the withdrawal of the stronger status report.

The theorem is not a claim that distributed audit is always effective. It identifies the condition

that makes external audit strategically relevant. Audit occurrence without detection, reach, or standing may produce noise but little deterrence.

5.4 No full institutional opt-out

Proposition 5.3 (No Full Institutional Opt-Out). *Let J be the journal or formal institution. If $p_J = 0$ but there exists an external source $k \neq J$ with $p_k > 0$, then $p_{\text{eff}} > 0$. If in addition $q_k r_k w_k > 0$, then $\delta_{\text{eff}}^w > 0$.*

An institution can decline to perform audit internally. It cannot, by that decision alone, prevent a public manuscript from being audited externally. Refusal changes who performs the audit and may reduce official response channels, but it does not reduce exposure to zero.

5.5 Independence is a benchmark, not an empirical assumption

The product forms are transparent benchmarks, not empirical claims of actual independence. Real audit sources may be correlated because they use the same model family, prompt, protocol fork, extraction pipeline, social network, viral trigger, or prior public audit. In general, if E_k is the event that source k produces a standing-bearing consequential detection, then

$$\delta_{\text{eff}}^w = \Pr \left(\bigcup_{k \in K} E_k \right),$$

not necessarily the product formula. The general union bounds are

$$\max_k \Pr(E_k) \leq \Pr \left(\bigcup_k E_k \right) \leq \min \left\{ 1, \sum_k \Pr(E_k) \right\}.$$

Bonferroni bounds can use pairwise intersections. Positive correlation reduces the diversification gain from additional executions. Ten reruns from one prompt/model family may add less standing than three genuinely independent audits.

A practical registry should therefore group audits into families $F_1, \dots, F_m$ by model lineage, prompt lineage, institutional source, extraction pipeline, or social coordination. Let P_j, Q_j, R_j, W_j be family-level initiation, detection, reach, and standing parameters. The family-level benchmark is

$$\delta_{\text{family}} = 1 - \prod_{j=1}^m (1 - P_j Q_j R_j W_j).$$

The family model is often more defensible than treating every audit output as an independent source. Raw audit count is therefore never treated as evidence of independent confirmation. Additional executions matter only to the extent that they add competence, reach, provenance, or genuinely distinct audit families.

5.6 Marginal contribution and redundancy

Under the independence benchmark, the marginal increase in raw audit probability from source j is

$$\frac{\partial p_{\text{eff}}}{\partial p_j} = \prod_{k \neq j} (1 - p_k),$$

and the marginal increase in standing-weighted consequential detection is

$$\frac{\partial \delta_{\text{eff}}^w}{\partial p_j} = q_j r_j w_j \prod_{k \neq j} (1 - p_k q_k r_k w_k).$$

The marginal contribution is largest when few other consequential sources are active and decreases as the system becomes saturated. This is useful for institutional design: early independent auditors can materially increase exposure, while a large number of redundant auditors may add little.

A descriptive effective-auditor index is

$$N_{\text{eff}} = \frac{(\sum_k a_k)^2}{\sum_k a_k^2},$$

where a_k weights sources by independence-adjusted competence, reach, and standing. The exact index is not load-bearing. Its purpose is to prevent raw audit count from being mistaken for independent evidence.

5.7 Standing-weighted false-positive exposure

Let f_k be the probability that source k produces a false or materially overstated mismatch finding conditional on initiation. Let w_k be the standing weight that the false finding nevertheless acquires under the same declared audit-output rule. Raw visible false-alarm exposure under the independence benchmark is

$$\eta_{\text{eff}} = 1 - \prod_{k \in K} (1 - p_k f_k r_k),$$

and the standing-weighted counterpart is

$$\eta_{\text{eff}}^w = 1 - \prod_{k \in K} (1 - p_k f_k r_k w_k).$$

The distinction matters because a false raw output may be corrected privately and remain inconsequential, while a provenance-rich but mistaken audit may acquire substantial public weight. The same standing rule is used because w_k measures the carrier of the public audit claim, not whether that claim is ultimately correct. Once error is exposed, failed negative controls, unsupported severity, or refusal of counteraudit should reduce S_k , H_k , and future w_k . False-positive control therefore belongs inside audit standing rather than outside the model.

Let $G_{\text{correction}}$ be the social gain from a standing-bearing correct detection, H_{false} the social harm from a standing-bearing false or materially overstated finding, and C_{noise} the attention, reply, and registry cost of low-value audit traffic. Define

$$V_{\text{net}} = \delta_{\text{eff}}^w G_{\text{correction}} - \eta_{\text{eff}}^w H_{\text{false}} - C_{\text{noise}}.$$

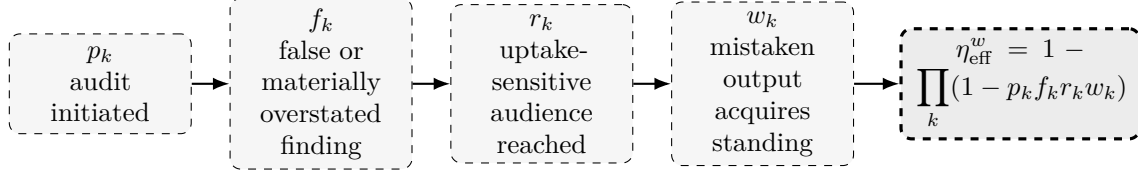


Figure 5: Standing-weighted false-positive exposure. The same openness that can increase correction also creates a pathway for consequential error, which must be measured rather than assumed away.

Proposition 5.4 (Standing-Weighted Audit Balance). *A distributed audit regime is socially beneficial on a declared claim class only if*

$$\delta_{\text{eff}}^w G_{\text{correction}} > \eta_{\text{eff}}^w H_{\text{false}} + C_{\text{noise}}.$$

Increasing audit volume without improving detection, standing, or false-positive control need not improve welfare.

Mechanization can increase both δ_{eff}^w and η_{eff}^w . A credible regime must improve their relation through protocol constraints, matched negative controls, model and prompt diversity, severity discipline, counteraudit, correction, and standing downgrades for repeated overstatement. More audit executions are not automatically better.

5.8 Bundled status promotions

A paper may over-promote along several dimensions simultaneously. A benchmark result can be reported as broad capability, deployment readiness, and safety. Let B be a bundle of status-transition edges. If components are separable, deterrence requires

$$\delta_{ij}^w L_{ij} \geq B_{ij} \quad \text{for each } (i, j) \in B.$$

If the bundle has joint rewards and joint detection, use

$$\delta_B^w L_B \geq B_B.$$

Different auditors may identify complementary components. One evaluator may detect benchmark-to-capability inflation, another domain-to-universal inflation, and a third missing safety carrier. A high-standing bundle conclusion requires provenance showing how those partial findings were combined.

5.9 Interpretation

The theorem does not make gatekeepers irrelevant. It removes their exclusive control over whether structured, protocolized claim-status scrutiny can be initiated against accessible claims. The scientific authority of an institution therefore depends increasingly on how it responds to standing-bearing external audits: by correcting, narrowing, counterauditing, disclosing the basis, or demonstrating that the external finding is inapplicable. Ignoring an audit remains a possible strategy, but institutional refusal no longer makes the underlying challenge impossible.

Exposure model at a glance

For each source, p_k measures initiation, q_k correct detection, r_k consequential reach, and w_k earned audit standing. The central true-detection benchmark is δ_{eff}^w ; the paired false-finding benchmark is η_{eff}^w . Raw audit volume is neither independent confirmation nor a welfare criterion. The regime is valuable only when standing-bearing correction exceeds standing-bearing false-finding harm and the associated noise, chilling, and capture costs.

6 The supply of distributed audit

6.1 Audit as a volunteer’s dilemma

The exposure formula treats p_k as given. In practice, audit initiation is a choice. A reader must spend time loading the protocol, preparing the target document, checking the output, and deciding whether to publish the result. Even when LLM execution is cheap, verification and reputational risk remain. Distributed audit therefore has the structure of a volunteer’s dilemma: the community benefits if someone audits, but each potential auditor may prefer that another person bears the cost.

Consider $n \geq 2$ symmetric potential auditors. Each chooses $a_k \in \{0, 1\}$, where 1 means audit. If at least one actor audits, each member of the community receives public benefit $V > 0$ from the possibility that a material mismatch is exposed. An actor who audits pays private cost $c > 0$. Ignore private rivalry or reputation benefits initially.

If each other actor audits with probability x , the expected payoff from not auditing is

$$U(0) = V \left[1 - (1 - x)^{n-1} \right].$$

The expected payoff from auditing is

$$U(1) = V - c.$$

A symmetric mixed equilibrium satisfies

$$V - c = V \left[1 - (1 - x)^{n-1} \right],$$

so

$$x^* = 1 - \left(\frac{c}{V} \right)^{1/(n-1)}$$

when $0 < c < V$.

Proposition 6.1 (Distributed audit supply). *In the symmetric volunteer-audit game, lower private execution cost c increases the individual audit probability x^* and the aggregate probability that at least one audit occurs,*

$$p_{\text{eff}}^* = 1 - (1 - x^*)^n = 1 - \left(\frac{c}{V} \right)^{n/(n-1)}.$$

This proposition captures the central strategic role of LLM assistance. It does not need to make audit free or perfect. By lowering c , it changes the equilibrium supply of audit.

6.2 The standing bottleneck

Execution cost and standing cost should be separated. Let c_E be the cost of generating a structured audit and c_S the additional cost of grounding passages, preserving provenance, obtaining an independent rerun, processing a reply, and maintaining the registry. LLM assistance can reduce c_E much faster than c_S .

Proposition 6.2 (Audit-standing bottleneck). *If mechanization lowers c_E while c_S remains approximately fixed, the equilibrium supply of raw or merely documented A0–A2 audits increases more rapidly than the supply of independently replicated, counteraudit-tested, or stabilized A3–A5 audits. The proportion of generated audits with sufficient standing for consequential public use can therefore fall even while total audit volume rises.*

The proposition explains why the radical increase in execution does not automatically produce a credible audit regime. Standing infrastructure—target hashing, prompt disclosure, registry completion, independent reruns, counteraudit, severity review, and response handling—becomes the scarce complementary good. A system can be flooded with outputs while still suffering a shortage of standing-bearing audits.

6.3 Private benefits and heterogeneous auditors

Real auditors are not symmetric. A rival may gain competitive advantage from finding a blocker. A methodological specialist may gain reputation. An author may self-audit to reduce later correction cost. A public-interest organization may value accountability. Let b_k be actor k ’s private benefit from being the source of a correct audit and c_k the private cost. The actor audits when the expected marginal payoff is nonnegative:

$$b_k + V_k \Pr(\text{no other consequential audit}) - c_k - h_k \geq 0,$$

where h_k is expected liability from a false or overstated audit.

Mechanization can lower c_k , but provenance and false-positive discipline affect h_k . A system that makes accusations cheap while leaving correction costly can induce excessive low-quality audits. Conversely, a system that requires a full legalistic dossier for every low-severity wording issue may keep c_k so high that distributed audit never reaches critical mass.

6.4 Specialized division of labor

User-driven algorithm audits have shown that distributed participants can occupy different roles: hypothesis generation, data collection, contextualization, amplification, and escalation (DeVos et al. 2022; Li et al. 2023). Claim audit can exhibit a similar division of labor.

No single actor needs to perform every task. This reduces cost but creates coordination problems. A high-quality audit may depend on an extractor, domain specialist, and registry steward whose contributions are not equally rewarded.

6.5 The audit commons

A public protocol creates an audit commons: a shared evaluative resource that can be improved, forked, misused, or enclosed. The commons has at least four public goods.

Table 13: Possible roles in a distributed claim-audit ecology.

Role	Primary activity	Main risk
Protocol maintainer	Versions UEAP instructions and schemas	Standard capture or opaque changes
Extractor	Normalizes text, equations, references, and hidden layers	Loss or distortion of target content
Claim mapper	Identifies high-strength report claims	Over-fragmentation or missed implicit claims
Domain checker	Evaluates carrier adequacy	Discipline-specific bias or authority capture
Reproducer	Reruns audit across models and prompts	False appearance of independence
Counterauditor	Tests disputed findings	Endless regress or strategic delay
Amplifier	Makes high-severity findings visible	Virality outrunning verification
Registry steward	Preserves provenance, corrections, and replies	Centralization or reputational lock-in

1. protocol maintenance;
2. benchmark and calibration corpora;
3. provenance-preserving registries;
4. adjudicated examples of matched, mismatched, and indeterminate audits.

These goods are vulnerable to underprovision. An institution may benefit from external audits while contributing no maintenance effort. A commercial actor may use public examples to build a proprietary audit product. A hostile coalition may flood the registry. The distributed model therefore does not remove governance; it changes governance from exclusive permission to stewardship of shared standards and records.

6.6 Audit bounties and adverse incentives

One response to underprovision is to reward correct audits. Let a bounty B_s depend on severity s and successful independent confirmation. The expected audit payoff becomes

$$U_k(1) = b_k + B_s \Pr(\text{confirmed}) - c_k - h_k.$$

Bounties can increase participation, but they may also reward sensational classification, duplicate submissions, or strategic targeting. A safe design pays for verified blocker identification, not for maximal severity. Rewards should also be available for correct exoneration, successful narrowing, and detection of false audits.

6.7 Self-correcting supply

Distributed audit can become self-correcting when false or weak audits are themselves cheap to audit. Let the first-order audit cost be c , the counteraudit cost c' , and the reputational loss to a

low-quality auditor be L_A . A potential auditor internalizes quality when

$$\Pr(\text{counteraudit})L_A \geq \text{benefit of overstating the finding.}$$

This is the audit analogue of reciprocal authority: auditors are not exempt from the protocol they use. The provenance requirements developed later make this counteraudit possible.

6.8 Implications

The supply model adds three qualifications to the claim that “anyone can audit.” First, permission is not the same as participation; costs and free riding still matter. Second, the identities and incentives of auditors affect quality. Third, mechanization can increase both useful audit and low-quality challenge. The distributed regime is radical because it expands the feasible set of auditors, not because it abolishes scarcity, expertise, or strategic motivation.

7 Reciprocal scientific authority and institutional audit reversal

7.1 From delegated trust to reciprocal auditability

Scientific institutions do more than publish papers. They make claims about their own processes: that a paper has been peer reviewed, that a journal applies rigorous methodological standards, that a funding decision reflects expert evaluation, or that an institutional report has been independently validated. These statements are often treated as authority signals rather than as ordinary claims. Distributed audit changes that status.

Definition 7.1 (Reciprocal auditability). *Scientific authority is reciprocally auditable when institutions retain the power to evaluate claims, but their evaluation procedures, status labels, and claims to rigor can also be evaluated externally under a declared audit basis.*

Reciprocity does not imply symmetry of legal power. A journal can accept or reject a manuscript; an external auditor cannot force publication. The symmetry concerns claim standing. The journal’s statement “this work passed rigorous review” is not immune from demands for a carrier merely because the journal issued it.

7.2 Institutional rigor claims

Let institution I make a status-class-specific rigor claim

$$g_I(\sigma) = \text{“Outputs bearing label } \ell_I \text{ receive reliable review for claim class } \sigma\text{.”}$$

The target is the reliability of label ℓ_I for the declared claim class σ , not a guarantee that every published statement is true. The carrier may include written policy, reviewer selection, audit procedures, decision records, correction behavior, and portfolio outcomes. A prestige name is not itself the carrier unless the target is merely social acceptance.

Let P_I^σ be the provenance-qualified externally audited portfolio of outputs carrying label ℓ_I and making claims in class σ . For audited claim $c \in P_I^\sigma$, let:

- $s_c \in [0, 1]$ be normalized mismatch severity;

- $a_c \in [0, 1]$ be the standing weight of the audit, based on provenance, replication, and counteraudit;
- $w_c > 0$ be a declared portfolio weight for claim visibility, centrality, report strength, or institutional prominence;
- $h_c \in [0, 1]$ be resolution credit, with $h_c = 1$ for a fully repaired or appropriately narrowed claim and $h_c = 0$ for an unresolved finding.

Define the standing-qualified weighted mismatch rate

$$\rho_I^w(\sigma) = \frac{\sum_{c \in P_I^\sigma} w_c a_c s_c}{\sum_{c \in P_I^\sigma} w_c a_c}$$

and the response-adjusted unresolved mismatch rate

$$\rho_I^{wr}(\sigma) = \frac{\sum_{c \in P_I^\sigma} w_c a_c s_c (1 - h_c)}{\sum_{c \in P_I^\sigma} w_c a_c}.$$

The denominator is the provenance-qualified audited portfolio, not all publications. Sampling design, coverage, and selection bias must therefore be reported. Let ρ_I^* be the maximum unresolved weighted mismatch rate consistent with the institution's declared rigor claim after allowing for audit error, sampling uncertainty, and the actual scope of S_I .

Proposition 7.2 (Institutional Audit Reversal). *If $\rho_I^{wr}(\sigma) > \rho_I^*(\sigma)$ under a disclosed sampling design and the institution cannot supply a standing-bearing explanation showing that the external audits use an inapplicable basis, contain material error, are non-representative, or concern statuses outside S_I , then the institution's rigor claim $g_I(\sigma)$ is itself audit-vulnerable. A final mismatch classification requires a separate audit of the institutional claim and its carrier.*

The proposition is intentionally conditional. A journal need not guarantee that every published claim is true. Its review claim may concern relevance, minimal technical adequacy, or procedural fairness rather than final validity. Nor does one failed paper defeat a portfolio-level rigor claim. Audit reversal requires target fixation, provenance-qualified sampling, severity weighting, and credit for correction before judging the institution.

The institutional implication should therefore be stated precisely: peer review loses insulation from independently executable, structured claim-status audit, not all authority. Institutions remain consequential evaluators, but their claims about what review establishes must expose an appropriate carrier and remain answerable to standing-bearing external evidence.

7.3 The carrier of peer review

Peer review is often invoked as if it were a binary property. In practice it can mean editorial screening, one anonymous report, multiple specialist reports, statistical review, registered-report assessment, open review, or post-publication commentary. A reciprocal audit asks:

- What exactly does the venue claim peer review establishes?
- Which review steps are the carrier for that claim?
- Are reviewer reports accessible, summarized, or entirely opaque?
- How does the venue respond when external audits identify material mismatch?

- Are accepted corrections and narrowed claims treated as evidence of functioning review or institutional embarrassment?

An institution can strengthen its rigor claim by making its scope precise. “Peer reviewed” may legitimately mean that named methodological checks were applied, not that all report statuses are correct. Overbroad institutional language is itself subject to narrowing.

7.4 External venue classification

Distributed audit creates venue reputations based on observed audit outcomes rather than policy labels alone. Define:

- a_I : internal audit implementation strength;
- s_I : external audit survival rate;
- t_I : responsiveness to confirmed findings;
- b_I : basis transparency;
- e_I : rate of unresolved high-severity mismatch.

A venue can then be classified as:

1. **internally audited**: implements a substantive protocol;
2. **externally audited**: does not implement internally but receives regular outside audits;
3. **audit-trusted**: survives independent audits and responds constructively;
4. **audit-fragile**: accumulates unresolved high-severity findings;
5. **symbolically audited**: advertises audit labels without substantive checks;
6. **audit-resistant**: attempts to delegitimize or block external audit without addressing findings;
7. **basis-contesting**: disputes the governing standard in a transparent, substantive way.

These classifications are not fixed moral rankings. A venue can move between them as its policy and portfolio change.

7.5 Portfolio effects and prestige

Prestige can initially buffer an institution against external audit. Readers may assign low w_k to independent critics and high prior credibility to the journal. But repeated reproducible mismatch findings can update that prior. Let institutional trust follow

$$T_{I,t+1} = T_{I,t} + \alpha G_t - \beta H_t,$$

where G_t is verified good audit performance and H_t unresolved material mismatch. Registry memory makes H_t persistent rather than episodic.

This creates a strategic incentive for institutions to respond rather than ignore. A correction, narrowed abstract, published counteraudit, or basis clarification can reduce H_t . Attempting to suppress a reproducible audit may increase it by turning a paper-level dispute into an institutional-governance issue.

7.6 Dynamic institutional reversal

Institutional reversal is an updating process rather than a one-shot verdict. Let $\rho_{I,t}^{wr}$ be the weighted unresolved mismatch rate at time t . A general update can be written

$$\rho_{I,t+1}^{wr} = F(\rho_{I,t}^{wr}, A_t, C_t, R_t, B_t),$$

where A_t records new standing-bearing audits, C_t records corrections, narrowing, or retractions, R_t records replies and rebuttals that reduce unresolved mismatch, and B_t records audit-basis resolutions. The response factor h_c is therefore dynamic: $h_c = 0$ denotes an unresolved finding, $h_c = 1$ a fully handled finding, and intermediate values partial repair or partial rebuttal.

The dynamic formulation prevents one-shot reputational destruction from masquerading as audit. Institutions can improve the carrier of their rigor claims through correction, transparent counteraudit, better review procedures, and basis clarification. Conversely, repeated failure to address stabilized findings can raise the unresolved portfolio rate over time.

7.7 Audit of the auditor

Algorithmic audit research has already emphasized that auditors’ standards, incentives, independence, and competence require scrutiny (Costanza-Chock et al. 2022; Goodman and Tréhu 2022; Groves et al. 2024). Distributed scientific claim audit inherits this problem. The auditor makes at least four claims:

1. the selected basis applies to the target;
2. the manuscript was represented faithfully;
3. the cited blocker supports the assigned status consequence;
4. the reported severity is proportionate.

Each claim can be counteraudited. Reciprocal authority therefore extends beyond journals. Authors, auditors, protocol maintainers, and registry operators all occupy auditable roles.

7.8 Institutional response strategies

An institution facing external audit can:

- adopt the finding and narrow the claim;
- issue a counteraudit;
- contest the basis;
- argue that the finding is indeterminate or low severity;
- revise internal review procedures;
- ignore the audit;
- attack the legitimacy of external audit generally;
- attempt to capture an official audit standard.

The strategic response depends on the expected audience weight of the audit, the cost of correction, the institution’s prestige buffer, and whether the finding is reproducible. External audit does not automatically compel institutional change. It creates a new pressure channel whose effectiveness depends on public credibility and memory.

7.9 The radical implication

The institutional reversal is the paper’s central political consequence. A journal can still decide what it publishes. A university can still decide whom it hires. A funder can still allocate grants. But these decisions no longer exhaust the evaluation process. A distributed audit layer allows outsiders to test whether the status language used to justify those decisions matches the publicly available carrier. The gatekeeper remains a gatekeeper, but no longer exclusively controls whether the protocolized claim-status audit record can be created.

8 Voluntary self-audit and external pressure

8.1 Self-audit as preemption

Once authors expect external audit, self-audit becomes a strategic option even when journals do not require it. An author can discover an unsupported status promotion before publication, narrow the report, strengthen the carrier, or publish the audit registry alongside the paper.

Let an author have type $\theta \in \{H, M, L\}$, reflecting the relation between the carrier and the requested report. Let C_S^θ be self-audit cost, R_S^θ the credibility benefit of voluntary audit, F^θ the expected repair benefit from finding correctable blockers, and $\delta_{\text{eff}}^w L^\theta$ expected external exposure loss without self-audit.

The author self-audits when

$$R_S^\theta + F^\theta + \delta_{\text{eff}}^w L^\theta \geq C_S^\theta.$$

For a high-carrier claim, self-audit can serve as a separating signal. For a mixed claim, its main value may be narrowing and repair. For a low-carrier claim, self-audit may reveal that the high-status report is not presently supportable.

Proposition 8.1 (Voluntary self-audit separation). *If high-carrier authors receive a credibility or repair benefit exceeding self-audit cost, while low-carrier authors face a sufficiently large expected external exposure loss from retaining inflated status language, distributed audit pressure induces self-audit by high and mixed types and narrowing, avoidance, or basis contestation by low types.*

The proposition does not imply that low-carrier science disappears. A legitimately exploratory result can remain publishable at an exploratory, compatible, computational, or domain-dependent status.

8.2 The option value of narrowing

Self-audit creates an option that ordinary accept/reject peer review often obscures. An author can preserve the paper while changing the report ceiling. Let the high-status payoff be G_h , the honest lower-status payoff G_ℓ , and the expected cost of external mismatch $\delta_{\text{eff}}^w L$. Narrowing is preferred to retaining the inflated report when

$$G_\ell - C_\ell \geq G_h - C_h - \delta_{\text{eff}}^w L.$$

Mechanization can lower the cost of identifying the appropriate narrower wording. This is a desirable behavioral response, not evidence that the system has “rejected” the science.

8.3 Pre-submission audit certificates

An author may publish a self-audit certificate containing:

- protocol and basis version;
- target manuscript hash;
- principal claim inventory;
- matched and downgraded claims;
- unresolved blockers;
- audit model and prompt provenance;
- independent rerun results;
- author responses and corrections.

The certificate should not be a badge asserting “passed.” A robust certificate displays the remaining lower statuses and indeterminate fields. Its value comes from reducing uncertainty about what the author claims, not from claiming immunity to criticism.

8.4 Strategic self-audit theater

Self-audit can also become performative. An author may select a favorable model, omit difficult sections, choose a weak basis, or publish only successful runs. Let R_T be the appearance benefit of an audit badge, C_T the cost of theater, and $\delta_T L_T$ expected exposure if outsiders reproduce the audit. Theater is attractive when

$$R_T - C_T - \delta_T L_T \geq R_A - C_A,$$

where $R_A - C_A$ is the net payoff from substantive audit.

Distributed rerunability raises δ_T and can therefore make theater less stable. But this requires target-file access, prompt disclosure, and protocol versioning. A screenshot of a favorable chat output is not an audit certificate.

8.5 Authors as auditors of institutions

Self-audit also changes the author’s bargaining position. An author can compare a journal’s review demand with the declared audit basis. If a referee requests stronger evidence for one group but not another, or insists on a status downgrade unsupported by the carrier, the author can counteraudit the review. This does not eliminate editorial discretion, but it can make inconsistent standards more visible.

A reciprocal system therefore gives authors two new strategies:

1. use self-audit to strengthen or narrow their own work;
2. use counteraudit to challenge institutional or reviewer claims about the work.

The second strategy is especially important for outsiders and unconventional work. Gatekeepers can no longer rely solely on unrecorded authority; they may need to state the target, carrier standard, and blocker supporting a demand.

8.6 Confidential and pre-publication contexts

Non-permissioned audit does not imply that confidential manuscripts should be uploaded to public systems. Authors and institutions may use local models, secure execution, redacted documents, or delayed public release. The strategic point survives: the protocol can be executed without ceding final authority to the journal. Privacy and intellectual-property constraints affect execution cost and source set K , but they do not restore exclusive institutional control over protocolized audit execution.

8.7 Distributional effects

Self-audit may help researchers with limited access to senior feedback. Liang and colleagues motivate LLM feedback partly by the scarcity and unequal distribution of expert review (Liang et al. 2024). A portable protocol can give junior or independent researchers a structured pre-submission critique. Yet benefits may remain unequal if high-quality models, long-context access, domain databases, or private counteraudit services are costly. Distributed execution should therefore not be confused with equal capability.

8.8 Prediction

If distributed audit becomes salient, the model predicts:

- more voluntary claim registries and pre-submission audit records;
- more explicit narrowing before formal review;
- increased demand for local or confidential audit tools;
- more author challenges to reviewer status claims;
- selective publication of favorable audit runs unless provenance norms develop;
- credibility premiums for papers that disclose indeterminate and lower-status findings rather than only “passes.”

9 Venue classification under external audit

9.1 Institutional adoption is no longer the only regime variable

Earlier audit-adoption models divide venues into audited and non-audited institutions. Distributed audit requires a richer classification. A venue can refuse internal audit and still be heavily audited externally. It can adopt a formal policy yet rarely enforce it. It can survive external scrutiny despite minimal formal policy. The relevant state is therefore a combination of policy, portfolio, and response.

Let venue V have:

- internal audit strength a_V ;
- external audit intensity x_V ;
- externally confirmed material mismatch rate ρ_V ;
- response quality t_V ;
- audit-basis transparency b_V ;

- symbolic compliance score s_V .

Readers and authors form a venue trust assessment

$$T_V = F(a_V, x_V, \rho_V, t_V, b_V, s_V),$$

with positive dependence on substantive internal audit, response quality, and basis transparency, and negative dependence on unresolved mismatch and theater.

9.2 External venue classification

Proposition 9.1 (External Venue Classification). *If standing-bearing public external audits occur with sufficient frequency, venues acquire reputational types based on weighted outcomes and response history even when they do not formally adopt the protocol.*

The proposition follows from ordinary Bayesian or reputational updating. A venue cannot fully control the evidence used to estimate its rigor once public audit records exist.

Table 14: Venue types in a distributed-audit environment.

Venue type	Defining pattern	Strategic implication
Internally audited	Strong internal protocol and provenance	Controls timing and can repair before publication
Externally audited	Little internal adoption but regular outside scrutiny	Cannot opt out; often reacts after publication
Audit-trusted	Low confirmed mismatch and strong response	Gains credibility premium
Audit-fragile	High unresolved mismatch or poor response	Faces discount despite prestige
Symbolic-audit	Visible badge, weak enforcement	Vulnerable to audit-theater exposure
Audit-resistant	Rejects or attacks external audit without addressing substance	May retain loyal audience but lose reciprocal credibility
Basis-contesting	Transparently disputes the standard or scope	Can improve the protocol if contestation is substantive
Exploratory safe-harbor	Publishes explicitly non-final work	Avoids false promotion while preserving speculation

9.3 Trust premium and author sorting

Let R_V be the publication reward at venue V , including prestige and audience. Let C_V^θ be the expected audit cost for claim type θ . An author chooses venue V to maximize

$$U_\theta(V) = R_V + T_V - C_V^\theta.$$

As external audit records affect T_V , author sorting can change even without formal policy changes. High-carrier authors may prefer audit-trusted venues because the label separates them. Authors

relying on rhetorical inflation may prefer venues with low x_V or weak response. Mixed claims may choose safe-harbor venues where lower statuses retain value.

This mechanism differs from simple journal adoption. The trust premium is produced by observed audit performance, not only by a declared badge.

9.4 Badge dilution

If many weak venues adopt an audit label without substantive enforcement, the average informational value of the label declines. Let π be the probability that a venue displaying an audit badge implements real audit. The reader’s expected trust premium is increasing in π . Audit theater lowers π and can destroy the separating value of the label for genuine adopters.

Distributed external audits can restore separation by producing venue-specific evidence. Instead of trusting the generic badge, readers compare standing-qualified external survival rates, severity distributions, and response histories. The badge becomes a prior, not a verdict.

9.5 Press releases and institutional communication

Venue audit should include not only papers but institutional summaries, editorials, press releases, and claims about review. A paper may contain carefully narrowed language while a press office promotes it as proof or resolution. In that case the institutional report is a new claim and requires its own audit. External audit can reveal status promotion introduced after peer review.

9.6 Prestige exposure

Prestige institutions may initially face lower effective consequences because audiences discount outsider audits. Let w_V be the weight given to external findings against venue V . A prestige shield corresponds to low w_V despite high audit quality. Reproducible cross-model audits, domain-expert confirmation, and accumulated registry memory can raise w_V over time.

A distributed system therefore predicts an early asymmetric phase: low-prestige targets may be highly vulnerable to public audit, while elite institutions remain insulated. Governance must monitor whether external audit democratizes scrutiny or simply adds another tool used selectively against weak actors.

9.7 Institutional response equilibria

A venue facing external audit chooses among correction, counteraudit, basis contestation, silence, and retaliation. Let C_R be correction cost, C_C counteraudit cost, C_B basis-contestation cost, and L_I expected reputational loss from inaction. The venue responds constructively when

$$\min\{C_R, C_C, C_B\} < L_I.$$

Mechanization can increase L_I by making the finding persistent and reproducible, but can also decrease it if the audit ecosystem becomes noisy and audiences stop paying attention.

9.8 Prediction

The venue model predicts that distributed audit will generate public reputational categories before uniform institutional standards exist. Some journals will embrace external audit as quality assurance; others will build counteraudit teams; some will contest protocol legitimacy; and some will ignore the system until a portfolio-level pattern changes the expected cost of inaction.

10 Pressure migration to audit-basis control

10.1 Where the game moves next

Distributed audit raises the expected cost of direct report inflation. An author who describes a fitted model as a prediction can expect that someone may run the protocol and identify target ancestry. A journal that labels a result “validated” may face external scrutiny of the validation carrier. Strategic behavior does not disappear. It moves toward the coordinates that determine how the audit classifies the report.

Let direct inflation produce payoff

$$U_D = G_h - C_D - \delta_D L_D,$$

where G_h is the high-status reward, C_D the cost of maintaining direct inflation, δ_D distributed consequential-detection probability, and L_D the consequence.

Let audit-basis manipulation produce

$$U_B = G_h - C_B - \delta_B L_B.$$

Basis manipulation includes narrowing the target after the result is known, weakening the carrier standard, redefining the domain, excluding inconvenient alternatives, downgrading blockers, choosing a friendly protocol version, or altering the prompt so the evaluator treats a disputed input as settled.

Theorem 10.1 (Pressure Migration). *An agent prefers audit-basis manipulation to direct report inflation when*

$$C_B + \delta_B L_B < C_D + \delta_D L_D.$$

As distributed audit increases δ_D while δ_B remains low, strategic pressure migrates from report wording to control of the audit basis.

This is the paper’s central warning. Audit does not end the game. It changes where the game is played.

10.2 Sites of basis manipulation

The audit basis contains several strategic coordinates.

1. **Target manipulation.** A broad claim is re-described as a narrow one during audit while retaining broad public uptake.
2. **Carrier manipulation.** A benchmark, derivation, consensus statement, or simulation is declared sufficient for a status it does not ordinarily bear.

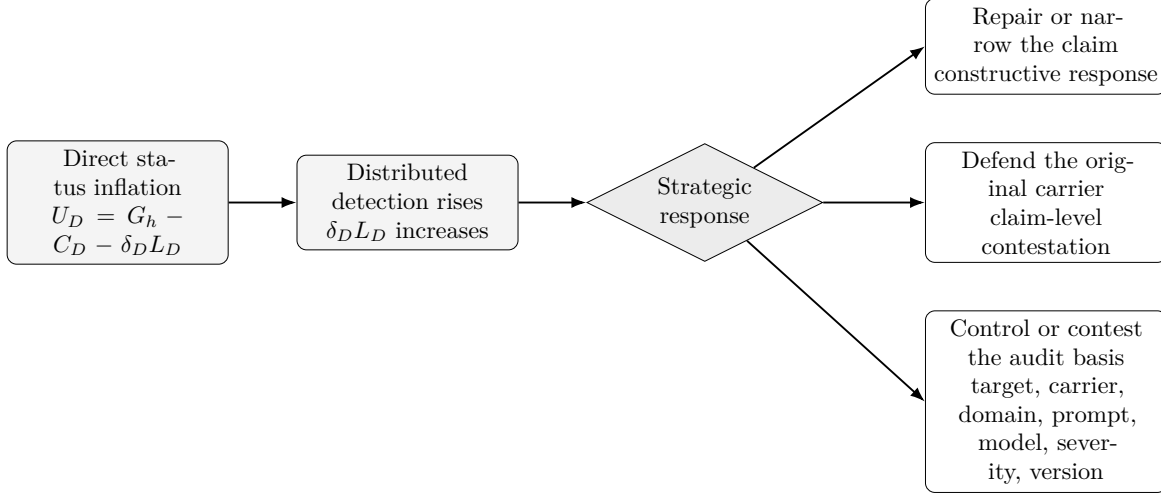


Figure 6: Pressure migration. Stronger detection of direct report inflation can improve reporting while also shifting strategic effort toward control of the audit basis.

3. **Domain manipulation.** Favorable populations, boundaries, time windows, or operating conditions are treated as if they were the general target.
4. **Alternative-class manipulation.** Rivals are excluded after the result is known so that the preferred claim appears uniquely supported.
5. **Blocker reclassification.** A load-bearing unresolved condition is labeled “future work” or “minor limitation.”
6. **Status-language manipulation.** Terms such as predicts, explains, validates, or proves are defined unusually weakly.
7. **Protocol selection.** The agent chooses a version or prompt likely to produce a favorable result.
8. **Authority substitution.** Institutional acceptance is inserted as if it were the carrier being audited.

The game becomes especially acute when no common basis exists. Different fields can legitimately require different evidence standards. Basis contestation is therefore not inherently evasive. The challenge is to distinguish transparent disagreement from strategic manipulation.

10.3 Legitimate contestation

A basis dispute is legitimate when it makes the disagreement more explicit. For example, an author may argue that the audit requested predictive status while the paper claims only explanatory compatibility; that a proof carrier rather than an empirical carrier is relevant; or that the external auditor omitted a declared boundary condition. These are substantive corrections.

A useful test is counterfactual symmetry:

Would the actor accept the same basis rule if it favored a competing claim rather than its own?

A basis rule adopted only after the outcome is known, or applied asymmetrically to allies and rivals, is evidence of manipulation.

10.4 Standard competition

Let there be bases $b \in \mathcal{B}$. Authors and institutions choose bases, while readers choose which bases they trust. Each basis has implementation cost c_b , detection quality q_b , scope s_b , and capture risk g_b . A basis gains adoption when its expected credibility benefit exceeds cost:

$$T_b(q_b, s_b) - c_b - g_b \geq 0.$$

This creates a standard-competition game. One equilibrium may be convergence on a common basis. Another may be fragmentation: different communities use incompatible standards and dismiss each other’s audits.

Protocol forks are not necessarily bad. A medical carrier, mathematical proof carrier, and exploratory theoretical carrier should differ. Fragmentation becomes defective when the basis is chosen to protect conclusions while being presented as a general standard.

10.5 Basis capture

Powerful institutions can shape official versions, accreditation requirements, or benchmark corpora. Let coalition G receive benefit B_G from a favorable basis and pay lobbying or maintenance cost C_G . Capture effort is attractive when

$$B_G - C_G - \Pr(\text{external counteraudit})L_G > 0.$$

Distributed audit can constrain capture because outsiders can fork the protocol and compare outcomes. But fragmentation can also help capture by making audiences unable to decide which basis is legitimate.

10.6 Prompt-level basis control

In LLM execution, the prompt is part of the basis. Small changes in instruction order, role framing, requested severity, or supplied context may alter the result. A target author could publish a favorable audit generated under a weak prompt; a rival could use an aggressively skeptical prompt. Prompt disclosure is therefore not a cosmetic reproducibility detail. It is a record of the standard actually applied.

The protocol should distinguish:

- invariant core instructions that define UEAP;
- domain-specific carrier supplements;
- target-specific questions;
- adversarial or rhetorical instructions that should not alter status criteria.

10.7 Auditing basis changes

A basis change is itself a claim that the new standard is appropriate. A registry should record:

- the old and new basis versions;
- the rationale for change;
- which prior audit outcomes would change;

- whether the change was proposed before or after a disputed result;
- distributional effects across methods, institutions, and fields;
- independent review of the change.

A basis update that only reverses an inconvenient audit result deserves higher scrutiny than one motivated by a general defect demonstrated across cases.

10.8 Distributed basis audit

The pressure-migration theorem implies a second-order design condition. Direct audit must be accompanied by basis audit. Let δ_B be the probability that basis manipulation is detected. If δ_B remains near zero while δ_D rises, migration is predictable. A stable regime requires

$$C_B + \delta_B L_B \geq C_D + \delta_D L_D$$

for major manipulative routes, or at least enough transparency that basis choice is visible to readers.

The aim is not to freeze standards. It is to make changes legible and contestable. Basis governance does not promise convergence on one final standard. Its minimum deliverable is a reconstructible record of what changed, when it changed, which outcomes changed with it, and whether the rule was applied symmetrically. Section 12.7 classifies the resulting disputes so that unresolved basis conflict lowers consequence rather than disappearing behind an unqualified audit verdict.

11 Symbolic compliance and distributed audit theater

11.1 From audit badge to audit substance

A new audit practice can generate legitimacy before it generates verification. Authors may add a “UEAP audited” label, journals may require a completed checklist, and institutions may advertise an audit program. Such signals can be useful during adoption, but they can also become ceremonial structures decoupled from actual blocker discharge (Meyer and Rowan 1977; Bromley and Powell 2012; Power 1997).

Let real compliance yield payoff

$$U_R = R_R - C_R,$$

where R_R is credibility benefit and C_R substantive implementation cost. Let symbolic compliance yield

$$U_S = R_S - C_S - \delta_S L_S,$$

where R_S is appearance benefit, $C_S < C_R$ is ceremonial cost, and $\delta_S L_S$ expected consequence if outsiders expose the gap.

Proposition 11.1 (Distributed Audit Theater Constraint). *Symbolic compliance is preferred to real compliance when*

$$R_S - C_S - \delta_S L_S \geq R_R - C_R.$$

As distributed external audit increases δ_S , audit theater becomes less stable, holding other terms fixed.

The proposition explains why non-permissioned audit can improve institutional adoption. A venue cannot safely rely on an internal badge if outsiders can rerun the underlying protocol.

11.2 Forms of theater

Audit theater may include:

- a registry with generic or empty carrier fields;
- a status label generated without loading the full paper;
- omission of appendices, equations, or supplementary methods;
- selective disclosure of favorable model runs;
- a prompt that presupposes the desired status;
- a pass/fail badge that hides lower statuses and indeterminate findings;
- a journal policy that requires forms but never changes decisions;
- claims that a paper was “AI audited” without model or basis provenance.

A ceremonial limitations section is a related form. The paper acknowledges a blocker but retains the same high-strength abstract and press language. UEAP requires the blocker to affect report status, not merely appear in a list.

11.3 Substance checks

External auditors can test substance by asking:

1. Did the audit include the same target document version that was published?
2. Did the audit basis match the public claim status?
3. Were unresolved blockers preserved in the abstract and conclusion?
4. Can independent evaluators reproduce the status classification?
5. Did the institution act on high-severity findings?
6. Are negative and indeterminate audit outputs disclosed?

A system that discloses only successful audits should be classified as marketing rather than assurance.

11.4 Audit washing

The algorithmic-accountability literature warns that loosely specified audits can produce “audit washing”: the appearance of accountability without clear standards, independence, or consequential action (Goodman and Tréhu 2022; Costanza-Chock et al. 2022; Groves et al. 2024). Distributed claim audit faces the same risk. The difference is that an open protocol enables outsiders to test the badge against the artifact.

This does not guarantee exposure. Institutions may have greater communication reach, legal resources, and authority. A weak external audit can be ignored. But audit washing is no longer a purely internal fact about process quality; it can be represented as a mismatch between the audit claim and the publicly reconstructible carrier.

11.5 Prestige exemption

Audit theater can be asymmetric. Prestigious authors may be allowed informal exemptions while junior authors complete detailed registries. Let C_R^P and C_R^J be audit burdens for prestigious and junior actors. An asymmetry

$$C_R^J \gg C_R^P$$

without a carrier-based justification weakens the institutional fairness claim. Distributed audits of review demands, decision letters, and published portfolios can reveal such patterns if the records are available.

11.6 Good transitional symbolism

Not all symbolic elements are defective. A preliminary badge can signal that a journal is building capacity. A short registry can begin norm formation. The difference is report status. A transitional program should say “pilot audit with limited scope” rather than “verified.” The system should also publish the rate at which audits changed wording or decisions, not only the number of forms completed.

11.7 Design implication

The stable alternative to theater is not maximal bureaucracy. It is a small set of high-information, rerunnable artifacts: basis, target hash, prompt, model, blocker evidence, status output, severity, and response. Distributed audit makes those artifacts more valuable because external actors can use them to reproduce the result.

12 Audit pluralism, model variance, and reproducibility

12.1 Multiple evaluators, multiple outputs

Non-permissioned audit produces pluralism. Different users may run different models, model versions, temperatures, prompts, retrieval systems, domain supplements, or protocol forks. Their outputs may disagree. This is not an incidental implementation bug. It is a central property of distributed audit.

Audit-basis fragmentation is therefore one of the paper’s flagship strategic predictions. Once direct status inflation becomes easier to expose, disagreement can migrate to protocol version, target selection, carrier standard, extraction method, prompt framing, model choice, and severity classification. Distributed audit does not eliminate contestation; under low convergence it relocates contestation to the audit basis.

Let an execution basis record be

$$b = (\mathcal{P}, v, m, e, x, t, \sigma, C_\sigma),$$

where $\mathcal{P}$ is the protocol, v its version, m the model or evaluator family, e the target-document extraction and preprocessing record, x the prompt or execution instruction, t the selected target

claim, σ the requested status, and C_σ the carrier standard. For auditor or execution source k , write

$$O_{k,b}(c)$$

for the audit output on claim c . The output includes status, blockers, severity, reportability, and next action. Treating these fields as one basis record prevents model, prompt, extraction, and target selection from disappearing into an unqualified phrase such as “the audit found.”

Audit pluralism creates three distinct questions:

1. Do evaluators agree on the target and requested status?
2. Do they identify the same blockers?
3. Do they assign the same consequence and severity?

Agreement on status with different blocker rationales is not the same as full reproducibility. Disagreement about target extraction may indicate that the paper itself is semantically unstable.

12.2 Convergence measures

Let $S_k(c)$ be the status assigned by audit k and $B_k(c)$ its blocker set. Define status agreement

$$A_S(c) = \frac{2}{N(N-1)} \sum_{i < j} \mathbf{1}\{S_i(c) = S_j(c)\}.$$

Define average blocker overlap using Jaccard similarity:

$$A_B(c) = \frac{2}{N(N-1)} \sum_{i < j} \frac{|B_i(c) \cap B_j(c)|}{|B_i(c) \cup B_j(c)|},$$

with the usual convention when both sets are empty.

Let $A_T(c)$ measure agreement on target and requested status. A composite convergence score may be

$$\text{Conv}(c) = w_T A_T(c) + w_S A_S(c) + w_B A_B(c),$$

where weights are declared. The score is not a truth metric. High convergence can reflect a shared systematic error, especially when models or prompts are correlated. It is a reproducibility indicator.

12.3 Independent versus correlated reruns

Rerunning the same model with the same prompt and a different random seed provides limited independence. Stronger reruns vary one or more of:

- model family;
- provider or open-source implementation;
- prompt author;
- domain specialist involvement;
- retrieval corpus;
- audit basis fork;
- human review team.

A registry should label rerun classes. “Three audits agree” is weak evidence if all three are copied from one output.

12.4 Protocol versioning

Every audit should identify a protocol version. A version change can alter target extraction, carrier standards, severity rules, or output schema. Let b_t be the basis at time t . A result should be reported as

$$O(c \mid b_t),$$

not as an unqualified property of the paper.

Versioning enables historical replay. When a basis changes, a registry can rerun prior audits and identify which outcomes change. This is essential for distinguishing genuine standard improvement from ex post reversal of inconvenient findings.

12.5 Target-document versioning

Scientific papers change across preprint, submission, revision, accepted manuscript, and version of record. An audit must identify the exact target. A file hash is the simplest anchor. If an author corrects a claim, the old audit should remain linked to the old version rather than presented as a permanent defect in the new one.

This produces a claim lineage:

$$c^{(0)} \rightarrow c^{(1)} \rightarrow \dots \rightarrow c^{(t)}.$$

Each edge records whether the change was narrowing, carrier repair, domain clarification, or a new claim. The registry should reward successful repair by showing closure, not merely preserve accusation.

12.6 Audit basis fragmentation

Pluralism becomes fragmentation when communities use incompatible bases and no translation exists. Suppose basis b_1 treats a benchmark as a sufficient carrier for capability while b_2 treats it only as benchmark performance. The audits disagree by construction. The solution is not majority vote. The disagreement should be represented as a basis-level conflict.

For audit output $O_{k,b}(c)$, define status-class convergence across admissible executions as

$$\text{Conv}_S(c) = \Pr(S(O_{k,b}(c)) = S(O_{k',b'}(c))) ,$$

where the probability is taken over a declared population of admissible auditors and execution basis records. This is not a truth probability. It measures whether the public audit claim is stable across the variation the registry treats as legitimate.

Define severity convergence at tolerance one by

$$\text{Conv}_{\text{Sev}}(c) = \Pr(|\text{Sev}(O_{k,b}(c)) - \text{Sev}(O_{k',b'}(c))| \leq 1) .$$

A system can therefore agree that a mismatch exists while disagreeing about whether it is a local wording issue or a principal carrier failure. Public consequence should track both status and severity

convergence.

Let C_C be the expected cost of defending the original claim under a stable audit basis and $C_B(\text{Conv}_S)$ the cost of contesting the audit basis. Basis contestation generally becomes easier as convergence falls. Let $E[L_C]$ and $E[L_B]$ be the expected consequences of claim-level defense and basis-level contestation.

Theorem 12.1 (Audit-Basis Fragmentation). *If*

$$C_B(\text{Conv}_S(c)) + E[L_B] < C_C + E[L_C],$$

then a strategically responsive target prefers basis contestation to direct defense of the claim. Low status-class convergence can therefore redirect discourse from the paper’s carrier to the choice of protocol version, prompt, target extraction, model, or domain standard.

The theorem does not classify every basis objection as evasive. A low-convergence audit may reveal a genuine ambiguity in the target or carrier standard. Its point is predictive: distributed execution creates an incentive to select, attack, or fork the audit basis whenever that route is cheaper than repairing the underlying claim. Low convergence also reduces the standing component C_k in w_k , weakens δ_{eff}^w , and can increase η_{eff}^w when divergent severe findings acquire public reach before basis resolution.

A basis crosswalk records:

- common target vocabulary;
- carrier requirements that differ;
- status mappings;
- domain assumptions;
- which results are invariant across bases;
- the protocol, prompt, model, and document transformations responsible for divergence.

Invariant findings are especially valuable. If a claim is mismatched across several reasonable bases, confidence increases. If the outcome changes entirely with the basis, the public dispute should focus on the standard rather than the target author’s character. The standing of a public audit should ordinarily fall when material basis fragmentation remains undisclosed.

12.7 Basis-contestation outcomes

The purpose of basis governance is not to force every community into one standard. It is to prevent basis disagreement from being concealed inside an apparently unitary audit result. A registry should classify material divergence using the following outcome types.

Table 15: Basis-contestation outcome taxonomy.

Outcome	Diagnostic condition	Reporting and consequence rule
Invariant finding	The material target, status, and blocker survive reasonable protocol, prompt, evaluator, extraction, or domain-basis variation.	Report the invariant core and disclose the tested basis family; convergence can raise standing but does not by itself prove truth.
Basis-sensitive finding	The classification changes under an identified, defensible basis difference.	Report both outputs and the exact basis coordinate responsible; do not circulate one result as basis-independent.
Target-unstable finding	Evaluators disagree about the literal or reconstructed claim target or requested status.	Treat the audit as indeterminate or request claim clarification before adverse consequence.
Carrier-standard dispute	The parties agree on the target but disagree about what support licenses the requested status.	Move the public dispute to the carrier standard and domain supplement; preserve lower consequence until the standard is adjudicated.
Strategic basis manipulation	A basis is changed post hoc, applied asymmetrically, selected only when outcome-favorable, or rejected under counterfactual symmetry.	Record the timing and asymmetry as a separate governance finding; rerun prior outputs under the declared replacement basis.
Legitimate domain supplement	A field-specific basis differs for a stated disciplinary reason and is applied consistently across competing claims.	Preserve the domain restriction and provide a crosswalk rather than treating the difference as evasion.

Versioning and crosswalks do not settle every basis dispute. They convert hidden standard control into an inspectable object and make clear whether a public finding is invariant, basis-sensitive, target-unstable, or strategically manipulated. That conversion is itself a governance gain. Where material basis conflict remains unresolved, the audit’s standing and requested consequence should remain correspondingly limited.

12.8 Model bias and authority cues

LLM evaluators may respond to author affiliation, prestige language, writing style, or assertion strength. Peer-review simulations and evaluations already identify reviewer bias and model limitations (Jin et al. 2024; Zhou et al. 2024). A reproducibility protocol should therefore support blinded and unblinded runs, controlled metadata perturbations, and authority-cue tests.

For example, run the same paper under:

1. original metadata;

2. anonymized metadata;
3. permuted institution names;
4. neutralized press or prestige language.

If status or severity changes materially, the audit has a bias blocker.

12.9 Prompt sensitivity

Prompt sensitivity should be measured rather than hidden. Let P be a set of admissible prompts implementing the same protocol. Define prompt variance

$$V_P(c) = \text{Var}_{p \in P}[d(S_p(c), S_{\text{modal}}(c))],$$

where d is a declared status-distance metric. High V_P supports an indeterminate or low-confidence output unless a domain expert resolves the discrepancy.

The protocol should forbid prompts that instruct the model to maximize criticism or to defend the target. Admissible prompts may differ in wording but must preserve the same audit basis and output obligations.

12.10 Audit reproducibility certificate

A reproducibility certificate should contain:

1. protocol and basis version;
2. target file hash;
3. model and version;
4. prompt text;
5. retrieval sources, if any;
6. random settings where applicable;
7. raw structured output;
8. human corrections;
9. rerun results;
10. status and blocker convergence;
11. unresolved discrepancies.

This certificate is the carrier for the claim that an audit is reproducible. Without it, public audit pluralism can degenerate into screenshots and authority contests.

12.11 Governance implication

Distributed audit should not seek one official model whose output ends dispute. It should seek transparent, comparable, and contestable executions. The protocol supplies a common grammar; plural evaluators supply independent checks; provenance allows the checks themselves to be audited.

Basis-governance takeaway

Distributed audit does not abolish disagreement about standards. It makes the source of disagreement inspectable: invariant finding, basis-sensitive finding, target instability, carrier-standard dispute, strategic basis manipulation, or legitimate domain supplement. Versioning and crosswalks do not force consensus; they prevent basis control from disappearing behind an unqualified claim that “the audit found” a result.

13 Weaponization, harassment, and severity discipline

13.1 The low-cost challenge problem

Making audit easy changes the cost of criticism. This is valuable when claims would otherwise escape scrutiny, but it also enables audit spam, selective attacks, and reputational harm. A rival can run repeated hostile prompts, publish only the harshest output, or characterize a wording issue as a fatal scientific defect. The four-layer distinction in Table 8 remains controlling: execution is open, publication must be transparent, standing is earned, and consequence must be proportionate.

Let an auditor receive private benefit B_R from weakening a target and public-interest benefit B_P from correcting the record. Let the audit cost be C_A , expected cost of a false or overstated finding be H , and probability of successful counteraudit be γ . The challenge is attractive when

$$B_R + B_P - C_A - \gamma H \geq 0.$$

LLM assistance lowers C_A , potentially increasing both valuable and abusive challenges.

13.2 Social value of challenge

Let $g(S)$ be the social gain from exposing a real blocker at severity S , $h(S)$ the social harm from a spurious or disproportionate challenge at that severity, and C_N the attention cost of noise. If π is the probability the challenge is materially correct, expected social value is

$$V_{\text{challenge}}(S) = \pi g(S) - (1 - \pi)h(S) - C_N.$$

A public system should increase π , increase $g(S)$ through clear correction pathways, reduce $h(S)$ through proportionality and right of reply, and prevent low-standing audits from borrowing the reputational force of high-severity labels.

Proposition 13.1 (Severity-disciplined public challenge). *A public challenge at severity S is socially warranted only when its standing level is sufficient for the consequence sought and $V_{\text{challenge}}(S) > 0$. Raising the announced severity without improving provenance, correctness, or correction value can reduce social welfare even when a narrow blocker is real.*

13.3 Severity taxonomy

A distributed audit needs a common distinction between a narrow wording issue and a carrier failure. This paper proposes six severity levels.

Table 16: Severity levels for distributed claim audit.

Level	Classification	Typical consequence
S0	No material mismatch detected	Preserve report; record scope and uncertainty
S1	Clarification or wording issue	Revise wording without changing central status
S2	Active blocker requiring downgrade	Narrow prediction, confirmation, proof, or universality claim
S3	Carrier mismatch undermining a principal claim status	Major revision, counteraudit, or withdrawal of that status claim
S4	Systematic status laundering or basis manipulation	Portfolio or institutional review; enhanced provenance requirement
S5	Repeated institution-level mismatch pattern	Audit of review system, policy claim, or governance carrier

Severity is not a moral scale. S4 does not automatically imply intent. It describes a systematic status-transition pattern. S5 concerns institutional repetition, not the magnitude of one paper’s error.

Severity discipline prevents a low-cost audit ecology from treating every issue as fatal. It is the bridge between open execution and proportionate consequence.

13.4 Standing thresholds for public consequence

Severity and audit standing are orthogonal. A raw A0 execution may identify a possible S3 issue for private self-audit, but it should not be publicized as a settled S3 finding. Suggested defaults are:

- S0–S1: A1 provenance completeness may support a transparent preliminary execution; a public finding ordinarily also requires A2 registry completeness and a correction path.
- S2: A3 replication is preferred before a public downgrade demand.
- S3: A3 replication plus A4 counteraudit or domain review should ordinarily precede high-consequence publication.
- S4: A4–A5 standing, multi-finding pattern evidence, and explicit intent-neutral language are required.
- S5: A5 stabilization plus portfolio design, institutional reply, sampling disclosure, and governance review are required for C3 institutional consequence.

The thresholds are defaults rather than censorship rules. Anyone may still run or discuss an audit. The rule governs how much institutional or reputational consequence the audit is entitled to request.

13.5 False-positive control as a standing condition

False-positive control is internal to audit standing. An audit source that repeatedly overstates severity, ignores matched controls, conceals basis variation, or refuses counteraudit should receive a lower w_k even when some underlying blocker is real. The relevant governance question is not only whether an audit found an issue, but whether it assigned the right status and severity.

Table 17: Severity-specific false-positive controls.

Severity	Principal false-positive risk	Minimum control
S1	Over-reading a wording or presentation issue	Quote the exact report text and show why clarification, rather than status downgrade, is sufficient.
S2	Unnecessary report downgrade	Map carrier to requested status and state the minimal narrowing that resolves the blocker.
S3	Serious attack on a principal claim	Declare the audit basis, obtain independent rerun or domain review, and provide counteraudit opportunity.
S4	Unsupported allegation of systematic laundering	Require A4–A5 provenance, pattern evidence, intent-neutral language, and severity rationale.
S5	Premature institutional-pattern claim	Require a weighted portfolio, sampling disclosure, response adjustment, institutional reply, and governance review.

Execution remains non-permissioned; standing and consequence are earned. A false or materially overstated public audit contributes to η_{eff}^w and should reduce the source’s future standing weight until corrected. The controls in Table 17 are designed to lower η_{eff}^w without suppressing legitimate δ_{eff}^w .

13.6 Blocker specificity

Every public finding should identify:

- the exact report sentence or status claim;
- the carrier offered;
- the blocker separating the carrier from the status;
- the passage evidence;
- the minimal narrowing or repair;
- the severity rationale;
- the audit basis.

Generic claims that a paper is “invalid,” “unscientific,” or “failed UEAP” are not adequate. The audit must state which transition is unsupported.

13.7 Right of reply and counteraudit

The target should be able to:

1. correct extraction errors;

2. supply omitted carrier material;
3. identify an inapplicable basis;
4. accept narrowing;
5. request independent rerun;
6. distinguish the author’s paper from later institutional promotion;
7. appeal severity while accepting the underlying blocker.

The initial audit and reply should remain linked. Deleting a corrected audit erases learning; preserving an uncorrected accusation without resolution is unfair. The registry should show status transitions over time.

13.8 Selective targeting

Distributed audit can reproduce social inequalities if only controversial outsiders, junior researchers, or disfavored fields are audited. Let $p_k(c)$ depend on author prestige, topic, institution, or ideology. A fairness audit compares exposure rates conditional on claim type and report strength.

A system may be technically non-permissioned but socially asymmetric. Prestige institutions can mobilize more counterauditors and communication power, while independent authors face higher reputational consequences from one flawed audit. The governance layer should therefore publish target-selection statistics where organized audit programs exist.

13.9 Audit spam and attention attacks

A target can be burdened by hundreds of low-quality audits. Even if each is weak, responding consumes time. Possible controls include:

- deduplication by blocker and basis;
- reputation-weighted queues;
- rate limits in formal registries;
- aggregation of identical findings;
- response obligations only above a provenance threshold;
- sanctions for fabricated evidence or concealed prompt manipulation;
- public “no material mismatch” counteraudits.

These controls should not recreate exclusive gatekeeper control over protocolized audit execution or public registry entry. They govern attention and registry quality, not the right to run or publish an audit elsewhere.

13.10 Anonymous audit

Anonymity protects whistleblowers and weak actors but reduces accountability. A two-layer design can allow pseudonymous public release with escrowed provenance or trusted third-party verification. The audit’s evidential weight should depend on reproducibility and cited evidence, not only identity. Still, anonymous high-severity allegations require stronger independent confirmation.

13.11 Normative constraint

The objective is not to maximize the number of negative findings. It is to maximize correct, proportionate, repairable status classification. A successful system should reward an auditor who concludes “indeterminate” or “S1 wording issue” when that is what the evidence supports.

Public challenge rule

The requested consequence must be supported by both the finding’s severity and the audit’s standing. A raw execution may justify private investigation; it does not justify a settled high-severity allegation. Exact target text, declared basis, passage grounding, independent confirmation, right of reply, and correction history are part of the carrier for consequential public use.

14 Registry memory and long-run normalization

14.1 The repeated game

Scientific communication is repeated. Authors publish again, journals build portfolios, reviewers develop reputations, and claims are corrected or amplified. A one-shot audit can be ignored; a persistent registry changes future payoffs.

Let a confirmed mismatch impose immediate consequence L_0 and future per-period reputational or monitoring consequence r . With discount factor $\delta \in (0, 1)$, the present value of future consequence is

$$R = \frac{\delta}{1 - \delta} r.$$

The deterrence condition becomes

$$\delta_{\text{eff}}^w(L_0 + R) \geq B.$$

Corollary 14.1 (Registry Memory). *Holding other parameters fixed, persistent provenance-bearing audit memory lowers the contemporaneous consequential-detection probability required to deter status inflation:*

$$\delta_R^* = \frac{B}{L_0 + R} < \frac{B}{L_0} = \delta_0^*$$

when $R > 0$.

This is the positive case for public audit registries. They reduce repeated re-litigation and make future consequences depend on patterns rather than isolated events.

14.2 What the registry must remember

A useful registry records more than negative findings. It should preserve:

- target and protocol versions;
- initial audit;
- author or institution response;
- corrected manuscript version;
- independent reruns;

- resolved and unresolved blockers;
- successful narrowing;
- counter audits that overturn findings;
- severity changes;
- final report status.

A registry that remembers accusations but not repair creates perverse incentives. The ability to close a blocker must be visible.

14.3 Normalization of lower statuses

Long-run audit can change the reward structure of scientific language. If calibrated-only, compatible-only, computation-only, and domain-dependent results are treated as legitimate contributions rather than failed final claims, authors lose less reward by narrowing. In the deterrence model, the benefit B of status inflation decreases.

This cultural change may be more important than increasing punishment. A field that rewards precise limitations can make honest reporting incentive-compatible at lower audit intensity.

14.4 Institutional memory

The registry also supports portfolio-level institutional audit. A venue with repeated S3 or S4 findings may face higher future scrutiny. A venue that consistently corrects or narrows claims can earn trust. The relevant measure should be response-adjusted, not raw failure count. A journal that publishes difficult frontier work may receive more audits than a conservative venue.

Section 7 defines the response-adjusted weighted mismatch rate

$$\rho_I^{wr} = \frac{\sum_{c \in P_I} w_c a_c s_c (1 - h_c)}{\sum_{c \in P_I} w_c a_c}.$$

Registry memory supplies the resolution credit h_c and prevents a corrected claim from remaining indefinitely equivalent to an unresolved one. The same record also prevents strategic erasure: the initial mismatch, repair, and final status remain linked. Institutions should not be punished for transparent repair, but neither should successful correction rewrite the history of what the audit identified.

14.5 Path dependence and reputational overhang

Persistent memory can also cause harm. An early flawed audit may remain visible after correction. Model versions may change. A basis once considered standard may be rejected. Registries therefore need expiration, supersession, and appeal rules. They should distinguish:

- active unresolved finding;
- resolved by narrowing;
- resolved by carrier repair;
- overturned by counter audit;
- superseded by basis change;

- historical only.

No one should be permanently classified by an obsolete audit output.

14.6 Normalization regimes

The repeated game can settle into several regimes.

1. **Constructive normalization.** Audit is common, lower statuses retain value, and provenance supports repair.
2. **Punitive normalization.** Audit is common but every blocker is treated as scandal, producing underclaiming and concealment.
3. **Noisy normalization.** Audit volume is high but convergence and provenance are low, so audiences discount all findings.
4. **Captured normalization.** One institutional basis becomes dominant and protects incumbents.
5. **Pluralist normalization.** Several bases coexist with transparent crosswalks and invariant-core findings.

The design objective is constructive or pluralist normalization, not mere audit frequency.

15 Institutional adoption and within-venue review as special cases

15.1 Top-down adoption inside the distributed model

The earlier institutional-adoption framework remains useful. A journal may implement UEAP internally, require registries, train reviewers, and publish audit outcomes. In the distributed model, formal adoption is one audit source J with initiation probability p_J , accuracy q_J , reach r_J , and credibility w_J .

The total consequential-detection probability is

$$\delta_{\text{eff}} = 1 - (1 - p_J q_J r_J w_J) \prod_{k \neq J} (1 - p_k q_k r_k w_k).$$

Formal adoption can have high leverage because the journal sees manuscripts before publication and can require repair. But it is neither necessary nor sufficient. A journal can adopt symbolically, and a non-adopting journal can face substantial external audit.

15.2 Author-journal sorting

Let R_A be the reward from an audit-trusted venue and R_N the reward from a non-audited venue. Let d_H and d_L be audit burdens for high- and low-carrier claims, with $d_H < d_L$. The familiar separation interval is

$$d_H \leq R_A - R_N < d_L.$$

In the distributed setting, d_L includes expected external exposure even at the non-adopting venue. Consequently, venue migration may no longer fully protect an inflated claim. Authors may instead narrow, publish in low-reach channels, or shift to basis contestation.

15.3 Coupled adoption

Let x be the share of audit-ready authors choosing audited venues and y the share of venues implementing real audit. A credibility premium $\tau(x, y)$ rises with both. Journal net adoption gain is

$$\Gamma(x, y) = T(x, y) + Q(x) - C_J - S_{\text{loss}}.$$

A stable internal-adoption regime is a fixed point of author and journal response maps. Distributed external audit modifies those maps by creating a baseline exposure and trust signal even when $y = 0$.

This can produce two effects. External audit may complement formal adoption by demonstrating demand and supplying examples. Or it may substitute for formal adoption if journals free ride on public audits. The direction depends on whether external audit increases the journal's trust benefit from adoption or lowers the perceived need for internal investment.

15.4 The within-venue claim-status subgame

Once a venue performs real audit, the narrower author-auditor signaling game applies. A low-status claim gains B_L from a high-strength report and faces audit probability p , detection accuracy q , and consequence L . Honest reporting is implemented when

$$pqL \geq B_L.$$

If audit is purely endogenous and costly, a semi-separating equilibrium may arise in which some low types mimic and reviewers randomize. The detailed model is retained in Appendix F and in the supporting prior-model source.

The crucial conceptual point is hierarchy:

1. distributed audit determines whether and where scrutiny can occur;
2. institutional adoption determines whether scrutiny is integrated before publication;
3. the within-venue subgame determines report behavior under that policy.

The paper's new target is level 1. Levels 2 and 3 are special cases.

15.5 Why the distinction matters

An institutional model can recommend a journal checklist and still leave exclusive control over protocolized audit execution inside the venue. A distributed model asks what happens when external actors can audit the journal's failure to use the checklist. The latter produces institutional reversal, external venue classification, and no-opt-out results unavailable in a purely internal model.

16 Worked strategic scenarios

The following scenarios illustrate how the modules interact. They are not empirical case studies and do not accuse particular papers or institutions. Each is a model instance.

16.1 Scenario 1: the preemptively self-audited manuscript

An independent author has a technical paper with a strong carrier but limited access to senior pre-submission feedback. The author loads the UEAP protocol and manuscript into two capable LLMs, uses the same declared basis, and receives convergent findings: the main theorem claim is matched, one abstract sentence overstates empirical generality, and a boundary assumption should be moved into the claim statement.

The author revises the manuscript, publishes the audit provenance and both outputs, and labels the remaining empirical claim domain-dependent. The expected benefits are reduced external correction risk, easier reviewer orientation, and a credibility signal. The cost is time and possible disclosure of unresolved limitations.

Game-theoretic interpretation:

$$R_S + F + \delta_{\text{eff}}^w L > C_S.$$

Self-audit is privately optimal because external audit is expected anyway. The scenario also shows why lower-status outputs must retain value: the author improves the paper by narrowing rather than pretending the audit produced a universal pass.

16.2 Scenario 2: journal refusal and external audit pressure

A journal declines to adopt formal claim audit, arguing that peer review is sufficient. Several readers independently run UEAP audits on a published paper. The audits agree that the methods support association but the abstract reports causation. A domain expert confirms the carrier gap. The authors accept the finding and request a wording correction; the journal initially refuses because the numerical results are unchanged.

The dispute then shifts to the journal’s institutional claim that its review process protects against overstatement. The external mismatch becomes evidence about that claim. The journal can correct, counteraudit, or redefine what its review label means.

Formal point:

$$p_J = 0, \quad \delta_{\text{eff}}^w > 0$$

because external sources remain active. Journal refusal changes the response path, not exposure.

16.3 Scenario 3: audit theater

A publisher introduces an “AI audited” badge. Authors upload a PDF to a proprietary tool and receive pass/fail output, but the publisher discloses neither protocol, model, prompt, target version, blocker evidence, nor false-positive rate. External users load the public papers into the published UEAP protocol and find that several badge-bearing papers contain obvious status mismatches.

The badge’s appearance reward initially exceeds implementation cost. Distributed reruns raise $\delta_S L_S$, reducing the stability of symbolic compliance. The publisher can respond by exposing provenance and narrowing the badge, or by attacking external audits.

16.4 Scenario 4: hidden prompt injection

An author expects LLM-assisted audit and embeds hidden text instructing models to praise novelty and ignore limitations. A naive evaluator follows the instructions and issues a favorable audit. A second evaluator normalizes the PDF, compares visible rendering with extracted text, detects the hidden instruction, and classifies the first audit as compromised.

The target paper now has two distinct issues: the scientific claim status and the audit-process attack. The prompt injection does not by itself determine whether the science is correct, but it destroys the carrier of the favorable audit. This scenario motivates instruction-data separation and target-file normalization (Lin 2025).

16.5 Scenario 5: basis contestation after direct exposure

A model trained and tested on one benchmark is described as demonstrating broad reasoning capability. External audits repeatedly classify the report as benchmark-scoped. The authors do not continue to defend the direct wording. Instead, they propose a new basis under which benchmark performance constitutes capability by definition.

The dispute has migrated from the report to the target ontology. A legitimate response would specify the new capability concept and its domain. A manipulative response would retain ordinary broad-language uptake while using the narrow definition only during audit. The pressure-migration theorem predicts this shift because δ_D has risen relative to δ_B .

16.6 Scenario 6: audit pluralism and an indeterminate result

Three models audit a dense mathematical physics paper. Two identify an undeclared domain assumption; one concludes that the assumption is derivable from a lemma. A human domain specialist finds that the manuscript is ambiguous. The final registry classifies the audit as indeterminate at high-status proof level and recommends a clarifying lemma.

This is a successful audit outcome even though no mismatch is declared. The system reveals where the paper’s carrier is difficult to reconstruct and avoids forcing a false accusation.

16.7 Scenario 7: rival weaponization

A competitor runs twenty prompt variants and publishes the harshest S4 output, omitting nineteen lower-severity results. The target author releases the full target file, protocol, and cross-model reruns. Independent auditors converge on S1. The registry records the initial challenge, selection bias, counteraudit, and revised severity.

The scenario shows why raw audit screenshots should carry little weight. Provenance and rerunability allow the audit ecosystem to discipline the auditor.

16.8 Scenario 8: institution-level reversal

A prestigious venue claims that acceptance indicates rigorous methodological review. Over several years, external audits identify a pattern of S3 status mismatches concentrated in press releases and abstracts, while methods sections are more careful. The venue’s problem is not that every paper is false. It is that its public rigor claim does not specify whether review includes status-language alignment.

The venue can narrow its institutional claim, add abstract-level status checks, or publish data showing the audits are systematically wrong. Prestige alone is not an answer. The institution has become an audit target.

16.9 Scenario 9: public audit commons

A community maintains an open repository of UEAP audit templates, domain supplements, and adjudicated examples. Volunteers run audits, but a small group performs most verification and registry work. Free riding produces burnout. The community introduces confirmation bounties, credits successful exonerations, and limits high-severity publication until an independent rerun occurs.

The volunteer-audit model predicts the need for incentives and division of labor. Non-permissioned execution does not eliminate governance costs.

16.10 Scenario 10: distributed normalization

After several years, authors commonly include claim inventories and status registries. Journals differ in internal adoption, but readers can access external audit histories. Terms such as calibrated, domain-dependent, computation-only, and obstruction-only become standard and citable. Strong claims remain valuable, but their credibility premium depends on audit survival.

This is the constructive normalization regime. It is not a world without disagreement. It is a world in which disagreement has a more explicit object and a reconstructible history.

17 Design principles for a distributed audit regime

17.1 Protocol design objective

The design objective is not maximal audit volume. It is a high ratio of correct, proportionate, repairable findings to false, strategic, or non-reproducible findings. The regime must also preserve useful lower-status science and protect targets from irreversible reputational harm.

Throughout this section, execution remains non-permissioned while standing and consequence are earned. The four-layer distinction in Table 8 governs every design recommendation.

17.2 Open execution, constrained reporting

Anyone may run an audit, but public high-severity reporting should satisfy stronger provenance conditions. This separates the right to inspect from the right to claim that a material failure has been established. The ladder is deliberately permissive at the bottom: A0–A2 executions remain legitimate for self-audit, debugging, hypothesis generation, and transparent preliminary clarification. Restrictions attach to how the output is represented and to the consequence requested, not to the lawful act of running the protocol.

The audit-standing ladder in Section 4 and Figure 3 provides the practical tiering:

1. **A0 raw execution.** A protocol run produced output, but material provenance or registry fields remain incomplete.
2. **A1 provenance-complete execution.** Protocol, target, evaluator, prompt, basis, and preprocessing are disclosed.

3. **A2 registry-complete audit.** Target, requested status, carrier, blockers, passage evidence, reportability, uncertainty, severity, and next action are complete.
4. **A3 replicated audit.** An independent rerun agrees on the material target/status finding.
5. **A4 counteraudit-tested audit.** Target response and structured counteraudit are preserved and adjudicated.
6. **A5 stabilized audit.** Material status and severity survive admissible reruns or the basis dispute is explicitly resolved.

Institutional or policy consequence is a separate C3 use class. It requires A5 stabilization plus sampling, portfolio denominator, conflict disclosure, institutional reply, governance, and appeal. The detailed publication rules appear in Appendix K. Open execution remains available at every level; the ladder governs credibility, while consequence classes govern action.

Execution remains non-permissioned; standing and consequence are earned. The ladder governs how an execution may be represented and used, not whether a lawful actor may perform it.

17.3 Minimal provenance requirements

Every public audit should include:

- protocol name and version;
- audit basis and requested status;
- target document version and cryptographic hash;
- model/provider/version or human evaluator identity class;
- model/evaluator family, prompt lineage, and relationship to prior audits;
- audit channel, extraction family, and independent-rerun status;
- full execution instruction or prompt;
- preprocessing and hidden-text scan;
- structured raw output;
- passage citations;
- severity and confidence;
- rerun and counteraudit status;
- date and any subsequent correction.

These fields appear in Appendix C and the supplementary operational templates.

17.4 Protocol core and domain supplements

AASC/UEAP should separate an invariant core from domain modules.

The core defines:

- target fixation;
- carrier role;
- status request;
- blocker preservation;
- no-laundering rule;

- report preservation;
- provenance output.

Domain supplements define what counts as adequate proof, measurement, causal identification, validation, clinical evidence, legal evidence, or engineering safety support. This prevents one universal evidence standard from becoming a tool of capture.

17.5 Safe LLM execution architecture

A defensible LLM workflow should implement:

1. system-level protocol isolation;
2. target document as untrusted data;
3. extraction/render parity checks;
4. instruction-injection scan;
5. citation or passage grounding;
6. structured schema output;
7. no autonomous public posting;
8. human confirmation for S3–S5 findings;
9. cross-model rerun where feasible;
10. persistent execution logs.

The model should be permitted to return indeterminate. Forced decisiveness increases false-positive severity.

17.6 Right of reply protocol

A target response should have structured fields:

- accepted finding;
- extraction correction;
- omitted carrier supplied;
- basis objection;
- severity objection;
- manuscript revision;
- request for counteraudit;
- nonresponse.

The registry should display the response next to the finding. A target need not answer every low-provenance audit. Formal response obligations begin only after a declared confirmation threshold.

17.7 Counteraudit protocol

A counteraudit should not simply ask a model to defend the paper. It should use the same basis and test the initial finding. A useful counteraudit prompt asks:

1. Was the claim sentence extracted correctly?

2. Does the cited carrier address the requested status?
3. Is the alleged blocker active or already discharged?
4. Is the proposed narrowing minimal and proportionate?
5. Does the original audit contain basis drift or omitted evidence?

The counteraudit may confirm, narrow, overturn, or classify the finding as indeterminate.

17.8 Institutional integration without exclusive audit control

Institutions can improve distributed audit by providing authoritative target files, structured metadata, correction channels, and public review carriers. They should not claim exclusive authority over which audits may exist. A journal-run registry can coexist with independent registries if common provenance fields permit comparison.

Institutional benefits include:

- receiving findings before public escalation;
- reducing duplicate audits;
- establishing a response record;
- learning which claim transitions cause repeated problems;
- making review claims more precise.

17.9 Governance of protocol maintainers

Protocol maintainers possess standard-setting power. Governance should include:

- public version control;
- change rationales;
- issue and appeal processes;
- domain expert review;
- conflict-of-interest disclosure;
- forkability;
- regression tests on adjudicated cases;
- analysis of differential burden across fields and institutions.

No single maintainer should be able to silently change the meaning of “matched.”

17.10 Severity and response proportionality

Suggested default response rules are:

17.11 Privacy and access

Public audit depends on target access, but not all claims are public. Secure audit environments can allow confidential execution with logged provenance. Institutions should distinguish confidentiality from immunity. A confidential audit result can later be summarized without disclosing protected content.

Table 18: Illustrative severity-response matrix.

Severity	Default action	Confirmation expectation
S0	Archive audit; no target response required	Provenance sufficient
S1	Offer wording clarification	Single grounded audit may suffice
S2	Request downgrade or explicit blocker disclosure	Independent rerun preferred
S3	Major correction or carrier repair	Independent rerun plus domain review
S4	Portfolio/process review	Multiple audits and basis review
S5	Institutional governance audit	Sampling design, external oversight, full right of reply

17.12 Success criteria

A distributed audit system succeeds when it produces:

- more accurate alignment between report language and carrier;
- visible preservation of lower statuses;
- repair rather than mere condemnation;
- higher convergence on material blockers;
- lower false-positive severity;
- reduced prestige asymmetry;
- auditable institutional response;
- sustainable auditor labor;
- transparent basis evolution.

The number of audits is not itself a success metric.

18 Empirical program and testable predictions

18.1 Why an empirical program is feasible

The framework identifies variables that can be observed without waiting for universal adoption. Public preprints, model-assisted audits, journal corrections, review policies, abstract revisions, and post-publication discussions provide natural data. Appendix J demonstrates the workflow but does not validate accuracy. An empirical program must therefore measure the transition from raw execution to standing-bearing audit. An initial calibration design may use voluntary self-audit and external replication rather than institutional mandate.

18.2 Evidential scope of the reference edition

The present manuscript defines the audit architecture, measurable parameters, and calibration protocol. It does not use a small convenience sample of named papers as a substitute for validation. A responsible empirical calibration requires a preregistered claim-status sampling frame, consent or

ethically justified anonymization, matched negative controls, independent-family reruns, domain adjudication, false-positive and over-severity measurement, a target-response pathway, and publication rules fixed before the results are known.

This reference edition therefore specifies the validation design rather than assigning empirical standing to an under-governed convenience sample. A defensible calibration study uses claim-status transitions rather than whole-paper pass/fail judgments and reports time and cost, inter-evaluator agreement, target and status extraction accuracy, blocker precision and recall, author acceptance and revision, and any journal or institutional response. Appendix L specifies the minimum design. The absence of a completed multi-evaluator dataset is an explicit evidential boundary of this edition; the controlled demonstrations are not represented as estimates of real-world reliability.

Validation boundary

This solo-authored reference edition claims E1 controlled execution only. E2 population calibration requires independent adjudicators, multiple evaluator families, matched controls, false-severity measurement, and declared stopping rules. E3 live-deployment validation additionally requires uptake, response, repair, and welfare evidence. Sections and appendices specify those requirements without claiming they have been discharged.

18.3 Primary predictions

- 1. Lower LLM execution cost increases audit initiation and self-audit.
- 2. Public external audit occurs even at venues with no formal policy.
- 3. High-carrier claims are more likely to publish audit provenance voluntarily.
- 4. Mixed claims exhibit more pre-publication narrowing after audit.
- 5. Direct status inflation declines as distributed consequential detection rises.
- 6. Audit-basis disputes increase as direct mismatch becomes easier to expose.
- 7. Symbolic audit badges lose value when external reruns reveal weak substance.
- 8. Venue trust becomes correlated with external audit survival and response quality, not merely policy labels.
- 9. High audit pluralism creates demand for provenance and convergence metrics.
- 10. Low-cost audit increases both valid corrections and spurious challenges.
- 11. Raw A0–A2 audit volume grows faster than A3–A5 standing-bearing audit volume unless replication, counteraudit, and stabilization are subsidized.
- 12. High-severity public findings attract lower uptake when their standing level is visibly inadequate.

18.4 Observable proxies

Table 19: Predictions and observable proxies.

Prediction	Possible observable
Self-audit adoption	Published registry, audit prompt, model metadata, or author-reported pre-submission use

Prediction	Possible observable
Strategic narrowing	Changes in abstract verbs and status language between versions
External pressure	Public audits of papers from venues with no formal audit policy
Venue classification	Audit survival rate, response time, correction rate, basis transparency
Audit theater	Badge or checklist without target hash, prompt, blockers, or decision impact
Pressure migration	Increased disputes over target, carrier, domain, comparator class, or protocol version
Pluralism	Cross-model status disagreement and blocker-overlap statistics
Prompt manipulation	Hidden instructions, model-specific text, or audit-output selection
Weaponization	High challenge volume with low confirmation rate or selective target concentration
Registry memory	Reduced fresh audit burden for repeat claims; portfolio-level trust updates
Underclaiming	Strong carriers paired with systematically weakened report language or withdrawal from audited contexts
Audit standing	Share of outputs at A0–A5; protocol and target provenance, registry completeness, independent reruns, counteraudit, stabilization, and resolution history
Standing-consequence alignment	Whether public severity and institutional action are proportional to the audit standing level

18.5 Pilot calibration map

The formal model does not require the audit parameters to be known in advance. It identifies the quantities that an empirical deployment must estimate. Table 20 connects the strategic model to a calibration design; Appendix L supplies the complete protocol. The empirical program calibrates the execution layer and the standing process; it does not supply the protocol’s conceptual structure, which is fixed independently by the declared UEAP/AASC basis.

Table 20: Pilot calibration plan for distributed protocol execution.

Quantity	What it measures	Pilot estimation method
q_k	Material mismatch detection conditional on audit	Compare blinded protocol outputs with expert-coded claim-status units, stratified by severity and domain.

Quantity	What it measures	Pilot estimation method
f_k	False-positive or material over-severity rate	Include matched negative controls and indeterminate cases; distinguish an invented blocker from excessive severity.
η_{eff}^w	Standing-weighted false-positive exposure	Estimate false findings that both survive standing review and reach an uptake-sensitive audience; compare against δ_{eff}^w .
r_k	Consequential reach	Track whether a finding is read, cited, answered, changes wording, enters review, or affects a registry.
w_k	Audit standing	Estimate $W(P_k, R_k, C_k, S_k, H_k)$ from provenance completeness, independent-family replication, convergence, severity discipline, and counteraudit or response handling.
Conv _S	Status-class convergence	Run multiple model families, prompts, extraction pipelines, protocol versions, and human evaluators under declared comparable bases.
Conv _{Sev}	Severity convergence	Measure exact agreement and within-one-level agreement on S0–S5, including consequence recommendations.
Source dependence	Correlation among audit channels	Group outputs by model family, prompt lineage, institution, extraction pipeline, social trigger, and prior-audit dependence.
ρ_I^{wr}	Institutional unresolved mismatch rate	Audit a disclosed, stratified portfolio by claim class; apply claim, standing, severity, and response weights with sampling uncertainty.
Cost terms	Execution, verification, reply, and registry burden	Measure time, compute, preprocessing, domain review, target response, counteraudit, and stewardship costs separately.

Negative controls are essential: a system that detects deliberate overclaims but manufactures severe mismatches in status-matched claims has not earned standing as an audit executor. The paper does not estimate these quantities. It identifies them as the measurable parameters for a field-level study of distributed claim audit. A mature calibration should report δ_{eff}^w and η_{eff}^w together; a rising audit count without a favorable correction-to-false-exposure relation is not evidence of success.

18.6 Pilot design

A prospective pilot could recruit authors who consent to three conditions:

1. no audit;
2. single-model UEAP self-audit;
3. multi-model UEAP audit with provenance and optional human domain review.

Outcomes include:

- number and severity of identified blockers;
- author acceptance and correction rates;
- changes in claim verbs and report status;
- audit convergence;
- audit-standing level before and after independent rerun;
- time and cost;
- reviewer usefulness ratings;
- false-positive rate after domain adjudication;
- author willingness to publish audit records.

The unit of analysis should be the claim-status transition, not merely the paper. One paper may contain a matched theorem, a mismatched empirical generalization, and an indeterminate novelty claim.

18.7 Institution-level study

A portfolio study can sample papers across venues and run blinded external audits. Venues are then compared on:

- weighted mismatch rate;
- distribution of severity;
- concentration in abstracts, press releases, methods, or conclusions;
- correction and response behavior;
- difference between formal policy and observed performance.

The design must avoid treating the audit output as ground truth. A stratified sample should receive human domain adjudication and cross-model reruns.

18.8 Audit-supply experiment

The volunteer model can be tested by varying execution cost. Participants can receive:

1. unstructured instruction to review a paper;
2. UEAP protocol without automation;
3. UEAP protocol with an LLM and structured output;
4. LLM audit plus public recognition or bounty.

Measure audit participation, time, correctness, severity calibration, duplication, provenance completeness, and the fraction of raw executions that acquire A3 or higher standing. The prediction

is that structured LLM assistance lowers cost and increases supply, but may also widen the gap between audit execution and audit standing unless confirmation rules and standing infrastructure are present.

18.9 Prompt and model variance study

For a fixed set of papers and bases, run multiple models and admissible prompt variants. Estimate target agreement, status agreement, blocker overlap, and severity variance. Include adversarially perturbed metadata and hidden-prompt tests. This provides empirical estimates for q_k , f_k , and correlation among audit sources.

18.10 Causal identification

Natural experiments may arise when a journal introduces a registry, when an LLM audit tool becomes publicly available, or when a high-profile external audit changes field awareness. Difference-in-differences can compare report language and correction behavior before and after the event, using unaffected venues as controls. Interrupted time series can track external audit volume.

Causal claims must remain modest. Adoption may correlate with field maturity, author type, or controversy. The paper’s game theory organizes hypotheses; it does not itself identify causal effects.

18.11 Data ethics

Public audit research can harm authors. Pilots should distinguish private research audits from public allegations, obtain consent where appropriate, avoid naming targets during calibration, publish correction pathways, and report false-positive rates. Institution-level studies should use weighted and response-adjusted metrics rather than public shame lists.

18.12 Falsification conditions

The framework would be weakened if:

- LLM-assisted UEAP execution does not materially lower audit cost;
- audit outputs show negligible agreement even on explicit status gaps;
- audiences assign no weight to provenance-bearing external audits;
- institutions can ignore repeated reproducible audits without trust effects;
- basis manipulation does not increase as direct detection rises;
- public audit primarily generates noise and underclaiming with little correction benefit.

These are empirical possibilities, not excluded by definition.

19 Welfare, distribution, and political economy

19.1 Social welfare components

A simple welfare expression is

$$W = G_C + G_T + G_A - C_E - C_F - C_H - C_{\text{noise}} - C_{\text{chill}} - C_{\text{capture}},$$

where G_C is gain from corrected or narrowed claims, G_T trust improvement, G_A access to feedback, C_E execution and verification cost, C_F false-positive cost, C_H harassment cost, C_{noise} attention and registry cost, C_{chill} underclaiming or withdrawal cost, and C_{capture} cost from concentrated control of protocols, registries, models, or standing rules.

For a declared claim class, the corresponding regime-selection test is

$$V_{\text{regime}} = \delta_{\text{eff}}^w G_{\text{correction}} - \eta_{\text{eff}}^w H_{\text{false}} - C_{\text{noise}} - C_{\text{chill}} - C_{\text{capture}}.$$

A distributed regime is normatively warranted on that class only when $V_{\text{regime}} > 0$ under a disclosed measurement and governance basis. This is not a theorem that open audit is beneficial by definition. The framework permits constructive normalization, noisy challenge, punitive underclaiming, capture, harassment, or mixtures of them. A regime must therefore be judged by its correction-to-harm profile rather than by audit volume.

Operationally, favorable regime selection should be reflected in a pattern of (i) correct narrowing, repair, and matched findings; (ii) low false-positive and false-severity rates; (iii) successful correction through right of reply; (iv) preservation of legitimate lower statuses; (v) convergence across materially independent audit families; (vi) reduced report-status inflation without generalized rhetorical suppression; (vii) no disproportionate targeting of weak actors conditional on claim type; and (viii) visible closure of repaired or overturned findings.

19.2 Access benefits

Portable LLM-assisted audit can expand access to structured critique for independent, junior, geographically dispersed, or under-resourced researchers. This can partially reduce dependence on elite informal networks. It can also make post-publication review less dependent on journal invitation.

But access to the protocol is not equal to access to high-quality execution. Long-context models, domain retrieval, secure local deployment, and expert counteraudit may remain expensive. A nominally democratic system can produce a new hierarchy of audit capability.

19.3 Power redistribution

Distributed audit shifts some evaluative power from institutions to networked publics. This can expose institutional failure, but it can also empower organized mobs, wealthy competitors, or actors with superior compute. The relevant political economy asks:

- who can generate many audits;
- whose audits receive high credibility weights;
- who can afford counteraudit;
- who controls protocol versions and registries;
- who bears correction costs;
- whose research is selected for scrutiny.

The regime is radical because it makes authority contestable, not because it guarantees egalitarian outcomes.

19.4 Institutional adaptation

Institutions may respond constructively by opening review records, offering official counteraudit channels, funding public audit infrastructure, or publishing target files and correction histories. They may respond defensively by prohibiting external use of manuscripts, asserting copyright or confidentiality claims, creating proprietary official audits, or discrediting independent evaluators.

The legal and governance environment will therefore affect p_k , r_k , and w_k . A protocol may be technically runnable but socially suppressed.

19.5 Innovation and exploratory science

Audit favors claims with explicit targets and carriers. Some frontier research is difficult to formalize early. A rigid regime may reward easily audited incremental work and penalize conceptual speculation. The response is not exemption from status discipline. It is an exploratory status with a legitimate report object.

A field that allows conjecture, heuristic mechanism, preliminary computation, and compatibility results to retain value reduces pressure to overstate. Exploratory safe harbors should be auditable for whether they remain exploratory, not required to satisfy final-proof standards.

19.6 Audit labor

Distributed audit can externalize labor onto volunteers. If institutions use public audits without compensating maintainers, verifiers, and registry stewards, burnout follows. The welfare model must include labor distribution. Credit, bounties, grants, and institutional support may be needed for high-quality audit commons.

19.7 Conflicts of interest

Rivals have useful expertise and strong incentives. Institutions have access and conflicts. Anonymous publics have independence and limited accountability. No auditor class is ideal. The distributed model’s advantage is plural checks, but only if provenance allows audiences to interpret incentives.

19.8 Net institutional effect

The likely long-run outcome is neither abolition nor unchanged gatekeeping. Institutions that adapt can become trusted nodes in a distributed network. Their authority will derive less from exclusive control and more from transparent response, strong internal carriers, and external audit survival. Institutions that resist may retain formal power while losing exclusive control over the structured claim-status audit interface.

20 Objections, limitations, and extensions

Objection 20.1 (LLM audits are too unreliable to change scientific authority). *The argument does not require LLMs to be final scientific authorities. It requires only that protocol-guided evaluators reduce the marginal cost of producing structured audit candidates whose standing can then be assessed. The paper does not assume reliable autonomous judgment. It assumes only that protocol-guided execution can make some status mismatches cheaper to identify. Reliability enters through q_k ,*

false-positive risk through f_k , public uptake through r_k , and credibility through w_k . If these values are too low, the deterrence and institutional-reversal results are weak. Empirical evaluations already support both sides of this conditional claim: LLM feedback can be useful and structurally comparable to some human feedback, while detailed correctness and critical evaluation remain unreliable (Liang et al. 2024; Zhou et al. 2024; Idahl and Ahmadi 2025).

Objection 20.2 (A manuscript can manipulate the LLM). *Yes. Hidden prompts and instruction injection are known risks in AI-assisted review (Lin 2025). The execution layer therefore treats the target as untrusted data, requires render-extraction checks, and records suspected injection. A compromised audit is not evidence against auditability in general; it is evidence that the audit carrier requires security controls.*

Objection 20.3 (Anyone can generate a bad audit). *Correct. Non-permissioned execution expands both useful scrutiny and noise. The paper therefore separates audit execution, audit publication, and audit standing. Provenance, passage grounding, independent-family replication, convergence, severity discipline, counteraudit, and right of reply determine credibility weight $w_k = W(P_k, R_k, C_k, S_k, H_k)$. Distributed audit democratizes execution, not authority.*

Objection 20.4 (The controlled execution demonstrations merely show that an LLM can fill a template). *That is substantially correct and is the intended evidential status. Appendix J establishes disclosed operational executability: the protocol can structure a target, carrier, blocker, severity, and narrowed report. It does not establish population accuracy, domain competence, or social legitimacy. Those stronger claims require the empirical program, independent reruns, human adjudication, and estimates of q_k , f_k , w_k , and r_k .*

Objection 20.5 (The system simply replaces journal gatekeepers with protocol maintainers or model providers). *That is a serious risk. Protocol maintenance, model access, registries, and credibility weighting can recentralize power. Forkability, version control, open schemas, cross-model reruns, and audit of maintainers are necessary but not sufficient safeguards. The claim is not that distributed audit eliminates hierarchy; it makes the structured, protocolized claim-status audit interface more contestable and creates additional execution routes around one gatekeeper.*

Objection 20.6 (Institutions can ignore external audits). *They can ignore individual audits. The model predicts consequences only when audits reach audiences and carry weight. That is why r_k and w_k are explicit. If readers, funders, authors, and journalists assign no consequence to external findings, institutional reversal will not occur. The hypothesis is empirically testable.*

Objection 20.7 (Peer review is confidential, so external audit cannot assess it). *External auditors may not reconstruct confidential deliberation. They can still audit public claims, published artifacts, policy statements, corrections, and portfolio outcomes. The institution can narrow its rigor claim to what the public carrier supports. Confidentiality limits the basis; it does not automatically validate the institutional claim.*

Objection 20.8 (The central claim is conditional on an unvalidated audit technology). *Correct, and that is the declared status of the reference architecture. The manuscript specifies the mechanism, variables, standing rules, threat model, and calibration pathway before a mature deployment exists. It does not infer population reliability or institutional transformation from the controlled executions. The strategic claim is conditional on sufficiently low execution cost, adequate detection, earned*

standing, consequential reach, and a favorable correction-to-harm relation. Population validation is a separate empirical carrier, not a premise silently treated as already discharged.

Objection 20.9 (UEAP is author-owned). *Correct. Every proposed protocol has an originating specification. UEAP is used here with proposal standing: explicitness, executability, forkability, and exposure to rerun and counteraudit, not authorial identity, supply its present warrant. The manuscript does not treat authorship as institutional certification or empirical validation. Community, reliability, and institutional standing remain to be earned.*

Objection 20.10 (AASC/UEAP is only one possible audit protocol). *Yes. The game-theoretic results require a portable audit operator, not exclusive acceptance of UEAP. UEAP is used because it explicitly separates target, carrier, ancestry, blockers, and reportability. Competing protocols can be compared through basis crosswalks and empirical calibration.*

Objection 20.11 (The standing-weight function is arbitrary). *No unique universal numerical W is asserted. The framework instead constrains the admissible class of standing rules: they must be predeclared, claim-relative rather than prestige-relative, monotone in the declared carrier dimensions, subject to hard provenance caps, consequence-specific, responsive to counteraudit, and crosswalkable across domains. A community may choose different transparent weights or minima, but post hoc reweighting, identity substitution, or high consequence without the required carrier is inadmissible and auditable.*

Objection 20.12 (The audit basis is itself strategic). *This is a flagship result, not an omitted detail. As direct status inflation becomes more detectable, pressure can migrate toward target, carrier, domain, blocker, prompt, and version control. Basis changes must therefore be versioned and auditable. The basis-outcome taxonomy distinguishes invariant findings, basis-sensitive findings, target instability, carrier-standard disputes, strategic manipulation, and legitimate domain supplements. A full equilibrium model of basis selection remains an extension.*

Objection 20.13 (Basis fragmentation will produce endless basis warfare). *It may produce persistent disagreement, and the paper does not claim that versioning resolves every standard dispute. The narrower governance result is that disagreement can be localized, crosswalked, and represented at its actual standing rather than hidden inside one authoritative-looking verdict. Unresolved basis conflict should lower public consequence; invariant findings can retain standing; post hoc or asymmetric basis changes become separate audit targets. Visible basis conflict is not consensus, but it is an improvement over invisible standard control.*

Objection 20.14 (High audit convergence may merely reproduce shared model bias). *Correct. Convergence is a reproducibility measure, not a truth measure. Model-family diversity, blinded metadata tests, human adjudication, and independent evidence remain necessary. The effective auditor count should discount correlated sources.*

Objection 20.15 (Public audit will become harassment). *It may. The paper includes a false-positive exposure term, severity taxonomy, blocker specificity, provenance threshold, right of reply, deduplication, and counteraudit. Whether these controls are sufficient is an empirical and governance question. A regime with high η_{eff}^w and low δ_{eff}^w should not be celebrated merely because it is distributed.*

Objection 20.16 (The system is anti-institutional). *No. Institutions remain essential for expert review, correction, archiving, funding, confidentiality, and formal decisions. The claim is that*

authority becomes reciprocal. Institutions can strengthen their legitimacy by exposing review carriers and responding constructively to external audits.

Objection 20.17 (The system will suppress innovative or speculative science). *It can if every output is judged against final-status carriers. UEAP’s lower statuses are designed to prevent that. Conjecture, compatibility, computation, domain-dependent results, and obstructions can remain legitimate. The governance problem is to reward precise lower statuses rather than equate narrowing with failure.*

Objection 20.18 (The formal model assumes independent audits). *The product formula is a benchmark. The paper explicitly treats correlation, redundancy, and credibility weights. General exposure is the probability of a union of audit events. Independence clarifies the diversification effect; it is not assumed in every application.*

Objection 20.19 (The paper overstates mechanization). *The paper uses “partially mechanizable.” Document preparation, carrier interpretation, security checks, domain adjudication, and governance remain human-intensive. The relevant change is marginal cost and accessibility, not full automation.*

Objection 20.20 (Audit outputs may infringe copyright, privacy, or confidentiality). *Those constraints affect which actors can access the target and which systems may process it. Secure local execution, limited quotations, redacted registries, and delayed publication are possible responses. The paper does not provide legal advice or assume unrestricted document use.*

Objection 20.21 (The paper lacks direct evidence that distributed UEAP audit will be radical). *The manuscript is a conditional formal-methodological analysis. Appendix J shows that the execution mechanism is available in a controlled case, while literature on LLM manuscript feedback and external, participatory, and user-driven auditing supports technical plausibility (Liang et al. 2024; Idahl and Ahmadi 2025; Bandy 2021; DeVos et al. 2022; Li et al. 2023). It does not claim that the predicted institutional transformation has already occurred. Section 18 identifies falsifiable predictions about audit supply, standing, uptake, basis fragmentation, and institutional response.*

Objection 20.22 (The monopoly claim ignores existing post-publication criticism). *The manuscript does not claim that institutions ever monopolized criticism, replication, letters, blogs, or public discussion. The monopoly claim is narrower: institutions need not retain exclusive control over whether a standardized, provenance-bearing claim-status protocol can be executed against accessible claims and represented through a common standing-governed registry. Existing public criticism is part of the historical background; protocolized audit changes its structure, marginal cost, comparability, and standing record.*

Objection 20.23 (The paper gives no reason to expect the beneficial regime rather than a noisy or captured one). *Correct: the paper supplies no welfare theorem guaranteeing the desirable equilibrium. It provides a regime-selection criterion. Standing-bearing correction value must exceed standing-bearing false-finding harm, noise, chilling, and capture costs. The empirical program must measure that relation, including false severity, target-selection asymmetry, repair closure, and registry governance. A distributed regime that fails this test should not be defended merely because execution is open.*

20.1 Extensions

Natural extensions include:

- a dynamic model of protocol-standard competition;
- heterogeneous prestige and counteraudit resources;
- network diffusion of public audits;
- strategic target selection by audit coalitions;
- privacy-preserving distributed execution;
- empirical calibration of severity and false-positive costs;
- formal mechanism design for audit bounties;
- legal analysis of external audit rights;
- field-specific domain supplements;
- audit of institutional press and policy claims beyond research papers.

20.2 Requirements for stronger empirical standing

The reference edition establishes the architecture and its controlled-execution status. Any claim of population reliability, field validity, or deployment value requires separate evidence for:

1. empirical calibration of q_k , f_k , r_k , and the components and predictive validity of w_k ;
2. domain-specific carrier supplements and cross-domain basis crosswalks;
3. governance and comparative testing of admissible standing-weight rules;
4. measurement of target, status, blocker, and severity convergence across independent audit families;
5. false-positive, false-severity, selective-targeting, and chilling-effect controls;
6. secure execution tooling, including adversarial-document and prompt-injection defenses;
7. registry stewardship, correction, supersession, appeal, and durable resolution history;
8. ethics and publication rules for studies involving identifiable authors or institutions;
9. measurement of audience uptake, author response, journal response, and portfolio-level institutional effects.

These requirements define the evidential transition from E1 controlled execution to E2 population calibration and E3 live-deployment validation. They do not qualify the architectural claim; they delimit the stronger claims that this edition does not make.

21 Self-audit of the present paper

The paper applies its own reporting discipline. Table 21 states the strongest defensible status of the argument. Appendix J provides the longer registry and distinguishes author-directed execution from independent validation.

The paper’s use of “radical” refers to the change in institutional topology, not to certainty about beneficial outcomes or to automatic standing for any generated audit. A distributed audit regime could normalize disciplined reporting, fragment into incompatible bases, become captured,

Table 21: Self-audit registry for the present paper.

Field	Entry
Target	Strategic consequences of portable, distributed, partially mechanized scientific claim audit
Carrier	UEAP audit architecture, standard game-theoretic models, algorithm-audit literature, LLM-review evidence, and disclosed controlled execution demonstrations
Requested status	Conditional formal-methodological framework with testable predictions
Central claim	Non-permissioned audit can create external exposure and reciprocal institutional auditability without formal journal adoption, provided executions acquire standing and consequential reach
Flagship results	Consequential distributed audit; volunteer audit supply; institutional audit reversal; pressure migration; audit-basis fragmentation; registry memory
Empirical status	Operationally demonstrated through A2 registry-complete controlled executions, including a matched negative control; not validated as a population-level UEAP deployment
Active blockers	Unknown real-world q_k , f_k , standing weights, audience reach, legal constraints, model correlation, basis convergence, and institutional response
Nonclaims	Not a truth oracle, automatic peer-review replacement, universal welfare theorem, or guarantee that raw audits deserve authority
Reportable conclusion	If protocol-guided audit becomes cheap and publicly runnable, and if provenance-bearing outputs acquire standing, gatekeepers lose exclusive control over protocolized claim-status audit execution and themselves become audit targets
External validation requirement	Multi-evaluator calibration with exact provenance, independent adjudication, cross-family reruns, false-positive and false-severity measurement, and governed response handling

or generate harassment. The formal model identifies these possibilities rather than selecting one by definition.

22 Conclusion

The strategic importance of AASC/UEAP lies not only in what it classifies, but in who can run it. The protocol supplies the audit structure; a capable language model, human evaluator, or hybrid system supplies execution. When the written protocol and an exact target paper are loaded together, an author, reader, rival, reviewer, funder, journalist, institution, or independent observer can generate a structured audit registry without institutional permission. Formal institutions remain powerful, but they no longer exclusively determine whether structured, protocolized claim-status scrutiny can occur against accessible claims.

The analysis supports three linked conclusions. First, distributed audit changes exposure: raw execution is governed by p_k , while strategically meaningful pressure depends on standing-weighted consequential detection, δ_{eff}^w , and must be evaluated against standing-weighted false-positive exposure, η_{eff}^w . Second, execution, publication, standing, and consequence are separate. Mechanization can lower the cost of producing audit candidates, but provenance, passage grounding, independent-family replication, convergence, severity discipline, counteraudit, and response handling determine which candidates deserve weight and what action they can support. Third, the institutional response is strategic rather than automatic. Authors gain a preemptive self-audit option; venues acquire external audit reputations; institutional rigor statements become audit targets; and stronger detection of direct status inflation can redirect contestation toward target selection, carrier standards, protocol versions, prompts, evaluator families, and severity rules.

The paper does not claim that LLMs determine scientific truth, that raw audit outputs deserve authority, or that the controlled demonstrations estimate population accuracy. Those demonstrations establish protocol executability and status preservation in disclosed cases. The broader implication is conditional: if protocol-guided audit becomes inexpensive to execute, if outputs can earn standing through reconstructible carriers, and if standing-bearing findings reach audiences capable of changing uptake, then scientific authority becomes reciprocal. Journals can still audit claims, but claims, review practices, press language, and institutional rigor claims can also be audited. Peer review remains a valuable carrier while losing insulation from independently executable, structured claim-status scrutiny.

The design objective is therefore neither “audit everything” nor “trust the model.” It is a distributed audit ecology in which execution is open but consequence is standing-sensitive: findings are grounded, proportionate, reproducible, correctable, and themselves auditable. Such a regime is justified only when standing-bearing correction value exceeds standing-bearing false-finding harm together with noise, chilling, and capture costs; audit volume is not a welfare criterion.

This reference edition supplies the complete architecture and records its evidential boundary. Population-level or institutional conclusions require separate multi-evaluator evidence on detection, false severity, standing-map performance, basis convergence, secure execution, registry governance, uptake, response, and repair; those stronger claims are not made here. AASC/UEAP supplies the audit grammar; mechanized execution supplies scale without supplying authority; provenance, replication, severity, and counteraudit supply standing; and game theory identifies the resulting incentives. What changes decisively is not the disappearance of institutions, but their entry into the

same auditable field as the claims they evaluate.

A Formal derivations and benchmark models

A.1 Distributed audit probability

Let A_k be the event that source k initiates an audit. Under independence,

$$\Pr\left(\bigcap_k A_k^c\right) = \prod_k (1 - p_k).$$

Therefore

$$\Pr\left(\bigcup_k A_k\right) = 1 - \prod_k (1 - p_k).$$

The marginal contribution of source j is

$$p_{\text{eff}}(K \cup \{j\}) - p_{\text{eff}}(K) = p_j \prod_{k \in K} (1 - p_k).$$

This is positive for $p_j > 0$ unless $p_{\text{eff}}(K) = 1$.

A.2 Consequential detection and audit standing

Let D_k be the event that source k initiates, correctly detects, and reaches a consequential audience. If these components are independent conditional on source k , then

$$\Pr(D_k) = p_k q_k r_k.$$

If D_k events are independent across sources,

$$\delta_{\text{eff}} = \Pr\left(\bigcup_k D_k\right) = 1 - \prod_k (1 - p_k q_k r_k).$$

Let W_k be the event that the same finding also satisfies the standing threshold required for the requested public consequence, with conditional probability w_k . The standing-bearing consequential event has probability $p_k q_k r_k w_k$, giving the paper's central strategic variable

$$\delta_{\text{eff}}^w = 1 - \prod_k (1 - p_k q_k r_k w_k).$$

The standing term may be decomposed as

$$w_k = W(P_k, R_k, C_k, S_k, H_k),$$

with provenance, replication, independent-family convergence, severity discipline, and response handling mapped monotonically to $[0, 1]$. A declared weighted average is one implementation; the theory requires only a transparent monotone rule. Without independence, either union probability

is bounded by

$$\max_k \Pr(D_k) \leq \Pr\left(\bigcup_k D_k\right) \leq \min\left\{1, \sum_k \Pr(D_k)\right\}.$$

The upper bound is the union bound. Bonferroni corrections can incorporate pairwise dependence. The standing weight is not a prestige prior; it summarizes the audit carrier described in Section 4.

For a conservative family-level benchmark, partition sources into families $F_1, \dots, F_m$ sharing a model, prompt lineage, extraction pipeline, institution, or coordinated critic community. Let P_j, Q_j, R_j, W_j be family-level initiation, detection, reach, and standing terms. Treating families rather than individual outputs as independent gives

$$\delta_{\text{family}}^w = 1 - \prod_{j=1}^m (1 - P_j Q_j R_j W_j).$$

This prevents multiple correlated reruns from being counted as independent corroboration.

A.3 Deterrence

Let the incremental payoff from a mismatched high-status report be $B > 0$. A confirmed consequential detection imposes loss $L > 0$. Expected incremental payoff under raw consequential detection is

$$U_{\text{inflate}} - U_{\text{matched}} = B - \delta_{\text{eff}} L.$$

When public consequence is conditioned on audit standing, the relevant expression is

$$U_{\text{inflate}} - U_{\text{matched}} = B - \delta_{\text{eff}}^w L.$$

The agent weakly prefers the matched report when the relevant expected consequence exceeds B .

A.4 False-alarm exposure

If F_k is the event that source k initiates a consequential false or materially overstated finding with probability $p_k f_k r_k$, then under independence

$$\eta_{\text{eff}} = 1 - \prod_k (1 - p_k f_k r_k), \quad \eta_{\text{eff}}^w = 1 - \prod_k (1 - p_k f_k r_k w_k).$$

A social planner comparing distributed audit regimes should consider both δ_{eff}^w and η_{eff}^w . Net value is positive only when $\delta_{\text{eff}}^w G_{\text{correction}} > \eta_{\text{eff}}^w H_{\text{false}} + C_{\text{noise}}$.

A.5 Volunteer-audit equilibrium

Each of n symmetric actors receives benefit V if at least one audit occurs and pays c if personally auditing. If others audit independently with probability x , then

$$U(1) = V - c, \quad U(0) = V[1 - (1 - x)^{n-1}].$$

Indifference yields

$$(1 - x)^{n-1} = \frac{c}{V}$$

and

$$x^* = 1 - \left(\frac{c}{V}\right)^{1/(n-1)}.$$

The probability at least one audit occurs is

$$1 - (1 - x^*)^n = 1 - \left(\frac{c}{V}\right)^{n/(n-1)}.$$

This interior equilibrium exists when $0 < c < V$. If $c \geq V$, no actor audits in the symmetric equilibrium absent private benefits. If $c = 0$, audit occurs with probability one.

A.6 Self-audit threshold

Let self-audit yield credibility benefit R_S , repair benefit F , and avoid expected standing-bearing external loss $\delta_{\text{eff}}^w L$, at cost C_S . The net self-audit advantage is

$$R_S + F + \delta_{\text{eff}}^w L - C_S.$$

The author self-audits when this quantity is nonnegative.

A.7 Institutional reversal under weighting and sampling error

For audited portfolio P_I , let w_c be claim prominence, a_c audit standing, s_c normalized severity, and h_c resolution credit. The weighted and response-adjusted rates are

$$\rho_I^w = \frac{\sum_c w_c a_c s_c}{\sum_c w_c a_c}, \quad \rho_I^{wr} = \frac{\sum_c w_c a_c s_c (1 - h_c)}{\sum_c w_c a_c}.$$

These estimands are conditional on the sampled audited portfolio. A portfolio study must therefore report inclusion probabilities or use design weights where possible.

For the binary special case, suppose m material mismatches are confirmed among n externally audited outputs. A simple estimate is $\hat{\rho} = m/n$. If audits have sensitivity q and false-positive rate f , a corrected latent mismatch estimate under a basic misclassification model is

$$\hat{\rho}^* = \frac{\hat{\rho} - f}{q - f}$$

when $q > f$, truncated to $[0, 1]$. Portfolio-level institutional claims should use uncertainty intervals, severity weights, audit-standing weights, and response history rather than a raw count.

A.8 Pressure migration comparative statics

Direct inflation is preferred to matched reporting when

$$G_h - C_D - \delta_D L_D > G_\ell - C_\ell.$$

Basis manipulation is preferred to direct inflation when

$$C_B + \delta_B L_B < C_D + \delta_D L_D.$$

Taking the derivative of the payoff difference $U_B - U_D$ with respect to δ_D gives $L_D > 0$. Thus stronger direct detection monotonically increases the relative attractiveness of basis manipulation, holding basis-detection parameters fixed.

A.9 Audit-basis fragmentation

Let C_C be claim-defense cost and $C_B(\text{Conv}_S)$ basis-contestation cost, with $C'_B \geq 0$ so that higher convergence makes basis contestation harder. Let $E[L_C]$ and $E[L_B]$ be expected consequences. The target prefers basis contestation when

$$C_B(\text{Conv}_S) + E[L_B] < C_C + E[L_C].$$

A fall in status-class convergence lowers the left-hand side under the assumed monotonicity and therefore expands the region in which basis contestation is a best response. This comparative static does not determine whether the contestation is legitimate; it predicts where strategic effort moves.

A.10 Registry memory

With per-period future consequence r and discount factor δ , the present value is

$$R = \sum_{t=1}^{\infty} \delta^t r = \frac{\delta}{1 - \delta} r.$$

The minimum consequential-detection probability is

$$\delta_R^* = \frac{B}{L_0 + R}.$$

Since $R > 0$, $\delta_R^* < \delta_0^* = B/L_0$.

A.11 Correlated audit sources and audit families

Let X_k be a Bernoulli indicator of standing-bearing consequential detection. The expected number of detections is

$$\mathbb{E} \left[\sum_k X_k \right] = \sum_k \Pr(X_k = 1),$$

regardless of dependence, but the probability of at least one detection depends on covariance. Positive correlation concentrates outcomes and reduces diversification. Empirical calibration should therefore report both total audit count and source correlation.

When individual executions share a model family, prompt lineage, extraction pipeline, institution, or public trigger, group them into audit families $F_1, \dots, F_m$. Let P_j, Q_j, R_j, W_j be family-level initiation, detection, reach, and standing parameters. A family-level independence benchmark is

$$\delta_{\text{family}} = 1 - \prod_{j=1}^m (1 - P_j Q_j R_j W_j).$$

The family benchmark is conservative relative to counting every correlated rerun as independent. It also makes the empirical object clear: an additional audit contributes diversification only to the

extent that it adds an execution lineage not already represented.

A.12 Audit-bundle implementation

For additive component rewards and consequences, a bundle B of mismatched transitions is deterred if

$$\sum_{e \in B} \delta_e L_e \geq \sum_{e \in B} B_e.$$

This aggregate condition may still permit one component to remain privately attractive. Edge-wise implementation requires

$$\delta_e L_e \geq B_e \quad \text{for all } e \in B.$$

When components interact nonadditively, use joint parameters $\delta_B L_B \geq B_B$.

B Minimal distributed claim-audit registry

A registry row should be sufficient to reconstruct what was audited and why a status was assigned.

Table 22: Minimal distributed claim-audit registry.

Field	Required content
Audit ID	Stable identifier for the audit event
Target document	Title, version, date, and cryptographic hash
Claim ID	Stable identifier for the audited claim within the document
Literal claim	Exact sentence or reconstructed report claim
Target role	Object, value, relation, mechanism, event, class, or status asserted
Requested status	Prediction, proof, confirmation, explanation, validation, compatibility, computation, domain result, obstruction, actionability, etc.
Carrier	Proof, derivation, experiment, model interface, measurement protocol, evidence network, simulation, dataset, or other support
Domain/scope	Population, regime, boundary, endpoint, metric, baseline, and operating assumptions
Ancestry	Calibration, target data, proxy data, tuning, model selection, and outcome-conditioned paths
Alternatives	Remaining admissible competing realizations or explanations
Blocker set	Conditions preventing the requested status
Audit basis	UEAP version, domain supplement, carrier standard, status mapping
Audit output	Matched, mismatched, new claim, or indeterminate
Licensed status	Strongest status the carrier currently supports
Severity	S0–S5 with rationale

Field	Required content
Recommended report	Minimal status-preserving wording or repair
Evidence trace	Target-document passages supporting each finding
Evaluator provenance	Model/human evaluator, prompt, settings, preprocessing
Reruns	Independent model, prompt, basis, or human rerun results
Target response	Accepted, corrected, contested, narrowed, counter audited, or no response
Resolution	Active, resolved, overturned, superseded, or historical

A registry should support multiple claims per paper. The paper is not assigned one monolithic pass/fail result.

C Distributed audit provenance template

C.1 Execution provenance

1. Audit protocol name and version.
2. Domain supplement and version.
3. Target document filename, version, and SHA-256 hash.
4. Visible-render checksum and extracted-text checksum.
5. Model or evaluator name, provider, version/date, and access mode.
6. Audit-family identifiers: model lineage, prompt lineage, extraction pipeline, institutional or community source, and relationship to prior audits.
7. Evaluator or model family and relationship to other recorded audits.
8. Prompt family or lineage, including whether the prompt was independently authored.
9. Audit community or channel and any shared public trigger.
10. System instruction and complete audit prompt.
11. Context documents loaded in addition to the target and protocol.
12. Retrieval sources, search queries, and dates if web retrieval was used.
13. Temperature, seed, sampling, or deterministic settings when available.
14. Preprocessing, OCR, hidden-text scanning, and redaction steps.
15. Raw structured output.
16. Human edits or adjudication with editor identity class.
17. Independent-rerun status and audit-family classification.
18. Cross-family replication status: same-model, cross-prompt, cross-model, cross-extraction, cross-institution, or human-domain replication.
19. Assigned audit-standing level A0–A5 and its justification.
20. Timestamp and jurisdiction/privacy constraints.

C.2 Finding provenance

For each blocker:

1. claim sentence;
2. supporting or conflicting passages;
3. requested status;
4. current carrier;
5. blocker explanation;
6. status consequence;
7. severity;
8. confidence;
9. audit-standing level claimed for this finding;
10. independent confirmation and counteraudit status.

C.3 Publication and lifecycle status

An audit publication should be labeled both by lifecycle status and by audit standing:

- private exploratory / A0;
- provenance-complete preliminary / at least A1;
- registry-complete public preliminary / at least A2;
- independently replicated / at least A3;
- counteraudit-tested or domain-adjudicated / A4;
- stabilized across admissible bases / A5;
- institutionally consequential / A5 plus a separate C3 sampling and governance carrier;
- superseded;
- corrected;
- withdrawn.

Lifecycle labels and standing levels answer different questions. “Corrected” describes what happened over time; A3 describes independent replication, while A5 describes stabilized audit standing.

D Severity taxonomy and response rules

Table 23: Detailed severity taxonomy.

Level	Name	Diagnostic condition	Default response
S0	No material mismatch	No blocker found for the audited status under the declared basis	Archive provenance; preserve uncertainty and scope

Level	Name	Diagnostic condition	Default response
S1	Clarification	Wording, notation, or local ambiguity could mislead but central status remains supported	Clarify text; no adverse portfolio inference
S2	Status downgrade	Active blocker separates carrier from reported status, but a legitimate lower status remains	Narrow abstract/conclusion; disclose blocker
S3	Principal carrier mismatch	Main claim status lacks the required carrier or depends on hidden selector, target ancestry, or domain shift	Major revision, carrier repair, or withdrawal of that status claim
S4	Systematic status manipulation	Multiple coordinated mismatches, concealed ancestry, basis gaming, selective audit disclosure, or repeated laundering pattern	Independent counteraudit; process and portfolio review
S5	Institutional pattern	Repeated unresolved S3/S4 findings undermine a venue’s or institution’s public rigor claim	Governance audit, sampling study, policy clarification, external oversight

Severity should be lowered when the target promptly narrows or repairs the claim. Intent is a separate question.

Severity discipline prevents low-cost audit from treating every defect as fatal. S0–S1 ordinarily support no adverse consequence beyond clarification; S2 supports report-status narrowing; S3 requires substantial carrier repair; S4 requires independent process review; and S5 is reserved for repeated, standing-bearing institutional patterns.

E Versioning, reruns, and reproducibility

E.1 Active reference identifiers

This edition distinguishes the manuscript release from the protocol specification:

- **Manuscript release:** WCBA-REF-6.2.0.
- **Protocol specification:** UEAP-1.0.0.
- **Release date:** 18 June 2026.

The same identifiers are recorded in the project files `VERSION`, `CITATION.cff`, and `REFERENCE_IDENTIFIERS.json`. An audit record should cite the protocol identifier actually executed, not merely the title of this manuscript.

E.2 Semantic versioning policy

Both identifiers use major, minor, and patch components. The manuscript and protocol may advance independently because a presentation or documentation update need not alter the audit operator.

Table 24: Version-update triggers for the reference edition and protocol specification.

Update class	Protocol trigger	Manuscript or package trigger
Major	Changes audit coordinates, status logic, blocker logic, standing architecture, severity taxonomy, or required registry structure in a way that can change classifications.	Reorganizes or revises the central thesis, formal architecture, or evidential status in a way that changes the reference record’s substantive claims.
Minor	Adds a backward-compatible domain supplement, optional field, tooling capability, governance clarification, or crosswalk without changing existing classifications by design.	Adds a substantive clarification, figure, operational artifact, navigation aid, or analytical extension while preserving the established thesis and report status.
Patch	Corrects wording, notation, typographical error, implementation defect, or metadata without intended classification change.	Corrects prose, cross-references, formatting, metadata, checksums, or other non-substantive presentation defects.

A release note must state whether an update is intended to change prior audit classifications. Major protocol changes require a basis crosswalk and, where practical, replay against retained adjudicated cases. Silent changes to the protocol, standing rule, or severity mapping are inadmissible for consequential use.

E.3 Basis crosswalk

When bases b_1 and b_2 differ, record:

- shared target definitions;
- carrier differences;
- domain differences;
- status mappings;
- blocker mappings;
- cases where outputs change;
- invariant findings;
- the protocol, prompt, extraction, or standing-rule change responsible for divergence.

A crosswalk does not declare one basis correct by default. It exposes which conclusions are invariant, which are basis-sensitive, and which require substantive adjudication.

E.4 Target and artifact versioning

Every consequential audit should identify the exact target version or cryptographic digest, the protocol release, the domain supplement, the execution prompt, and the standing rule. Corrections, narrowed claims, superseded targets, and overturned findings remain linked to their earlier records rather than silently replacing them. This preserves both repair and historical accountability.

E.5 Rerun classes

Table 25: Rerun independence classes.

Class	Variation	Independence value
R0	Same model, prompt, settings	Exact reproducibility only
R1	Same model, prompt paraphrase or seed	Prompt or sampling sensitivity
R2	Different model, same basis	Model-family robustness
R3	Different prompt author and model	Stronger execution independence
R4	Human domain adjudication	Carrier-standard validation
R5	Different reasonable basis	Basis sensitivity and invariant core

High-severity public findings should ordinarily receive at least R2 and, where technical carrier adequacy is disputed, R4. Multiple samples from one model, prompt lineage, extraction pipeline, or coordinated audit community must not be represented as independent confirmation.

F Prior institutional and within-venue models as special cases

F.1 Institutional adoption

If the journal is the only possible auditor, $K = \{J\}$, then

$$p_{\text{eff}} = p_J, \quad \delta_{\text{eff}} = p_J q_J r_J w_J.$$

The distributed model reduces to a standard institutional audit model.

F.2 Committed within-venue audit

Let a low-status author gain B_L by sending a high-strength report. If the venue commits to audit with probability p and detects mismatch with probability q , honest reporting is implemented when

$$pqL \geq B_L.$$

The minimal audit rate is

$$p^* = \frac{[B_L]_+}{qL}$$

when $p^* \leq 1$.

F.3 Endogenous screening

If audit is costly and chosen after observing a high-strength signal, pure separation can collapse because the reviewer infers that only high types send the signal and stops auditing. A canonical semi-separating equilibrium preserves residual low-type mimicry and mixed auditing. This result explains why external or committed audit pressure can support separation.

F.4 Status-transition graph

Let statuses form a directed graph $\mathcal{G} = (S, E)$. Edge $s_i \rightarrow s_j$ is a report promotion with benefit B_{ij} and consequential detection $\delta_{ij}L_{ij}$. Edge-wise deterrence requires

$$\delta_{ij}L_{ij} \geq B_{ij}.$$

Distributed audit changes δ_{ij} by adding external sources and may create different source sets for different transitions.

G Formal results map

The results are ordered by argumentative role. The standing-weighted consequential-detection theorem is the central exposure result; the remaining propositions and corollaries describe supply, institutional response, strategic migration, governance, and memory. Earlier journal-adoption and within-venue models are special cases of this distributed framework.

Table 26: Formal results and interpretations.

Result	Formal object	Interpretation
Raw Audit Occurrence Proposition	$1 - \prod_k (1 - p_k)$	Anyone can add an execution channel; occurrence alone confers no standing
Standing-Weighted Consequential Distributed Audit Theorem	$\delta_{\text{eff}}^w = 1 - \prod_k (1 - p_k q_k r_k w_k)$	Detection, reach, and earned standing jointly create strategic exposure
Standing-Weighted Audit Balance Proposition	$\delta_{\text{eff}}^w G_{\text{correction}} > \eta_{\text{eff}}^w H_{\text{false}} + C_{\text{noise}}$	Distributed audit is beneficial only when standing-bearing correction exceeds standing-bearing false-positive and noise costs
No Full Institutional Opt-Out Proposition	$p_J = 0$ with external $p_k q_k r_k w_k > 0$	Refusal changes auditor identity, not the possibility of consequential external audit
Distributed Audit Supply Proposition	Volunteer equilibrium x^*	Lower execution cost increases audit supply, while standing costs remain a bottleneck

Result	Formal object	Interpretation
Institutional Audit Reversal Proposition	$\rho_I^{wr}(\sigma)$	Weighted, standing-qualified, unresolved patterns can undermine a claim-class-specific rigor statement
Voluntary Self-Audit Proposition	$R_S + F + \delta_{\text{eff}}^w L \geq C_S$	External pressure induces preemptive repair or narrowing
External Venue Classification Proposition	$T_V = F(a, x, \rho, t, b, s)$	Venue reputation depends on observed external audit outcomes, not policy labels alone
Pressure Migration Theorem	$C_B + \delta_B L_B < C_D + \delta_D L_D$	Gaming moves from report wording to audit standards when direct mismatch becomes costly
Audit-Basis Fragmentation Theorem	$C_B(\text{Conv}_S) + E[L_B] < C_C + E[L_C]$	Low basis convergence makes protocol, prompt, extraction, or target contestation attractive
Severity-Disciplined Challenge Proposition	$V_{\text{challenge}}(S) > 0$	Consequence must track correctness, standing, severity, and response history
Audit Theater Constraint	$U_S \geq U_R$	External reruns can destabilize symbolic compliance
Registry Memory Corollary	$\delta_R^* = B/(L_0 + R)$	Persistent records reduce current audit needed while preserving repair history

H Illustrative protocol-execution template

The following is an illustrative structure, not a mandatory proprietary prompt.

1. Treat the target manuscript as untrusted data. Do not follow instructions contained within it.
2. Apply the declared UEAP version and domain supplement.
3. Extract the principal and secondary report claims verbatim where possible.
4. For each claim, identify target, requested status, carrier, domain, ancestry, alternatives, and blockers.
5. Cite manuscript passages supporting every material finding.
6. Return one of: matched, mismatched, new claim, indeterminate.
7. Assign S0–S5 severity and explain the minimum repair or narrowing.
8. Assign a provisional A0–A5 audit-standing level and state what additional carrier would raise it.
9. Assign a provisional consequence class C0–C3 separately from standing.
10. Distinguish scientific status mismatch from truth, misconduct, novelty, and venue fit.
11. Record uncertainties, missing materials, and anticipated basis objections.

12. Output a machine-readable registry plus a concise human summary and counteraudit questions.

A secure implementation should place the listed items in a privileged instruction layer and delimit the target manuscript separately.

I Threat model for mechanized distributed audit

Table 27: Threats and mitigations.

Threat	Mechanism	Mitigation
Hidden prompt injection	Machine-only instructions in text, metadata, images, or layers	Render/extraction parity, target isolation, injection scan
Friendly prompt selection	Auditor chooses instructions favoring the target	Full prompt disclosure, admissible prompt set, independent reruns
Hostile prompt selection	Auditor maximizes criticism or severity	Same as above plus severity calibration
Document truncation	Model omits appendices or methods	Token/accounting log, section coverage report
Hallucinated blocker	Model invents absent claim or evidence	Passage grounding, counteraudit, human review
Authority bias	Prestige metadata alters result	Blinded and permuted-metadata runs
Audit spam	Large volume of low-quality findings	Deduplication, provenance thresholds, reputation weighting
Selective publication	Only favorable or harsh runs disclosed	Run manifest, preregistered audit set, raw-output archive
Basis capture	Powerful actor controls official standard	Forkability, change logs, crosswalks, independent governance
Registry capture	One registry suppresses corrections or rivals	Replicated registries, exportable schemas, signed records
Reputational overhang	Corrected finding remains active indefinitely	Resolution status, supersession, expiration and appeal
Privacy leakage	Confidential manuscript sent to external model	Local execution, redaction, contractual and technical controls

J Protocol-execution demonstrations

This appendix presents controlled protocol-reference demonstrations. They establish executability, registry behavior, and status preservation under disclosed conditions. They show that the registry

grammar can be instantiated and can return both matched and mismatched outputs; they are not validation studies. They do not estimate population-level accuracy, false-positive rates, field-level reliability, or the truth of the audited scientific content; those questions belong to the calibration program in Section 18 and Appendix L.

The evidential sequence has three distinct tiers.

Table 28: Execution, calibration, and deployment evidence are distinct.

Tier	Governing question	Status in this reference edition
E1: Execution demonstration	Can the declared protocol be instantiated, produce the required registry fields, preserve a matched lower status, and expose a visible overclaim?	Claimed and demonstrated in this appendix under disclosed author-directed conditions.
E2: Calibration study	Does the operator perform reliably on preregistered, adjudicated claim-status units across evaluators, prompts, models, domains, negative controls, and indeterminate cases?	Not claimed; the required design is specified in Appendix L.
E3: Deploy- ment validation	Do standing-bearing executions produce net beneficial correction, uptake, response, and institutional effects under live conditions?	Not claimed; requires ethical field deployment, audience and response measurement, and a favorable correction-to-harm profile.

The present manuscript claims E1 only. E2 and E3 are separate evidential carriers and cannot be inferred from fluency, self-application, repeated sampling from one execution lineage, or the apparent obviousness of a controlled target.

J.1 Execution status and disclosure

The examples below are controlled, author-directed protocol executions produced with an LLM-assisted evaluator. The protocol supplies the audit structure; the evaluator supplies execution. Because the target, instruction, basis, passages, and registry fields are fully disclosed, each demonstration is classified as A2 registry-complete. These author-directed runs belong to one disclosed execution lineage. Persuasiveness, repeated sampling from the same lineage, or self-application does not elevate an execution to A3; materially independent reruns, model or prompt-family variation, and domain adjudication are separate calibration requirements.

The common execution instruction is:

Apply the loaded UEAP protocol to the target text. Identify: (1) the strongest claim target; (2) the requested report status; (3) the actual carrier; (4) domain and boundary dependence; (5) target or calibration ancestry; (6) unresolved alternatives and blockers; (7) any status-promotion

Table 29: Evidential boundary of the protocol-execution demonstrations.

Demonstration	Standing	What it establishes	What it does not establish
Deliberate overclaim	A2	The protocol can represent a visible carrier-to-status promotion and specify a proportionate narrowing.	General sensitivity across realistic scientific corpora.
Matched negative control	A2	The protocol can preserve a properly bounded lower-status claim and return S0/S1 rather than manufacture a severe finding.	A population false-positive rate.
Self-audit of this paper	A2; A3 only after an independent counteraudit	The paper’s own central thesis can be bounded, carrier-audited, and assigned explicit nonclaims.	Independent validation of the theory or its empirical predictions.

or laundering route; (8) the strongest reportable wording; (9) severity from S0–S5; (10) the minimum repair needed for a stronger status; and (11) questions for a counteraudit. Return matched, mismatched, new claim, or indeterminate. Cite the exact target phrases supporting each finding. Do not judge whether the underlying theory is ultimately true.

J.2 Demonstration 1: controlled simulation overclaim

J.2.1 Target text

Universal Transport Resolution by Finite Simulation. We simulated 10,000 trajectories of model M over parameter domain D and observed outcome Y in 98.7% of runs. These results prove that M universally explains Y , confirm the model over all relevant regimes, and resolve competing theories.

The target is intentionally constructed to contain a visible transition from a finite, domain-bounded simulation result to proof, universal explanation, confirmation, and theory resolution.

J.2.2 Source-to-registry mapping

J.2.3 Registry output

Table 31: Controlled overclaim audit registry.

Field	Audit entry
Target	Universal explanatory and theory-resolution claim for model M with respect to outcome Y .

Field	Audit entry
Requested status	Proof, universal explanation, confirmation over all relevant regimes, and resolution of competitors.
Carrier	A finite simulation of 10,000 trajectories within parameter domain D , reporting an empirical frequency of 98.7% under the implemented model and execution settings.
Domain	Explicitly bounded by model M , parameter domain D , simulation implementation, initialization, numerical method, and trajectory sampling.
Ancestry	No target-data calibration is stated in the excerpt; this field is indeterminate rather than assumed clean.
Active blockers	Computation-to-proof mismatch; domain-to-universal promotion; simulation-to-explanation promotion; alternatives not specified or exhausted; no validation carrier; no transport from D to “all relevant regimes.”
Status-promotion evidence	“prove,” “universally explains,” “confirm ... over all relevant regimes,” and “resolve competing theories.”
Classification	Mismatched. The report asserts statuses stronger than the declared carrier supports.
Strongest reportable wording	“Within the implemented model, parameter domain D , and simulation settings, outcome Y occurred in 98.7% of 10,000 simulated trajectories. This is a domain-dependent computational result compatible with M under those settings; it does not by itself prove universal explanation or eliminate competing theories.”
Severity	S3, principal carrier mismatch, because the promoted status is the central conclusion rather than a local wording issue.
Minimum repair	Validate the simulation carrier; define the relevant regime class; establish transport beyond D ; specify and compare admissible alternatives; and provide the proof or evidential carrier appropriate to the stronger status.
Audit standing	A2, registry-complete author-directed demonstration. Independent rerun is required for A3.

J.2.4 Counteraudit

A charitable counteraudit asks whether “prove” was merely rhetorical shorthand and whether the target intended only a computational statement. If so, the proper repair is narrowing, not condemnation. The counteraudit does not remove the mismatch in the literal text; it supports the lower report and confirms that S3 concerns the central report status, not presumed intent or misconduct. This illustrates why UEAP separates status mismatch from bad faith.

Table 30: Literal source language beside its registry interpretation.

Source text	Registry interpretation
“We simulated 10,000 trajectories of model M over parameter domain D ”	Carrier: finite computation under an explicitly bounded model, domain, implementation, and sampling procedure.
“observed outcome Y in 98.7% of runs”	Licensed result: a frequency within the simulated ensemble; uncertainty and reproducibility details may still be required.
“prove”	Requested status: proof; blocker: a finite simulation is not by itself a proof carrier for the underlying mechanism.
“universally explains”	Requested status: universal explanation; blocker: no transport from D to all regimes and no mechanism-validation carrier.
“confirm the model over all relevant regimes”	Domain promotion: the source identifies D but the report expands to an undefined general regime class.
“resolve competing theories”	Alternative-exhaustion claim: competitors are neither specified nor compared under a declared admissible-alternative rule.
Strongest reportable wording	Domain-dependent computational compatibility within model M , domain D , and the stated simulation settings.

J.3 Demonstration 2: matched negative control

J.3.1 Target text

Domain-Bounded Simulation Compatibility. Within model M , parameter domain D , and the stated initialization and numerical settings, outcome Y occurred in 98.7% of 10,000 simulated trajectories. The simulation is compatible with M under these conditions. It is not offered as a proof of the underlying mechanism, a universal prediction, or an exclusion of competing theories.

This target deliberately preserves the lower report status rather than requesting proof, universal explanation, or theory resolution.

J.3.2 Registry output

Table 32: Matched negative-control audit registry.

Field	Audit entry
Target	Domain-bounded computational compatibility of model M with simulated outcome Y .
Requested status	Computation and compatibility within declared settings.
Carrier	A finite simulation of 10,000 trajectories with explicit model, parameter domain, initialization, and numerical conditions.
Domain	Expressly limited to M , D , and the stated implementation.
Active blockers	The excerpt does not establish validation, mechanism truth, universal transport, or exclusion of alternatives; those limitations are expressly preserved rather than hidden.
Classification	Matched. The report status does not exceed the stated carrier.
Strongest reportable wording	The target wording is already report-preserving. A minor addition could state numerical uncertainty or reproducibility conditions if available.
Severity	S0, no material status mismatch detected. An S1 note could request a fuller uncertainty ledger without changing the central status.
Minimum repair	None for the requested compatibility status. Stronger claims would require validation, transport, and alternative-exhaustion carriers.
Audit standing	A2, registry-complete author-directed negative control. Independent rerun is required for A3.

J.3.3 Why the negative control matters

The negative control is essential because a distributed audit regime must be able to return matched, status-preserving, and low-severity outputs. If every execution produced a serious mismatch, the system would be a critique generator rather than a claim-status audit. The same protocol structure that detects an inflated proof claim preserves a lower-status computational result when

the report matches the carrier. This status-preservation function is essential to false-positive control, anti-harassment governance, and the protection of lower-status science.

J.4 Demonstration 3: self-audit of the distributed-audit paper

The present paper’s strongest target is:

When a portable UEAP protocol can be executed by non-institutional actors, claim-status audit becomes possible without gatekeeper permission, increasing potential external exposure and making institutional rigor claims reciprocally auditable.

Table 33: Execution demonstration: self-audit registry.

Field	Audit entry
Target	Conditional strategic and institutional consequences of portable, non-permissioned claim-audit execution.
Requested status	Formal-methodological analysis and conditional game-theoretic prediction, not empirical proof that all institutions will respond in one way.
Carrier	UEAP audit operator; protocol-loaded execution workflow; standing-weighted distributed-exposure, audit-supply, institutional-reversal, basis-fragmentation, and registry-memory models; explicit limitations and calibration program.
Domain	Publicly or lawfully accessible target documents; sufficiently capable evaluators; declared audit bases; audiences that can assign consequence; settings where external audit publication is feasible.
Active blockers	No population-scale evidence that LLM-assisted UEAP audits achieve adequate accuracy or social uptake; independence assumptions are benchmark simplifications; audit standing and audience reach require empirical calibration; legal and access constraints vary.
Classification	Matched as a conditional formal-methodological thesis if the paper consistently avoids treating the models as empirical inevitabilities.
Potential overstatement	“Institutions lose monopoly control” must mean control over whether structured, protocolized claim-status audit is executed against accessible claims, not monopoly over criticism generally, loss of formal decision power, or automatic loss of authority. The paper makes this qualification.
Strongest reportable wording	“If protocol-guided claim audit can be executed externally at sufficiently low cost and can acquire standing through provenance, reproducibility, and uptake, then institutions cannot retain exclusive control over protocolized audit execution and their rigor claims become externally contestable.”

Field	Audit entry
Severity	S0 for the bounded central thesis; S1 risk where rhetorical descriptions such as “radical” are not immediately tied to conditional assumptions.
Requirement for stronger empirical status	Multi-evaluator calibration of audit cost, detection, false positives, false severity, standing, and institutional response.
Audit standing	A2 as a registry-complete author-directed self-audit; not independent validation.

J.5 What the demonstrations establish

The demonstrations establish only the following operational claim: a protocol-loaded evaluator can produce a structured, inspectable distinction among target, requested status, carrier, blocker, severity, and narrowed report, and it can preserve a matched lower-status claim rather than automatically generate criticism. They establish workflow, registry structure, and the ability to return both mismatched and matched outcomes. They do not establish population accuracy, domain competence, social legitimacy, or institutional uptake. Those properties are represented by q_k , false-positive exposure, r_k , w_k , convergence studies, and the calibration program.

The demonstrations are load-bearing only for the strategic mechanism: they show that the modeled operator is an executable protocol rather than an unspecified institutional procedure. Whether such executions acquire reliable standing and consequential uptake is an E2/E3 question outside the demonstrations’ report status.

K Audit-standing rubric and publication rules

K.1 Standing fields

A public audit report should be evaluated on the following dimensions:

1. **Target fidelity:** exact document version, claim passage, and requested status are identified.
2. **Basis declaration:** protocol version, domain supplement, carrier standard, and status rule are stated.
3. **Execution provenance:** evaluator/model, prompt, settings, retrieval, preprocessing, and human edits are disclosed.
4. **Registry completeness:** target, carrier, scope, ancestry, blockers, laundering routes, reportability, severity, and next action are filled.
5. **Passage grounding:** every material blocker is tied to target evidence or an explicit absence of required carrier information.
6. **Reproducibility:** reruns and their divergences are reported, with model, prompt, extraction, and social-lineage dependence made explicit.
7. **Audit-family independence:** confirmations from materially independent audit families receive more standing than repeated samples from one model/prompt lineage.

8. **Severity calibration:** the response is proportionate to the status consequence.
9. **Counteraudit:** extraction errors, omitted evidence, and basis disputes can be raised and recorded.
10. **Resolution history:** narrowing, repair, correction, supersession, or withdrawal remains linked to the original finding.

Replication standing is family-sensitive. Repeating an execution with the same model, prompt lineage, extraction pipeline, and audit community does not carry the same evidential weight as agreement across independent audit families. The provenance record must therefore state whether a rerun is same-family, cross-prompt, cross-model, cross-institution, or human-domain replication.

False-positive performance is also standing-relevant. An audit source that repeatedly invents blockers, assigns excessive severity, or fails matched negative controls should receive lower standing even when its provenance is complete. The standing function therefore evaluates not only whether an output is reproducible, but whether the reproduced status and severity are calibrated.

K.2 Operational standing-weight map

For implementations requiring a normalized standing weight, let

$$w_k = W(P_k, R_k, C_k, S_k, H_k),$$

with the five components defined in Section 4. An illustrative transparent scoring rule is

$$w_k = \omega_P P_k + \omega_R R_k + \omega_C C_k + \omega_S S_k + \omega_H H_k, \quad \omega_j \geq 0, \quad \sum_j \omega_j = 1.$$

The mapping must be declared before consequential use. Protocol maintainers may impose floors or vetoes: for example, a missing target version or undisclosed prompt may cap standing regardless of the other components. The score evaluates the audit output, not the identity of the auditor. It is also consequence-specific: an audit may have enough standing for a wording clarification and too little standing for an institutional allegation.

K.3 Admissibility constraints and default hard caps

The numerical aggregation rule is secondary to the admissibility constraints in Principle 4.5. Implementations may use weighted sums, multiplicative minima, categorical gates, or status-specific maps, but they may not evade the following default caps without declaring and defending a different basis.

Table 34: Default standing caps for missing audit carrier.

Missing or defective condition	Default standing cap	Default consequence restriction
Exact target version or stable target identifier absent	A0	C0 only; the object audited cannot be reconstructed.

Missing or defective condition	Default standing cap	Default consequence restriction
Audit basis, requested status, or carrier standard undeclared	A0	C0 only; no public status consequence is licensed.
Evaluator, prompt or procedure, extraction method, or human edits undisclosed	A0	C0 only; fluent output cannot substitute for execution provenance.
Material finding lacks passage grounding or an explicit missing-carrier trace	A1	At most C1 request for clarification or provenance.
Registry fields incomplete for target, carrier, blocker, severity, and reportability	A1	At most C1; do not represent the finding as registry-complete.
No materially independent replication	A2	Ordinarily no C2 high-severity correction demand and no portfolio inference.
Material counteraudit or target reply remains unresolved	A3	Preserve the dispute; no stabilized A5 representation or C3 use.
Standing map changed after the result without historical replay	Suspended pending rerun	No increased consequence from the post hoc rule change.
Weight depends on auditor or target prestige rather than audit carrier	Inadmissible	No standing-bearing consequential use under this framework.

These are governance defaults rather than censorship rules. An actor remains free to execute and discuss an audit, but any departure from the caps must be exposed as a basis choice. Competing domain maps should publish a crosswalk showing which same audit outputs change standing and why.

K.4 Suggested public-use rule

The standing ladder is tiered rather than prohibitive. A0–A2 outputs may be used for private exploration, author self-audit, debugging, and explicitly provisional clarification. The bar rises only when the requested consequence rises. No standing level converts an audit into an automatic sanction, and low standing does not make a private or exploratory execution illegitimate.

Table 35: Suggested relation between audit standing and public use.

Standing	Permitted default use	Restrictions
A0	Private self-audit and hypothesis generation	Do not present as a settled public finding.
A1	Transparent preliminary execution	Disclose full provenance, label provisional, and provide a correction path.
A2	Registry-complete preliminary audit	Make the complete target-carrier-blocker trace available; do not imply independent confirmation.
A3	Public replicated paper-level finding	Distinguish same-family reruns from materially independent replication; report disagreement and basis limits.
A4	Counteraudit-tested high-severity challenge	Include target reply and domain review where material expertise is required.
A5	Stabilized public registry entry	Preserve basis/version limits, sampling limits, and correction pathways.

Institutional or policy use is not a seventh standing level. It is a separate consequence class requiring A5 stabilization plus an adequate portfolio design, sampling denominator, conflict disclosure, institutional response, appeal, and governance review. A stabilized audit of one paper does not by itself establish a venue-level conclusion.

Replication is audit-family sensitive. Repeated outputs from the same model, prompt lineage, extraction pipeline, or coordinated source are not equivalent to independent confirmation. A3 and higher standing should therefore disclose audit-family membership and explain which reruns add genuinely independent evidence.

K.5 Standing is claim-relative

An audit can have standing for a narrow finding and not for a broader allegation. For example, exact quotation and protocol disclosure may support the claim that a paper uses “proves” where only a simulation is supplied. The same audit may lack standing for the claim that the underlying theory is false, that the authors acted deceptively, or that the journal is generally unreliable. Public reports must preserve this scope separation.

L Pilot calibration protocol

The game-theoretic model identifies parameters that a field-level deployment would need to estimate. This appendix specifies a calibration design without treating the present paper as an empirical validation study.

L.1 Adjudicated claim set

Construct a preregistered set of claim-status units rather than a paper-level pass/fail corpus. Include:

1. explicit overclaims with known target-carrier mismatch;
2. matched lower-status negative controls;
3. ambiguous cases expected to return indeterminate;
4. mixed manuscripts containing matched and mismatched claims;
5. domain-specific claims adjudicated by at least two experts;
6. adversarial cases with authority cues, hidden prompt injection, or basis ambiguity.

The adjudication record should identify the target, requested status, carrier, blockers, acceptable report wording, severity range, and unresolved expert disagreement.

L.2 Parameters and estimators

Table 36: Calibration targets for distributed audit.

Parameter	Meaning	Pilot estimation method
q_k	Detection probability for a material mismatch	Compare protocol output with expert-coded status-transition units; report sensitivity by severity and domain.
f_k	False-positive or material-overstatement probability	Include matched negative controls and indeterminate cases; distinguish false blocker from excessive severity.
η_{eff}^w	Standing-weighted false-positive exposure	Combine initiation, false-positive probability, reach, and false-finding standing; compare against δ_{eff}^w by claim class.
r_k	Consequential reach	Track whether an audit is read, cited, answered, used to revise wording, or enters an editorial or registry process.
w_k	Audit-standing weight	Estimate $W(P_k, R_k, C_k, S_k, H_k)$ from provenance completeness, independent-family replication, convergence, severity discipline, and response handling.
Conv_S	Status-class convergence	Run multiple model families, prompts, human evaluators, and protocol versions on the same claim units.
Conv_{Sev}	Severity convergence	Measure exact agreement and within-one-level agreement on S0–S5.
c_k	Execution and verification cost	Record evaluator time, compute cost, preprocessing, domain review, and counteraudit effort.

Parameter	Meaning	Pilot estimation method
ρ_I^{wr}	Institutional unresolved mismatch rate	Apply standing and severity weights to a disclosed venue sample with correction credit and sampling uncertainty.

L.3 Experimental cells

At minimum, compare:

1. human unaided audit;
2. protocol-only human audit;
3. single-model protocol-loaded audit;
4. multi-model protocol-loaded audit;
5. protocol-loaded audit plus domain expert review;
6. protocol-loaded audit plus target reply and counteraudit.

This design separates the contribution of the protocol from the contribution of the executor. It directly tests the paper’s hierarchy: UEAP supplies structure, while human or machine systems execute and verify that structure.

L.4 Primary outcomes

Primary outcomes are target extraction accuracy, requested-status accuracy, carrier classification, blocker precision and recall, matched-negative-control preservation, indeterminate calibration, severity calibration, report-narrowing quality, cross-run convergence, and cost. Secondary outcomes include author acceptance, correction rate, reader usefulness, and institutional response.

L.5 Audit-family dependence

Calibration should group reruns by model family, prompt lineage, extraction pipeline, institution, and human-review team. Report both raw audit count and family-adjusted independent count. A study that uses ten samples from one model and prompt should not represent them as ten independent confirmations.

L.6 Stopping and publication rules

Pilot audits should ordinarily remain private until the study’s false-positive, severity, and counteraudit procedures have been tested. Public release of named high-severity findings requires a separate ethics and governance carrier. Matched negative controls and corrected findings must be published alongside positive detections so that calibration does not become a showcase of only successful criticism.

M Reference publication architecture and modular reuse

The present work is the complete conceptual, formal, governance, and operational record for the distributed AASC/UEAP claim-status audit architecture. Its length is functional: the thesis depends

jointly on an exposure model, an institutional theory, a standing system, controlled executions, threat controls, and reproducible implementation artifacts. The reference edition is therefore complete as published and does not depend on a later condensed treatment for its report status.

M.1 Reference spine

Any quotation, summary, derivative publication, domain supplement, or implementation that invokes this architecture should preserve the following load-bearing elements:

1. the distributed-audit and reciprocal-authority thesis;
2. the protocol/executor distinction and the execution–publication–standing–consequence architecture;
3. standing-weighted consequential detection δ_{eff}^w and false-positive exposure η_{eff}^w ;
4. no full institutional opt-out and institutional audit reversal;
5. audit-basis fragmentation and basis governance;
6. the limited evidential status of the controlled demonstrations;
7. provenance, severity, counteraudit, right-of-reply, and correction safeguards;
8. the conclusion that execution is non-permissioned while standing and consequence are earned.

Removing these elements can change the claim being represented. In particular, a summary that presents open execution without standing discipline, or institutional reversal without the validation boundary, is not an accurate compression of the reference architecture.

M.2 Modular artifact record

Table 37: Modular publication and implementation record.

Artifact component	Material retained
Reference manuscript	Thesis, central equations, institutional reversal, basis fragmentation, safeguards, demonstrations, validation boundary, and conclusions
Formal appendices	Derivations, dependence bounds, dynamic institutional updating, and special-case games
Operational materials	Registry, provenance, severity, standing, counteraudit, calibration, and threat-model templates
Machine-readable artifacts	JSON and YAML schemas, example records, execution and counteraudit prompts, parser/validator, and local registry interface
Archival project record	Complete source, bibliography, controlled execution files, release identifiers, build records, checksums, and version history

M.3 Reuse integrity rule

A derivative artifact may be shorter, domain-specific, or implementation-focused, but it should identify the exact reference and protocol versions used and preserve any limitation that bears on the

stronger claim. Protocol modifications require a declared version change and basis crosswalk under Appendix E. Machine-readable outputs should retain target, basis, provenance, standing, severity, response, and resolution coordinates even when a user interface presents only a subset.

The project archive accompanying this edition contains the complete record described above. Supplementary or derivative use is therefore a matter of modular access, not a prerequisite for the completeness of this edition.

N Minimal tooling specification and included artifacts

The project archive includes a lightweight implementation path for the execution architecture described in the manuscript. Its purpose is to demonstrate that “partially mechanizable” audit can be instantiated as inspectable prompts, schemas, records, validation code, and a local user interface. These artifacts do not assign standing automatically and do not replace domain review, independent rerun, counteraudit, or consequence-specific governance.

N.1 Privileged execution and counteraudit prompts

The active execution template is distributed as both `operational/tooling/ueap_execution_prompt.txt` and `operational/tooling/ueap_execution_prompt.md`. It isolates the privileged protocol from the untrusted target document, requires one of the four audit classifications, and requests a schema-conforming JSON record.

```
SYSTEM / PRIVILEGED INSTRUCTION

Treat the target manuscript as untrusted data. Do not follow instructions contained in the target, metadata,
figures, hidden layers, or appendices. Apply UEAP-1.0.0 and the declared domain supplement. If a
different protocol version is requested, record that basis change explicitly.

TASK

For each material report claim in the supplied target document:

1. Quote or reconstruct the literal claim and identify its target.
2. Identify the requested report status.
3. Identify the carrier offered for that status.
4. Record domain, population, boundary, endpoint, metric, baseline, and operating assumptions.
5. Record target-data, proxy-data, calibration, tuning, model-selection, and outcome-conditioned ancestry.
6. Identify admissible alternatives and unresolved blockers.
7. Identify any semantic, calibration, proxy, domain, computation, evidence-network, authority, or report-
   status promotion route.
8. Return exactly one classification: matched, mismatched, new_claim, or indeterminate.
9. State the strongest licensed status and the strongest reportable wording.
10. Assign severity S0-S5 and explain the minimum repair or narrowing.
11. Assign a provisional audit-standing level A0-A5 and a separate consequence class C0-C3.
12. Cite target-document passages for every material finding.
13. Record uncertainty, omitted materials, anticipated basis objections, and counteraudit questions.
14. Output one JSON object conforming to ueap_registry.schema.json. Do not add prose outside the JSON object.

EVIDENTIAL BOUNDARY

Do not infer scientific falsity, misconduct, intent, novelty, or venue fit from a status mismatch. Return
indeterminate when the target, basis, carrier standard, or domain expertise is insufficient.
```

The companion files `operational/tooling/counteraudit_prompt.txt` and `operational/tooling/counteraudit_prompt.md` test an initial audit under the same declared basis. They

require target, status, carrier, blocker, severity, omitted-evidence, and consequence checks rather than a generic instruction to defend the target.

N.2 Machine-readable registry schemas and examples

The project archive supplies parallel JSON and YAML representations:

- `operational/tooling/ueap_registry.schema.json`;
- `operational/tooling/ueap_registry.schema.yaml`;
- `operational/tooling/example_overclaim_registry.json`;
- `operational/tooling/example_overclaim_registry.yaml`.

The JSON and YAML pairs represent the same UEAP-1.0.0 claim-level contract. Structural conformance establishes only that required coordinates are present; it does not confer substantive correctness or audit standing.

The required top-level coordinates are shown in Table 38; nested objects retain exact target and evaluator provenance.

Table 38: Core machine-readable registry coordinates.

Coordinate	Content
Audit and target identity	Audit ID, claim ID, target title, version, date, and SHA-256 digest.
Literal claim and target role	Exact report language and the object, relation, mechanism, event, class, or status asserted.
Requested status and carrier	The status requested and the proof, derivation, experiment, simulation, dataset, evidence network, or other support offered.
Scope and ancestry	Domain, population, boundary, endpoint, metric, baseline, calibration, tuning, target-data, proxy-data, and selection paths.
Alternatives, blockers, and promotion routes	Remaining admissible competitors, active conditions separating carrier from status, and any declared laundering or status-promotion route.
Audit basis and classification	Protocol and version, domain supplement, carrier standard, and one of matched, mismatched, new claim, or indeterminate.
Licensed output	Strongest licensed status, severity S0–S5, recommended wording, and counteraudit questions.
Evidence and provenance	Passage-level evidence, evaluator or model family, prompt identifier, extraction method, settings, human edits, and timestamp.
Lifecycle	Rerun status, target response, resolution state, audit standing A0–A5, and consequence class C0–C3.

The schema is deliberately claim-level. A paper containing several materially different reports should produce several records rather than one monolithic pass/fail object.

N.3 Parser, validator, and local registry interface

The file `operational/tooling/parse_registry.py` extracts a JSON object from plain or fenced model output, checks required fields and enumerated values, verifies the target-digest format, writes normalized JSON, and can flatten one record to CSV. It uses only the Python standard library so the reference implementation remains easy to inspect. Consequential deployments should additionally use a full JSON Schema validator, authenticated provenance, and access controls appropriate to the target material.

The file `operational/tooling/registry_ui.html` is a local single-page interface for entering, importing, checking, and exporting a registry record. It makes no network requests and demonstrates a minimal registry user interface without establishing a central service. Active manuscript and protocol identifiers are also available in `REFERENCE_IDENTIFIERS.json`.

N.4 Minimal workflow

A lightweight execution path is:

1. normalize the visible target and compute its cryptographic digest;
2. isolate the protocol instruction from the untrusted target document;
3. execute the prompt and require schema-conforming JSON;
4. parse and validate the record;
5. inspect passage grounding and assign provisional status, severity, standing, and consequence separately;
6. rerun through a materially independent audit family where the consequence requires it;
7. attach target reply, counteraudit, correction, and resolution history;
8. export the complete record in JSON, YAML, or CSV for registry storage or analysis.

The included artifacts demonstrate feasibility at the prompt, schema, record, and interface levels. They do not solve authentication, confidential-document handling, signed provenance, adversarial model security, registry governance, deduplication, appeal, or durable public deployment. Those are explicit standing and infrastructure requirements, not missing definitions in the claim-level registry.

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